

# MASTER PHYSICAL CHEMISTRY

## ELECTROCHEMISTRY

COMPREHENSIVE MASTER TEXTBOOK & PROBLEM COMPENDIUM

**JEE Main • JEE Advanced • Chemistry Olympiad (INChO/IChO)**

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*Synthesized from 8 Definitive Classical Authorities:*

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Pearson • Essential Physical Chemistry

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# Chapter 1

## Electrolytic Conduction & Ionic Migration

### 1.1 Mechanisms of Electrical Conduction

Electrical conduction represents the net directional transfer of electric charge under the influence of an applied electric potential gradient ( $-\nabla\phi$ ). All conductors are classified into two fundamental categories based on their charge carriers and the thermodynamic nature of the transport process:

| Property              | Metallic (Electronic) Conduction                                                                                                              | Electrolytic (Ionic) Conduction                                                                                                                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Charge Carriers       | Delocalized valence electrons moving through a fixed positive lattice (electron gas model).                                                   | Solvated cations and anions migrating toward electrodes of opposite polarity.                                                                                                                                                                                             |
| Chemical Change       | No chemical decomposition occurs; purely physical phenomenon.                                                                                 | Accompanied by chemical redox reactions at the electrode-electrolyte interfaces (electrolysis).                                                                                                                                                                           |
| Matter Transport      | No macroscopic transport of matter.                                                                                                           | Direct stoichiometric transport of matter in the form of ions toward the electrodes.                                                                                                                                                                                      |
| Effect of Temperature | Resistance <b>increases</b> with temperature ( $\frac{dR}{dT} > 0$ ) due to increased thermal vibrations (phonon scattering) of lattice ions. | Resistance <b>decreases</b> (conductance increases) with temperature ( $\frac{dG}{dT} > 0$ ) due to: (i) decrease in solvent viscosity ( $\eta$ ), (ii) increase in degree of dissociation ( $\alpha$ ) for weak electrolytes, and (iii) enhanced ionic thermal velocity. |
| Ohm's Law Compliance  | Strictly obeys Ohm's law ( $V = IR$ ) under all ordinary electric fields.                                                                     | Obeys Ohm's law ( $V = IR$ ) at moderate field strengths and frequencies; deviations occur at extremely high electric fields (Wien effect) or high frequencies (Debye-Falkenhagen effect).                                                                                |
| Resistance Magnitude  | Extremely low resistivity ( $\rho \approx 10^{-8}$ to $10^{-6} \Omega \cdot \text{m}$ ).                                                      | Moderate to high resistivity ( $\rho \approx 10^{-2}$ to $10^2 \Omega \cdot \text{m}$ ).                                                                                                                                                                                  |

Table 1.1: Comprehensive comparison of Metallic and Electrolytic Conduction.

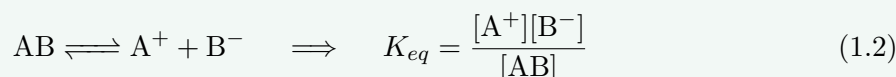
### Arrhenius Theory of Electrolytic Dissociation

According to Svante Arrhenius (1887):

1. When an electrolyte is dissolved in water, it spontaneously dissociates into positively charged ions (cations) and negatively charged ions (anions).
2. The total positive charge carried by the cations exactly equals the total negative charge carried by the anions, preserving macroscopic electrical neutrality:

$$\sum_i c_i z_i = 0 \quad (1.1)$$

3. For weak electrolytes, an equilibrium exists between the un-ionized molecules and the liberated ions:



4. The fraction of total electrolyte molecules that dissociate into ions is defined as the **degree of dissociation** ( $\alpha$ ):

$$\alpha = \frac{\text{Number of dissociated molecules}}{\text{Total number of molecules taken}} \quad (1.3)$$

## 1.2 Fundamental Quantities and Definitions

### 1.2.1 Resistance and Resistivity

When a uniform direct or alternating potential difference  $V$  (in volts) is applied across an electrolytic solution contained between two parallel electrodes of cross-sectional area  $A$  (in  $\text{cm}^2$ ) placed at a distance  $l$  (in cm) apart, the resistance  $R$  (in ohms,  $\Omega$ ) of the electrolyte column obeys Ohm's law:

$$R = \rho \cdot \frac{l}{A} \quad (1.4)$$

where  $\rho$  is the **resistivity** (or specific resistance) of the electrolytic solution (in  $\Omega \cdot \text{cm}$  or  $\Omega \cdot \text{m}$ ).

### 1.2.2 Conductance and Conductivity (Specific Conductance)

1. **Conductance** ( $G$ ): The ease with which electric current flows through the conductor:

$$G = \frac{1}{R} \quad \text{Unit: } \Omega^{-1} = \text{mho} = \text{Siemens (S)} \quad (1.5)$$

2. **Conductivity or Specific Conductance** ( $\kappa$ , **kappa**): Defined as the reciprocal of resistivity ( $\rho$ ):

$$\kappa = \frac{1}{\rho} = \frac{1}{R} \cdot \left( \frac{l}{A} \right) = G \cdot G^* \quad (1.6)$$

where  $G^* = \frac{l}{A}$  is the **Cell Constant** (in  $\text{cm}^{-1}$  or  $\text{m}^{-1}$ ).

- **Physical Meaning:** Conductivity  $\kappa$  represents the electrical conductance of a column of electrolyte solution having a unit length (1 cm) and a unit cross-sectional area ( $1 \text{ cm}^2$ ), i.e., the conductance of exactly  $1 \text{ cm}^3$  (1 mL) of the solution between two parallel plates.
- **SI Unit:**  $\text{S} \cdot \text{m}^{-1} = \Omega^{-1} \cdot \text{m}^{-1}$ .
- **CGS Unit:**  $\text{S} \cdot \text{cm}^{-1} = \Omega^{-1} \cdot \text{cm}^{-1}$ .
- **Conversion Factor:**

$$1 \text{ S} \cdot \text{cm}^{-1} = 100 \text{ S} \cdot \text{m}^{-1} \quad \Longleftrightarrow \quad 1 \text{ S} \cdot \text{m}^{-1} = 10^{-2} \text{ S} \cdot \text{cm}^{-1} \quad (1.7)$$

### 1.2.3 Cell Constant ( $G^*$ )

Since the exact geometric distance  $l$  and active plate surface area  $A$  of platinum electrodes in a conductivity cell cannot be measured accurately with a ruler due to non-uniform current field fringing at the plate edges, the cell constant  $G^* = \frac{l}{A}$  is determined experimentally using a standard standard solution of known conductivity—typically standardized aqueous KCl at 25°C (298 K):

$$G^* = \kappa_{\text{known}} \times R_{\text{measured}} \quad (1.8)$$

Once  $G^*$  is calibrated for a specific cell, the conductivity of any unknown electrolyte solution placed in the same cell is evaluated as:

$$\kappa_{\text{unknown}} = \frac{G^*}{R_{\text{measured}}} \quad (1.9)$$

## 1.3 Molar Conductivity and Equivalent Conductivity

### 1.3.1 Molar Conductivity ( $\Lambda_m$ )

**Definition:** Molar conductivity ( $\Lambda_m$ ) is defined as the conducting power of all the ions produced by dissolving exactly 1 mole of an electrolyte in a given volume  $V$  (in mL) of solution placed between two sufficiently large electrodes held 1 cm apart.

Let 1 mole of electrolyte be dissolved in  $V$  mL of solution. Since the conductance of 1 mL (1 cm<sup>3</sup>) of solution is equal to its conductivity  $\kappa$ , the conductance of  $V$  mL of solution containing 1 mole of electrolyte is:

$$\Lambda_m = \kappa \times V_{\text{in mL/mole}} \quad (1.10)$$

If the molar concentration of the solution is  $M$  (molarity in mol · L<sup>-1</sup>):

$$M \text{ moles are present in } 1000 \text{ mL} \implies V = \frac{1000}{M} \text{ mL} \cdot \text{mol}^{-1} \quad (1.11)$$

Substituting  $V$  into the equation gives:

$$\boxed{\Lambda_m = \frac{\kappa \times 1000}{M}} \quad \text{where } \kappa \text{ is in } \text{S} \cdot \text{cm}^{-1} \text{ and } M \text{ is in } \text{mol} \cdot \text{L}^{-1} \quad (1.12)$$

**Units of  $\Lambda_m$ :**

$$[\Lambda_m] = \frac{\text{S} \cdot \text{cm}^{-1}}{\text{mol} \cdot \text{cm}^{-3}} = \text{S} \cdot \text{cm}^2 \cdot \text{mol}^{-1} = \Omega^{-1} \cdot \text{cm}^2 \cdot \text{mol}^{-1} \quad (1.13)$$

**In SI Units:** If  $\kappa$  is in S · m<sup>-1</sup> and concentration  $c$  is in mol · m<sup>-3</sup> ( $c = 1000 \times M$ ):

$$\boxed{\Lambda_m = \frac{\kappa}{c} = \frac{\kappa}{1000 \times M}} \quad (\text{S} \cdot \text{m}^2 \cdot \text{mol}^{-1}) \quad (1.14)$$

**Unit Interconversion:**

$$1 \text{ S} \cdot \text{m}^2 \cdot \text{mol}^{-1} = 10^4 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1} \iff 1 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1} = 10^{-4} \text{ S} \cdot \text{m}^2 \cdot \text{mol}^{-1} \quad (1.15)$$

### 1.3.2 Equivalent Conductivity ( $\Lambda_{eq}$ )

**Definition:** Equivalent conductivity ( $\Lambda_{eq}$ ) is defined as the conducting power of all the ions produced by dissolving exactly 1 gram-equivalent of an electrolyte in a volume  $V$  of solution placed between two large electrodes spaced 1 cm apart.

$$\boxed{\Lambda_{eq} = \frac{\kappa \times 1000}{N}} \quad (\text{in } \text{S} \cdot \text{cm}^2 \cdot \text{eq}^{-1} \text{ or } \Omega^{-1} \cdot \text{cm}^2 \cdot \text{eq}^{-1}) \quad (1.16)$$

where  $N$  is the normality of the solution (eq · L<sup>-1</sup>).

### 1.3.3 Exact Relationship Between $\Lambda_m$ and $\Lambda_{eq}$

Recall the fundamental definition relating normality and molarity:

$$N = M \times n\text{-factor} \quad (\text{or } z) \quad (1.17)$$

where  $n$ -factor ( $z$ ) is the total positive or negative valency per formula unit of the electrolyte (e.g., for NaCl,  $z = 1$ ; for CaCl<sub>2</sub>,  $z = 2$ ; for Al<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub>,  $z = 6$ ). Therefore:

$$\Lambda_{eq} = \frac{\kappa \times 1000}{M \times z} = \frac{\Lambda_m}{z} \implies \boxed{\Lambda_m = z \times \Lambda_{eq}} \quad (1.18)$$

#### Illustration 1.1: Comprehensive Interconversion of Conductivity Units

The resistance of a 0.05 M solution of an electrolyte M<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> in a conductivity cell is 240  $\Omega$ . The distance between the electrodes is 1.5 cm and their cross-sectional area is 3.75 cm<sup>2</sup>. Calculate:

- The cell constant ( $G^*$ ).
- The conductivity ( $\kappa$ ) in both S · cm<sup>-1</sup> and S · m<sup>-1</sup>.
- The molar conductivity ( $\Lambda_m$ ) in S · cm<sup>2</sup> · mol<sup>-1</sup> and S · m<sup>2</sup> · mol<sup>-1</sup>.
- The equivalent conductivity ( $\Lambda_{eq}$ ).

**Step 1: Cell Constant ( $G^*$ ):**

$$G^* = \frac{l}{A} = \frac{1.5 \text{ cm}}{3.75 \text{ cm}^2} = 0.40 \text{ cm}^{-1} = 40.0 \text{ m}^{-1}$$

**Step 2: Conductivity ( $\kappa$ ):**

$$\kappa = \frac{G^*}{R} = \frac{0.40 \text{ cm}^{-1}}{240 \Omega} = 1.667 \times 10^{-3} \text{ S} \cdot \text{cm}^{-1}$$

$$\kappa_{\text{SI}} = \kappa \times 100 = 0.1667 \text{ S} \cdot \text{m}^{-1}$$

**Step 3: Molar Conductivity ( $\Lambda_m$ ):**

$$\Lambda_m = \frac{\kappa \times 1000}{M} = \frac{(1.667 \times 10^{-3} \text{ S} \cdot \text{cm}^{-1}) \times 1000}{0.05 \text{ mol} \cdot \text{L}^{-1}} = 33.33 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$$

$$\Lambda_{m,\text{SI}} = 33.33 \times 10^{-4} \text{ S} \cdot \text{m}^2 \cdot \text{mol}^{-1} = 3.333 \times 10^{-3} \text{ S} \cdot \text{m}^2 \cdot \text{mol}^{-1}$$

**Step 4: Equivalent Conductivity ( $\Lambda_{eq}$ ):** For the salt M<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub>, the total cationic charge is 2 × (+3) = +6, so the valence factor is  $z = 6$ .

$$\Lambda_{eq} = \frac{\Lambda_m}{z} = \frac{33.33}{6} = 5.556 \text{ S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$$

## 1.4 Effect of Dilution on Conductance Parameters

Understanding the variation of conductance parameters with dilution (decreasing concentration  $C$ ) is one of the most critical conceptual foundations in physical electrochemistry:

### 1. Conductance ( $G$ ): Increases on dilution.

- For *weak electrolytes*: Dilution increases the degree of dissociation  $\alpha$  (Ostwald's law), generating a larger absolute count of conducting ions.

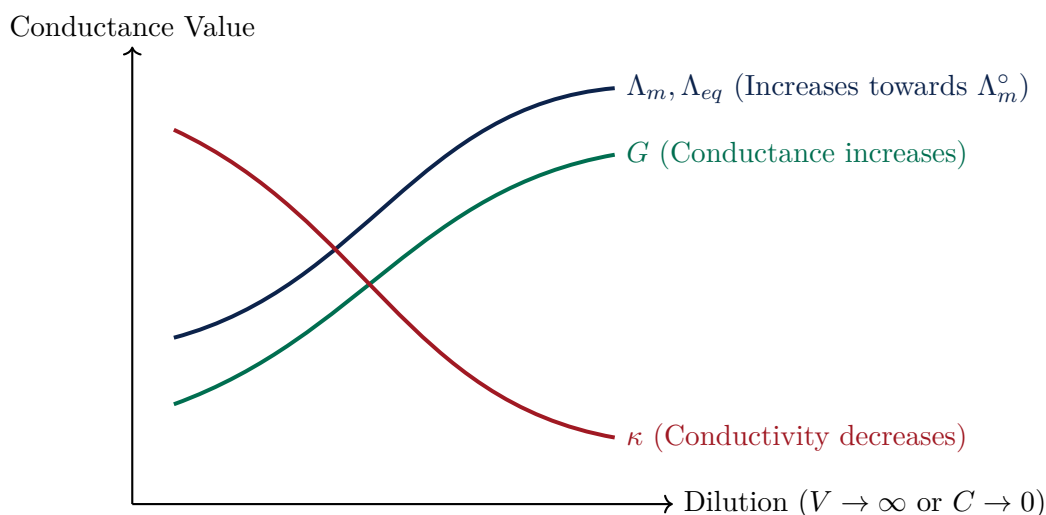


Figure 1.1: Contrasting dilution behavior of  $G$ ,  $\Lambda_m$ , and  $\kappa$ .

- For *strong electrolytes*: Dilution weakens the interionic electrostatic attractions, reducing ionic friction and enabling ions to move with greater mobility.

## 2. Conductivity ( $\kappa$ ): Decreases on dilution.

- Conductivity is the conductance of exactly  $1 \text{ cm}^3$  (1 mL) of solution.
- On adding solvent, although the total number of ions in the entire bulk increases or remains constant, the number of current-carrying ions per unit volume (1 mL) strictly decreases. Hence,  $\kappa$  inevitably decreases upon dilution for both strong and weak electrolytes.

## 3. Molar Conductivity ( $\Lambda_m$ ) and Equivalent Conductivity ( $\Lambda_{eq}$ ): Increases on dilution.

$$\Lambda_m = \kappa \times V \quad (1.19)$$

Upon dilution, while  $\kappa$  decreases, the dilution volume  $V$  (volume containing 1 mole of electrolyte) increases significantly. The dramatic increase in volume  $V$  completely overwhelms the decrease in  $\kappa$ , resulting in a net increase in  $\Lambda_m$  and  $\Lambda_{eq}$ . At infinite dilution ( $C \rightarrow 0$  or  $V \rightarrow \infty$ ), the molar conductivity reaches a maximum limiting value termed the **limiting molar conductivity** ( $\Lambda_m^\circ$  or  $\Lambda_m^\infty$ ).

—

## 1.5 Debye-Hückel-Onsager (DHO) Theory for Strong Electrolytes

Strong electrolytes are completely dissociated into ions even at moderate concentrations ( $\alpha \approx 1$ ). Therefore, the increase in  $\Lambda_m$  upon dilution cannot be attributed to an increase in ionization. Instead, the modern theory of electrolytes developed by Peter Debye, Erich Hückel (1923), and Lars Onsager (1927) explains this variation through interionic electrostatic interactions and the dynamics of the **ionic atmosphere**.

### Atkins Deep Dive: The Ionic Atmosphere and Relaxation Time

In an electrolytic solution, every central ion of charge  $z_i e$  is surrounded on average by a spherical cloud of ions possessing a net opposite charge—the **ionic atmosphere**.

- The effective radius of this ionic cloud is the **Debye screening length** ( $\kappa_D^{-1}$ ):

$$\kappa_D = \left( \frac{2e^2 N_A I}{\epsilon_0 \epsilon_r k_B T} \right)^{1/2} \implies \kappa_D^{-1} \propto \frac{1}{\sqrt{I}} \quad (1.20)$$

where  $I$  is the ionic strength of the medium.

- Under stationary equilibrium conditions, the charge density of the ionic atmosphere is spherically symmetric around the central ion, producing zero net horizontal electrostatic force.

When an external electric field is applied, two retarding effects arise that slow down the motion of the central ion:

- The Asymmetry or Relaxation Effect:** Under the external field, the central ion moves in one direction (e.g., cation toward cathode), while its oppositely charged ionic atmosphere migrates in the opposite direction (anion cloud toward anode).
  - The ionic atmosphere does not reform instantly ahead of the migrating ion, nor does it dissipate instantaneously behind it. A finite time is required for charge redistribution, known as the **relaxation time** ( $\tau \approx 10^{-10}$  s).
  - Consequently, the ionic atmosphere becomes distorted and asymmetric, leaving a surplus of counter-charge behind the moving central ion. This lagging charge cloud exerts an electrostatic backward drag, retarding the ionic velocity.
- The Electrophoretic Effect:** Ions in aqueous solutions are heavily solvated. When the counter-ions forming the ionic atmosphere move in the direction opposite to the central ion, they carry their solvation shells with them.
  - The central ion is therefore compelled to swim upstream against a counter-current flow of the viscous solvent.
  - This additional hydrodynamic resistance is the electrophoretic drag, analogous to electrophoresis of charged colloidal particles.
- Normal Viscous Drag:** The macroscopic hydrodynamic friction governed by Stokes' Law:

$$F_{\text{visc}} = 6\pi\eta r_i v_i \quad (1.21)$$

where  $\eta$  is solvent viscosity,  $r_i$  is the effective hydrodynamic (solvated) radius, and  $v_i$  is the ionic drift velocity.

#### 1.5.1 The Debye-Hückel-Onsager (DHO) Equation

By combining the relaxation drag and the electrophoretic drag with Stokes' law, Onsager derived the rigorous mathematical relation for univalent electrolytes:

$$\Lambda_m = \Lambda_m^\circ - (A + B\Lambda_m^\circ) \sqrt{C} \quad (1.22)$$

where  $C$  is the molar concentration, and  $A$  and  $B$  are constants depending solely on the temperature  $T$ , solvent dielectric constant  $\varepsilon_r$ , and viscosity  $\eta$ :

$$A = \frac{z^2 e F}{3\pi\eta} \left( \frac{2e^2 N_A}{\varepsilon_0 \varepsilon_r k_B T} \right)^{1/2} \quad (\text{Electrophoretic term}) \quad (1.23)$$

$$B = \frac{q z^3 e^2}{24\pi \varepsilon_0 \varepsilon_r k_B T} \left( \frac{2e^2 N_A}{\varepsilon_0 \varepsilon_r k_B T} \right)^{1/2} \quad (\text{Relaxation term}) \quad (1.24)$$

For a given solvent and fixed temperature, the equation simplifies to the classic linear form:

$$\Lambda_m = \Lambda_m^\circ - b\sqrt{C} \quad (1.25)$$

where  $b = A + B\Lambda_m^\circ$  is an empirical constant governed by electrolyte type (1 : 1, 2 : 1, etc.).

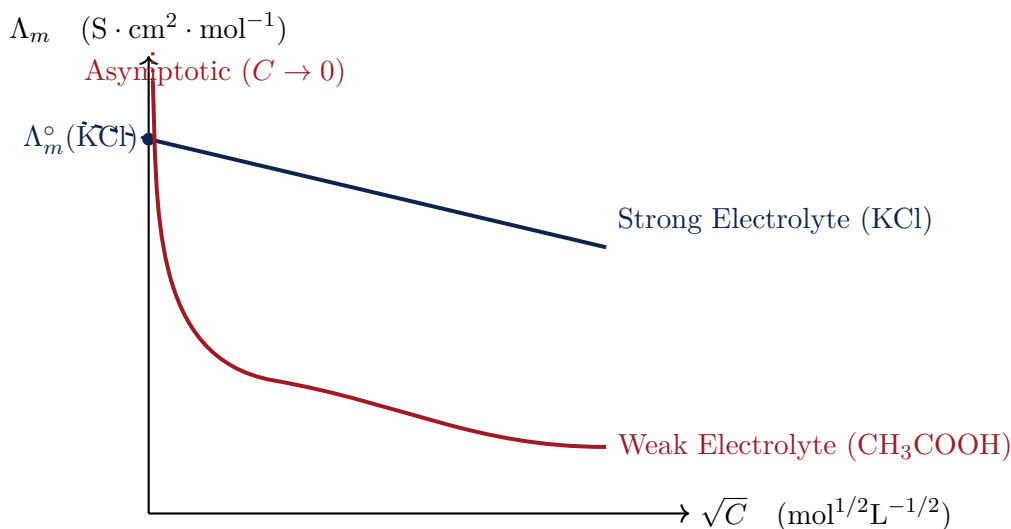


Figure 1.2: DHO plot of  $\Lambda_m$  versus  $\sqrt{C}$  for Strong vs. Weak Electrolytes.

### Crucial Comparison: Strong vs. Weak Electrolyte Plots

- **Strong Electrolyte** (KCl, HCl, NaCl): The curve is a straight line with slope  $-b$ . At infinite dilution ( $\sqrt{C} \rightarrow 0$ ), the plot can be extrapolated linearly to the vertical axis to accurately yield the intercept  $\Lambda_m^\circ$ .
- **Weak Electrolyte** (CH<sub>3</sub>COOH, NH<sub>4</sub>OH): As  $C \rightarrow 0$ , the degree of dissociation  $\alpha \rightarrow 1$ , causing  $\Lambda_m$  to shoot up asymptotically almost parallel to the vertical axis. Hence,  $\Lambda_m^\circ$  for weak electrolytes **cannot be determined by experimental graphical extrapolation!** It must be computed indirectly via **Kohlrausch's Law**.

## 1.6 Debye-Hückel Limiting Law (DHLL) & Ionic Strength

In non-ideal electrolyte solutions, concentrations must be replaced by chemical activities:

$$a_i = \gamma_i \cdot \frac{c_i}{c^\circ} \quad (1.26)$$

where  $\gamma_i$  is the activity coefficient. For a binary electrolyte  $M_{\nu+} X_{\nu-}$  dissociating into  $\nu_+$  cations and  $\nu_-$  anions ( $\nu = \nu_+ + \nu_-$ ), the mean ionic activity coefficient  $\gamma_\pm$  is defined as:

$$\gamma_\pm = (\gamma_+^{\nu_+} \gamma_-^{\nu_-})^{1/\nu} \quad (1.27)$$

### 1.6.1 Ionic Strength ( $I$ )

Introduced by Lewis and Randall (1921), the ionic strength accounts for the total electrostatic intensity of the ionic environment:

$$I = \frac{1}{2} \sum_i c_i z_i^2 \quad (1.28)$$

where  $c_i$  is the molar or molal concentration of ion  $i$ , and  $z_i$  is its ionic valence (signed or absolute).

#### Illustration 1.2: Calculation of Ionic Strength for Electrolyte Mixtures

Calculate the ionic strength ( $I$ ) of an aqueous solution containing 0.02 M  $\text{Fe}_2(\text{SO}_4)_3$  and 0.05 M  $\text{CaCl}_2$ .

##### Dissociation Stoichiometry:

- 0.02 M  $\text{Fe}_2(\text{SO}_4)_3 \implies [\text{Fe}^{3+}] = 2 \times 0.02 = 0.04 \text{ M}$ ,  $[\text{SO}_4^{2-}] = 3 \times 0.02 = 0.06 \text{ M}$ .
- 0.05 M  $\text{CaCl}_2 \implies [\text{Ca}^{2+}] = 0.05 \text{ M}$ ,  $[\text{Cl}^-] = 2 \times 0.05 = 0.10 \text{ M}$ .

##### Ionic Strength Formulation:

$$\begin{aligned} I &= \frac{1}{2} \left[ [\text{Fe}^{3+}] \cdot (+3)^2 + [\text{SO}_4^{2-}] \cdot (-2)^2 + [\text{Ca}^{2+}] \cdot (+2)^2 + [\text{Cl}^-] \cdot (-1)^2 \right] \\ &= \frac{1}{2} [0.04 \times 9 + 0.06 \times 4 + 0.05 \times 4 + 0.10 \times 1] \\ &= \frac{1}{2} [0.36 + 0.24 + 0.20 + 0.10] = \frac{1}{2} \times 0.90 = 0.45 \text{ M} \end{aligned}$$

### 1.6.2 Debye-Hückel Limiting Law (DHLL)

At very low concentrations ( $I < 0.01 \text{ M}$ ), the Debye-Hückel limiting law states:

$$\log_{10} \gamma_{\pm} = -A |z_+ z_-| \sqrt{I} \quad (1.29)$$

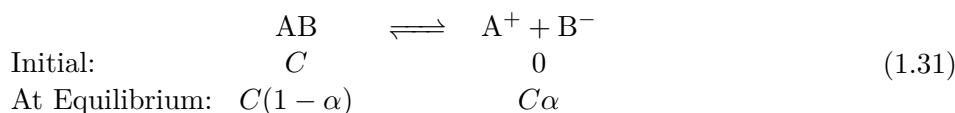
For water at 25°C (298 K), dielectric constant  $\epsilon_r = 78.54$ , the constant  $A$  evaluates to:

$$A = 0.509 \text{ mol}^{-1/2} \cdot \text{kg}^{1/2} \quad (\approx 0.51) \quad (1.30)$$

—

## 1.7 Ostwald's Dilution Law for Weak Electrolytes

Consider a binary weak electrolyte AB (such as  $\text{CH}_3\text{COOH}$ ) at initial analytical concentration  $C$  with degree of dissociation  $\alpha$ :



The equilibrium ionization constant  $K_a$  (or  $K_c$ ) is:

$$K_a = \frac{[\text{A}^+][\text{B}^-]}{[\text{AB}]} = \frac{(C\alpha)(C\alpha)}{C(1 - \alpha)} = \frac{C\alpha^2}{1 - \alpha} \quad (1.32)$$

Since for weak electrolytes  $\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$ :

$$K_a = \frac{C \left( \frac{\Lambda_m}{\Lambda_m^\circ} \right)^2}{1 - \frac{\Lambda_m}{\Lambda_m^\circ}} = \frac{C \Lambda_m^2}{\Lambda_m^\circ (\Lambda_m^\circ - \Lambda_m)} \quad (1.33)$$

If the electrolyte is very weak,  $\alpha \ll 1 \implies 1 - \alpha \approx 1$ :

$$K_a \approx C\alpha^2 \implies \alpha = \sqrt{\frac{K_a}{C}} \quad (1.34)$$

Rearranging the exact equation into linear form gives the famous **Ostwald-Kraus plot**:

$$\frac{1}{\Lambda_m} = \frac{1}{\Lambda_m^\circ} + \frac{\Lambda_m \cdot C}{K_a(\Lambda_m^\circ)^2} \quad (1.35)$$

Plotting  $\frac{1}{\Lambda_m}$  against  $C\Lambda_m$  yields a straight line with intercept  $\frac{1}{\Lambda_m^\circ}$  and slope  $\frac{1}{K_a(\Lambda_m^\circ)^2}$ , providing an experimental route to evaluate both  $\Lambda_m^\circ$  and  $K_a$  simultaneously.

## 1.8 Kohlrausch's Law of Independent Migration of Ions

Friedrich Kohlrausch (1875–1879) deduced from exhaustive experimental measurements that at infinite dilution, where interionic attractions become completely negligible, every ion makes a definite, independent contribution to the total conductance of the electrolyte, regardless of the identity of the counter-ion with which it is associated.

### Mathematical Formulations of Kohlrausch's Law

**1. In Terms of Molar Conductivities:** The limiting molar conductivity of an electrolyte is the sum of the limiting molar conductivities of its constituent ions, multiplied by their stoichiometric coefficients:

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ \quad (1.36)$$

where  $\lambda_+^\circ$  and  $\lambda_-^\circ$  are the limiting molar ionic conductivities of the cation and anion, and  $\nu_+$ ,  $\nu_-$  are the number of cations and anions released per formula unit.

**2. In Terms of Equivalent Conductivities:** The limiting equivalent conductivity of an electrolyte is simply the direct sum of the limiting equivalent ionic conductivities:

$$\Lambda_{eq}^\circ = \lambda_{eq,+}^\circ + \lambda_{eq,-}^\circ \quad (1.37)$$

Note the crucial relationship:

$$\lambda_{eq,+}^\circ = \frac{\lambda_+^\circ}{z_+} \quad \text{and} \quad \lambda_{eq,-}^\circ = \frac{\lambda_-^\circ}{z_-} \quad (1.38)$$

### 1.8.1 Fundamental Applications of Kohlrausch's Law

#### 1. Determination of $\Lambda_m^\circ$ for Weak Electrolytes

Weak electrolytes like acetic acid cannot have their  $\Lambda_m^\circ$  determined by direct extrapolation. However, by combining the  $\Lambda_m^\circ$  values of strong electrolytes containing the same ions:

$$\Lambda_m^\circ(\text{CH}_3\text{COOH}) = \Lambda_m^\circ(\text{CH}_3\text{COONa}) + \Lambda_m^\circ(\text{HCl}) - \Lambda_m^\circ(\text{NaCl}) \quad (1.39)$$

**Proof:**

$$[\lambda^\circ(\text{CH}_3\text{COO}^-) + \lambda^\circ(\text{Na}^+)] + [\lambda^\circ(\text{H}^+) + \lambda^\circ(\text{Cl}^-)] - [\lambda^\circ(\text{Na}^+) + \lambda^\circ(\text{Cl}^-)] = \lambda^\circ(\text{CH}_3\text{COO}^-) + \lambda^\circ(\text{H}^+)$$

#### 2. Determination of Degree of Dissociation ( $\alpha$ ) and $K_a/K_b$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{\Lambda_{eq}}{\Lambda_{eq}^\circ} \quad (1.40)$$

Once  $\alpha$  is evaluated at concentration  $C$ :

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{C\Lambda_m^2}{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)} \quad (1.41)$$

### 3. Determination of Solubility and $K_{sp}$ of Sparingly Soluble Salts

Salts such as  $\text{AgCl}$ ,  $\text{BaSO}_4$ ,  $\text{PbSO}_4$  are so sparingly soluble in water that their saturated solutions are extremely dilute. Hence, the solution can be treated as being at **infinite dilution**:

$$\Lambda_m = \Lambda_m^\circ \quad (1.42)$$

The experimental conductivity of the salt  $\kappa_{\text{salt}}$  is obtained by subtracting the solvent (water) conductivity from the solution conductivity:

$$\kappa_{\text{salt}} = \kappa_{\text{solution}} - \kappa_{\text{H}_2\text{O}} \quad (1.43)$$

The molar solubility  $S$  (in  $\text{mol} \cdot \text{L}^{-1}$ ) is then:

$$\Lambda_m^\circ = \frac{\kappa_{\text{salt}} \times 1000}{S} \implies S = \frac{\kappa_{\text{salt}} \times 1000}{\Lambda_m^\circ} \quad (1.44)$$

The solubility product  $K_{sp}$  is then calculated according to the dissolution stoichiometry:

- For 1 : 1 salt ( $\text{AgCl}$ ):  $K_{sp} = S^2$ .
- For 1 : 2 or 2 : 1 salt ( $\text{PbCl}_2$ ,  $\text{Ag}_2\text{CrO}_4$ ):  $K_{sp} = 4S^3$ .
- For 1 : 3 or 3 : 1 salt ( $\text{Al}(\text{OH})_3$ ):  $K_{sp} = 27S^4$ .
- For 2 : 3 salt ( $\text{Fe}_2(\text{SO}_4)_3$ ):  $K_{sp} = 108S^5$ .

### 4. Determination of Ionic Product of Water ( $K_w$ )

Extremely pure water dissociates weakly:  $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$ . The conductivity of pure water at  $25^\circ\text{C}$  is  $\kappa_{\text{H}_2\text{O}} = 5.5 \times 10^{-8} \text{ S} \cdot \text{cm}^{-1}$ . The limiting molar conductivity of water is:

$$\Lambda_m^\circ(\text{H}_2\text{O}) = \lambda^\circ(\text{H}^+) + \lambda^\circ(\text{OH}^-) = 349.8 + 198.6 = 548.4 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$$

The concentration of dissociated ions  $[\text{H}^+] = [\text{OH}^-] = c$  is:

$$c = \frac{\kappa_{\text{H}_2\text{O}} \times 1000}{\Lambda_m^\circ(\text{H}_2\text{O})} = \frac{5.5 \times 10^{-8} \times 1000}{548.4} \approx 1.003 \times 10^{-7} \text{ mol} \cdot \text{L}^{-1} \quad (1.45)$$

Therefore, the ionic product  $K_w$  at 298 K is:

$$K_w = [\text{H}^+][\text{OH}^-] = (1.003 \times 10^{-7})^2 \approx 1.0 \times 10^{-14} \text{ mol}^2 \cdot \text{L}^{-2} \quad (1.46)$$

—

## 1.9 Ionic Mobility, Drift Speed, and Transport Numbers

### 1.9.1 Ionic Mobility ( $u$ )

Under an applied electric field  $E = \frac{V}{l}$  (potential gradient in  $\text{V} \cdot \text{cm}^{-1}$ ), an ion attains a terminal constant drift velocity  $v$  (in  $\text{cm} \cdot \text{s}^{-1}$ ) due to balance between the electric driving force ( $F_{el} = z_i e E$ ) and Stokes' frictional drag ( $F_{drag} = 6\pi\eta r_i v$ ).

**Definition:** The **ionic mobility** ( $u$ ) is defined as the drift speed of an ion under a unit potential gradient ( $1 \text{ V} \cdot \text{cm}^{-1}$  or  $1 \text{ V} \cdot \text{m}^{-1}$ ):

$$u = \frac{v}{E} = \frac{\text{Drift velocity}}{\text{Potential gradient}} \quad \left( \text{Unit: } \frac{\text{cm} \cdot \text{s}^{-1}}{\text{V} \cdot \text{cm}^{-1}} = \text{cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1} \right) \quad (1.47)$$

In SI units:  $\text{m}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$ . Note that:

$$1 \text{ m}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1} = 10^4 \text{ cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1} \quad (1.48)$$

### 1.9.2 Fundamental Relation Between $\lambda_m^\circ$ and Ionic Mobility $u^\circ$

Consider a cross-section of solution with 1 mole of ions of charge  $z_i$  moving across an area under unit electric field. The total charge transported per second corresponds to the limiting ionic conductivity:

$$\boxed{\lambda_m^\circ = z_i u_i^\circ F} \quad \text{and} \quad \boxed{\lambda_{eq}^\circ = u_i^\circ F} \quad (1.49)$$

where  $F = 96485 \text{ C} \cdot \text{mol}^{-1}$  (Faraday's constant).

### 1.9.3 Transport Number (Transference Number, $t_+$ and $t_-$ )

**Definition:** The fraction of the total electric current carried by a particular ionic species is termed its **transport number**:

$$t_+ = \frac{I_+}{I_{\text{total}}} = \frac{u_+}{u_+ + u_-} = \frac{\lambda_+^\circ}{\lambda_m^\circ} \quad \text{and} \quad t_- = \frac{I_-}{I_{\text{total}}} = \frac{u_-}{u_+ + u_-} = \frac{\lambda_-^\circ}{\lambda_m^\circ} \quad (1.50)$$

**Fundamental Identity:**

$$\boxed{t_+ + t_- = 1} \quad (1.51)$$

### 1.9.4 Hittorf's Rule and Experimental Methods

**Hittorf's Rule:** The loss of concentration around any electrode compartment during electrolysis is directly proportional to the speed (mobility) of the ion that is migrating *away* from that compartment:

$$\frac{\text{Loss of electrolyte around anode}}{\text{Loss of electrolyte around cathode}} = \frac{u_+}{u_-} = \frac{t_+}{t_-} \quad (1.52)$$

$$t_+ = \frac{\text{Fall in concentration around anode}}{\text{Total fall in concentration around both compartments}} = \frac{\text{Gram-equivalents lost at anode}}{\text{Total gram-equivalents deposited at electrode}} \quad (1.53)$$

### 1.9.5 The Moving Boundary Method

The moving boundary method is the most precise absolute method for determining transport numbers. A sharp optical boundary between two electrolyte solutions having a common ion (e.g., HCl as principal electrolyte and  $\text{CdCl}_2$  as indicator electrolyte) is observed in a calibrated vertical tube.

If the boundary sweeps through a volume  $V$  (in mL) across distance  $x$  in a tube of cross-sectional area  $S$  ( $V = S \cdot x$ ), containing electrolyte of normality  $N$ , while a constant current  $I$  passes for time  $t$  seconds (total charge  $Q = I \cdot t$  coulombs):

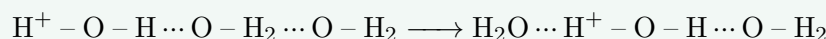
$$\boxed{t_+ = \frac{V \cdot N \cdot F}{1000 \cdot Q} = \frac{V \cdot N \cdot F}{1000 \cdot I \cdot t}} \quad (1.54)$$

### 1.9.6 Abnormal Mobilities of $\text{H}^+$ and $\text{OH}^-$ : The Grotthuss Mechanism

In aqueous solution, the limiting molar conductivity of  $\text{H}^+$  ( $\approx 349.8 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ ) and  $\text{OH}^-$  ( $\approx 198.6 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ ) are extraordinarily higher than all other monovalent cations and anions (e.g.,  $\lambda^\circ(\text{Na}^+) = 50.1$ ,  $\lambda^\circ(\text{K}^+) = 73.5$ ,  $\lambda^\circ(\text{Cl}^-) = 76.3$ ).

### The Grotthuss Proton-Jumping Mechanism

Protons do not migrate through bulk water primarily by standard hydrodynamic translational diffusion as hydrated hydronium ions ( $\text{H}_3\text{O}^+$  or  $\text{H}_9\text{O}_4^+$ ). Instead, charge transfer occurs by an ultrafast **proton-relay or proton-hopping mechanism**:



A hydrogen bond between adjacent water molecules rapidly converts into a covalent bond as a proton hops across, transferring positive charge across macroscopic distances almost instantaneously without requiring the individual oxygen atoms to undergo bodily translation. An identical reverse-hop mechanism accounts for the anomalous mobility of the hydroxide ion ( $\text{OH}^-$ ).

## 1.10 Conductometric Titrations

Conductometric titrations determine the equivalence point of a neutralization or precipitation reaction by monitoring the continuous variation of solution conductance as a titrant is added from a burette.

### Advantages over Indicator Titrations:

- Can be performed in turbid, colored, or highly fluorescent solutions where visual indicators fail.
- Highly accurate for extremely weak acids and weak bases where traditional indicators do not give sharp color changes.
- Eliminates subjective human error in visual endpoint detection.

### 1.10.1 Comprehensive Analysis of Conductometric Titration Curves

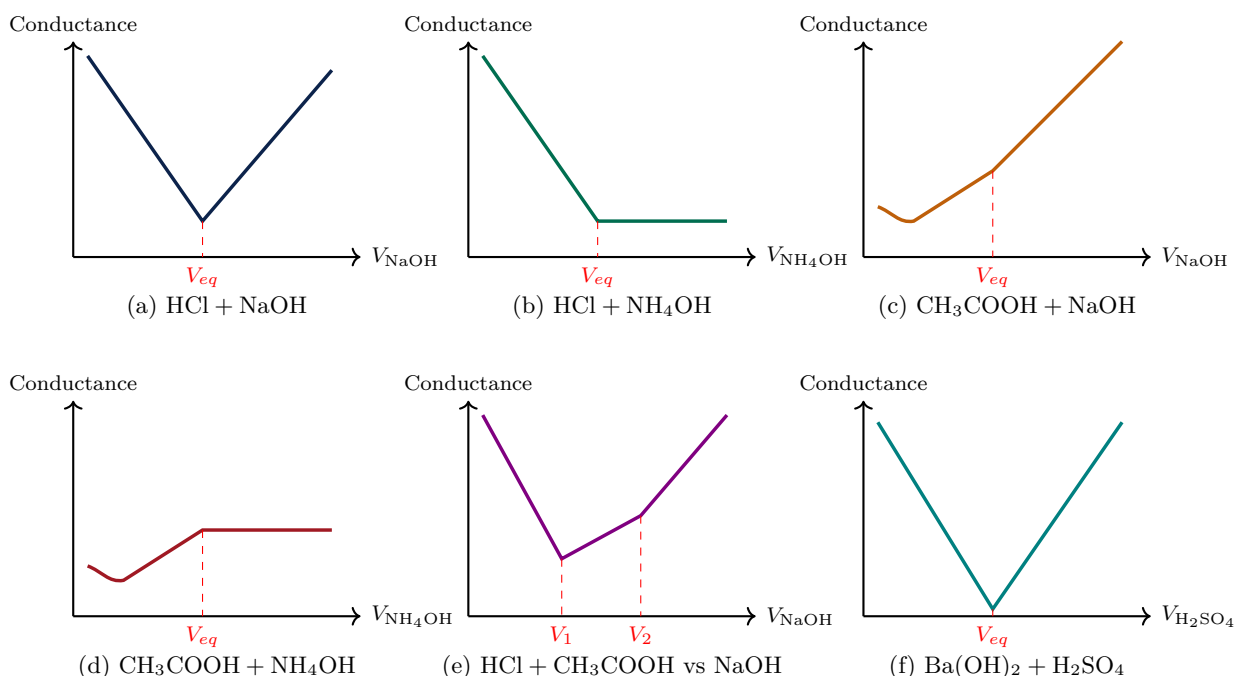


Figure 1.3: Master Reference: Conductometric Titration Curves across all classical reaction types.

#### 1. Strong Acid vs. Strong Base ( $\text{HCl} + \text{NaOH}$ ):

- *Reaction:*  $\text{H}^+ + \text{Cl}^- + \text{Na}^+ + \text{OH}^- \longrightarrow \text{Na}^+ + \text{Cl}^- + \text{H}_2\text{O}$

- *Before Equivalence:* Highly mobile  $\text{H}^+$  ions ( $\lambda^\circ = 349.8$ ) are progressively replaced by much slower  $\text{Na}^+$  ions ( $\lambda^\circ = 50.1$ ). Conductance drops steeply.
- *After Equivalence:* Addition of excess  $\text{NaOH}$  introduces unneutralized  $\text{Na}^+$  and fast  $\text{OH}^-$  ions ( $\lambda^\circ = 198.6$ ), causing conductance to shoot upward steeply.
- *Curve Shape:* Symmetrical sharp **V-shape**.

## 2. Strong Acid vs. Weak Base ( $\text{HCl} + \text{NH}_4\text{OH}$ ):

- *Before Equivalence:*  $\text{H}^+$  ions are replaced by  $\text{NH}_4^+$  ions ( $\lambda^\circ = 73.5$ ), causing the conductance to drop steeply.
- *After Equivalence:* Excess added  $\text{NH}_4\text{OH}$  is a weak electrolyte whose dissociation is suppressed by the common ion effect of  $\text{NH}_4\text{Cl}$  already present. The curve becomes nearly **horizontal**.

## 3. Weak Acid vs. Strong Base ( $\text{CH}_3\text{COOH} + \text{NaOH}$ ):

- *Initial Region:* A small initial drop occurs due to common ion suppression of  $\text{CH}_3\text{COOH}$  dissociation by the newly formed  $\text{CH}_3\text{COO}^-$ .
- *Buffer/Neutralization Region:* As more  $\text{NaOH}$  is added, highly conducting strong electrolyte  $\text{CH}_3\text{COONa}$  accumulates, increasing the ion count and raising conductance moderately.
- *After Equivalence:* Excess  $\text{NaOH}$  delivers highly mobile  $\text{OH}^-$  ions, resulting in a very steep upward break.

## 4. Weak Acid vs. Weak Base ( $\text{CH}_3\text{COOH} + \text{NH}_4\text{OH}$ ):

- Conductance rises moderately up to the equivalence point due to formation of  $\text{CH}_3\text{COONH}_4$ , and then remains virtually constant (flat line) because excess  $\text{NH}_4\text{OH}$  does not dissociate appreciably.

## 5. Mixture of Strong Acid and Weak Acid vs. Strong Base ( $\text{HCl} + \text{CH}_3\text{COOH}$ vs. $\text{NaOH}$ ):

- *First Stage (0 to  $V_1$ ):*  $\text{HCl}$  is neutralized preferentially because its high  $[\text{H}^+]$  completely suppresses the dissociation of  $\text{CH}_3\text{COOH}$ . Conductance drops sharply until all  $\text{HCl}$  is consumed at  $V_1$ .
- *Second Stage ( $V_1$  to  $V_2$ ):*  $\text{CH}_3\text{COOH}$  is neutralized, generating  $\text{CH}_3\text{COONa}$ . Conductance rises moderately until equivalence point  $V_2$ .
- *Third Stage (beyond  $V_2$ ):* Excess  $\text{NaOH}$  is added, and conductance rises sharply due to free  $\text{OH}^-$ .
- *Key Formula:* Volume of  $\text{NaOH}$  for  $\text{HCl} = V_1$ ; Volume of  $\text{NaOH}$  for  $\text{CH}_3\text{COOH} = V_2 - V_1$ .

## 6. Precipitation Titration: $\text{Ba}(\text{OH})_2 + \text{H}_2\text{SO}_4$ :



Both ions of the titrand and titrant form an insoluble precipitate ( $\text{BaSO}_4$ ) and unionized water ( $\text{H}_2\text{O}$ ). At the exact equivalence point, virtually all ions are removed, and conductance drops almost to **zero**! Beyond equivalence, excess  $\text{H}_2\text{SO}_4$  rapidly shoots the conductance upward.

## 1.11 Comprehensive Master Solved Illustrations

**Illustration 1.3: Determination of Sparingly Soluble Salt Solubility &  $K_{sp}$** 

The conductivity of a saturated solution of AgCl at 298 K is found to be  $3.40 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$ , while that of the pure water used to prepare the solution is  $1.60 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}$ . Given the limiting ionic conductivities:  $\lambda^\circ(\text{Ag}^+) = 61.9 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$  and  $\lambda^\circ(\text{Cl}^-) = 76.3 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ . Calculate:

- The net conductivity of the dissolved AgCl.
- The molar solubility  $S$  of AgCl in  $\text{mol} \cdot \text{L}^{-1}$  and in  $\text{g} \cdot \text{L}^{-1}$  ( $M_{\text{AgCl}} = 143.5 \text{ g} \cdot \text{mol}^{-1}$ ).
- The solubility product constant  $K_{sp}$  of AgCl.

**Part (a): Net Conductivity of Salt:**

$$\begin{aligned}\kappa_{\text{AgCl}} &= \kappa_{\text{solution}} - \kappa_{\text{H}_2\text{O}} \\ &= 3.40 \times 10^{-6} - 1.60 \times 10^{-6} = 1.80 \times 10^{-6} \text{ S} \cdot \text{cm}^{-1}\end{aligned}$$

**Part (b): Molar Solubility ( $S$ ):** By Kohlrausch's law:

$$\Lambda_m^\circ(\text{AgCl}) = \lambda^\circ(\text{Ag}^+) + \lambda^\circ(\text{Cl}^-) = 61.9 + 76.3 = 138.2 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$$

Since the solution is extremely dilute:

$$\begin{aligned}S &= \frac{\kappa_{\text{AgCl}} \times 1000}{\Lambda_m^\circ(\text{AgCl})} = \frac{1.80 \times 10^{-6} \times 1000}{138.2} \\ &= 1.302 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}\end{aligned}$$

In grams per liter:

$$S_{\text{g/L}} = S \times M_{\text{AgCl}} = 1.302 \times 10^{-5} \times 143.5 = 1.869 \times 10^{-3} \text{ g} \cdot \text{L}^{-1}$$

**Part (c): Solubility Product ( $K_{sp}$ ):** For the 1 : 1 dissolution  $\text{AgCl(s)} \rightleftharpoons \text{Ag}^+ + \text{Cl}^-$ :

$$K_{sp} = [\text{Ag}^+][\text{Cl}^-] = S^2 = (1.302 \times 10^{-5})^2 = 1.696 \times 10^{-10} \text{ mol}^2 \cdot \text{L}^{-2}$$

**Illustration 1.4: Moving Boundary Transport Number Calculation**

In a moving boundary experiment to determine the transport number of  $K^+$  in a 0.020 M solution of KCl, a current of 0.0016 A (1.6 mA) was passed through a tube of internal diameter 4.0 mm for 45 minutes (2700 seconds). The boundary swept through a distance of 5.20 cm. Calculate:

- The transport number of the potassium ion ( $t_{K^+}$ ).
- The transport number of the chloride ion ( $t_{Cl^-}$ ).
- The ionic mobility ( $u_{K^+}$ ) if the limiting molar conductivity  $\Lambda_m^\circ(KCl) = 149.9 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ .

**Part (a): Calculation of  $t_{K^+}$ :** 1. Cross-sectional area of tube ( $S$ ): Radius  $r = \frac{4.0 \text{ mm}}{2} = 2.0 \text{ mm} = 0.20 \text{ cm}$ .

$$S = \pi r^2 = \pi \times (0.20)^2 = 0.04\pi \approx 0.12566 \text{ cm}^2$$

2. Volume swept by boundary ( $V$ ):

$$V = S \times x = 0.12566 \text{ cm}^2 \times 5.20 \text{ cm} = 0.65345 \text{ cm}^3 \text{ (mL)}$$

3. Total charge passed ( $Q$ ):

$$Q = I \times t = 1.6 \times 10^{-3} \text{ A} \times 2700 \text{ s} = 4.32 \text{ C}$$

4. Applying the moving boundary equation ( $N = 0.020 \text{ eq} \cdot \text{L}^{-1}$ ):

$$\begin{aligned} t_{K^+} &= \frac{V \cdot N \cdot F}{1000 \cdot Q} = \frac{0.65345 \times 0.020 \times 96485}{1000 \times 4.32} \\ &= \frac{1260.95}{4320} \approx 0.4919 \quad (\approx 0.492) \end{aligned}$$

**Part (b): Transport Number of  $Cl^-$ :**

$$t_{Cl^-} = 1 - t_{K^+} = 1 - 0.4919 = 0.5081 \quad (\approx 0.508)$$

**Part (c): Ionic Mobility of  $K^+$ :** The limiting ionic conductivity of  $K^+$  is:

$$\lambda^\circ(K^+) = t_{K^+} \times \Lambda_m^\circ(KCl) = 0.4919 \times 149.9 = 73.74 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$$

Since  $\lambda^\circ = u^\circ F$ :

$$u^\circ(K^+) = \frac{\lambda^\circ(K^+)}{F} = \frac{73.74}{96485} = 7.643 \times 10^{-4} \text{ cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$$

## 1.12 Exhaustive Practice Problem Bank (Chapter 1)

### Organization of Practice Problems

All questions are rigorously organized by **Sub-topic**. Each question explicitly indicates its originating source book(s). Identical or cross-referenced problems from multiple books are co-located with combined source citations. In the Complete Solutions Edition, full mathematical and mechanistic derivations appear directly beneath each problem.

#### 1.12.1 Sub-Topic 1.1: Resistance, Resistivity, Conductivity & Cell Constant

##### Problem 1.1 [Sources: Neeraj Kumar Ex-I Q106 / Pearson Ex-I Q106]

Which of the following aqueous solutions of NaCl has the highest electrical resistance?

- |                 |                |
|-----------------|----------------|
| (a) 1 N NaCl    | (c) 2 N NaCl   |
| (b) 0.05 N NaCl | (d) 0.1 N NaCl |

##### Problem 1.2 [Source: Narendra Avasthi Level-1 Q125]

The relation among electrical conductance ( $G$ ), specific conductance ( $\kappa$ ), and cell constant ( $l/A$ ) is:

- |                                    |                                    |
|------------------------------------|------------------------------------|
| (a) $G = \kappa \cdot \frac{l}{A}$ | (c) $\kappa = G \cdot \frac{l}{A}$ |
| (b) $G = \frac{\kappa}{l/A}$       | (d) $G = \kappa \cdot A \cdot l$   |

##### Problem 1.3 [Source: Narendra Avasthi Level-1 Q128]

The resistance of a decimolar (0.1 M) solution between two electrodes placed 0.02 m apart and having a cross-sectional area of 0.0004 m<sup>2</sup> was found to be 50  $\Omega$ . The specific conductance ( $\kappa$ ) of the solution is:

- |                                   |                                                  |
|-----------------------------------|--------------------------------------------------|
| (a) 0.1 S $\cdot$ m <sup>-1</sup> | (c) 10 S $\cdot$ m <sup>-1</sup>                 |
| (b) 1.0 S $\cdot$ m <sup>-1</sup> | (d) $4 \times 10^{-4}$ S $\cdot$ m <sup>-1</sup> |

##### Problem 1.4 [Sources: Narendra Avasthi Level-1 Q129 / Neeraj Kumar Ex-I Q118]

The resistance of a 0.1 M KCl solution in a conductivity cell is 300  $\Omega$  and its conductivity is 0.013 S  $\cdot$  cm<sup>-1</sup>. The value of the cell constant is:

- |                          |                         |
|--------------------------|-------------------------|
| (a) 3.9 cm <sup>-1</sup> | (c) 3.9 m <sup>-1</sup> |
| (b) 39 m <sup>-1</sup>   | (d) None of these       |

##### Problem 1.5 [Sources: Neeraj Kumar Ex-I Q124 / Cengage DPP-3.1 Q5]

The resistance of solution A is 50  $\Omega$  and that of solution B is 100  $\Omega$ , both solutions being measured in the same conductivity cell. If equal volumes of solutions A and B are mixed together, what will be the resistance of the mixture using the exact same cell? (Assume no change in the degree of dissociation of A and B on mixing).

- |                    |                    |
|--------------------|--------------------|
| (a) 75 $\Omega$    | (c) 150 $\Omega$   |
| (b) 66.67 $\Omega$ | (d) 33.33 $\Omega$ |

**Problem 1.6****[Source: Cengage DPP-3.1 Q13]**

When a certain conductivity cell was filled with 0.01 M KCl solution, it had a resistance of  $160\ \Omega$  at  $25^\circ\text{C}$ . When filled with 0.005 M NaOH, it had a resistance of  $190\ \Omega$ . If the specific resistance of the 0.01 M KCl solution is  $700\ \Omega \cdot \text{cm}$ , the conductivity ( $\text{S} \cdot \text{cm}^{-1}$ ) of the NaOH solution is:

- (a) 0.00120 (c) 0.00180  
(b) 0.00170 (d) 0.00190

**1.12.2 Sub-Topic 1.2: Molar & Equivalent Conductivity, Dilution & Temperature Effects****Problem 1.7 [Sources: Narendra Avasthi Level-1 Q126 / Neeraj Kumar Ex-I Q111]**

If  $x$  is the specific resistance (in  $\Omega \cdot \text{cm}$ ) of an electrolyte solution and  $y$  is the molarity of the solution, then its molar conductivity  $\Lambda_m$  (in  $\text{S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ ) is given by:

- (a)  $\frac{1000x}{y}$  (c)  $\frac{1000y}{x}$   
(b)  $\frac{1000}{xy}$  (d)  $\frac{xy}{1000}$

**Problem 1.8****[Source: Narendra Avasthi Level-1 Q132]**

The molar conductivity of a solution of an electrolyte  $\text{AB}_3$  is  $150\ \text{S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ . If it ionizes as  $\text{AB}_3 \longrightarrow \text{A}^{3+} + 3\text{B}^-$ , its equivalent conductivity ( $\Lambda_{eq}$ ) will be:

- (a)  $150\ \text{S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$  (c)  $50\ \text{S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$   
(b)  $75\ \text{S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$  (d)  $450\ \text{S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$

**Problem 1.9****[Source: Narendra Avasthi Level-1 Q133]**

Equivalent conductivity of  $\text{Fe}_2(\text{SO}_4)_3$  is related to its molar conductivity by the expression:

- (a)  $\Lambda_{eq} = \Lambda_m$  (c)  $\Lambda_{eq} = 3\Lambda_m$   
(b)  $\Lambda_{eq} = \Lambda_m/3$  (d)  $\Lambda_{eq} = \Lambda_m/6$

**Problem 1.10 [Sources: Neeraj Kumar Ex-I Q107 / Narendra Avasthi Level-1 Q142]**

When a concentrated aqueous solution of an electrolyte is progressively diluted:

- (a) Its specific conductance increases, while equivalent conductance decreases.  
(b) Both specific conductance and equivalent conductance increase.  
(c) Its specific conductance decreases, while equivalent conductance increases.  
(d) Both specific conductance and equivalent conductance decrease.

## 1.12.3 Sub-Topic 1.3: DHO Theory, Ionic Mobility &amp; Debye-Hückel Limiting Law

**Problem 1.11****[Source: Cengage DPP-3.1 Q19]**

An ion travels a distance of 0.4 cm in 1 minute (60 s) under the influence of an electric potential difference of 2 V maintained between two electrodes placed 3 cm apart. Which of the following statements is/are correct?

- (a) Speed of ion =  $6.67 \times 10^{-3} \text{ cm} \cdot \text{s}^{-1}$
- (b) Ionic mobility =  $1.0 \times 10^{-6} \text{ m}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$
- (c) Ionic mobility =  $1.0 \times 10^{-2} \text{ cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$
- (d) Ionic mobility =  $4.44 \times 10^{-3} \text{ cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$

**Problem 1.12** **[Sources: Neeraj Kumar Ex-I Q109 / Narendra Avasthi Level-1 Q147]**

The correct order of limiting molar conductivities ( $\Lambda_m^\circ$ ) of alkali metal chlorides LiCl, NaCl, KCl in aqueous solution at 25°C is:

- (a) LiCl > NaCl > KCl
- (b) KCl > NaCl > LiCl
- (c) NaCl > KCl > LiCl
- (d) LiCl > KCl > NaCl

1.12.4 Sub-Topic 1.4: Kohlrausch's Law & Equilibrium Applications ( $K_a$ ,  $K_{sp}$ ,  $K_w$ )**Problem 1.13****[Source: Narendra Avasthi Level-1 Q134]**

The limiting equivalent conductivities of NaCl, KCl, and KBr are 126.5, 150.0, and 151.5  $\text{S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$  respectively at 298 K. If the limiting equivalent ionic conductance for  $\text{Br}^-$  is  $78.0 \text{ S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$ , the limiting equivalent ionic conductance for  $\text{Na}^+$  ions is:

- (a)  $128.0 \text{ S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$
- (b)  $125.0 \text{ S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$
- (c)  $49.0 \text{ S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$
- (d)  $50.0 \text{ S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}$

**Problem 1.14****[Source: Narendra Avasthi Level-1 Q135]**

The specific conductance of a saturated solution of silver bromide (AgBr) is  $\kappa \text{ S} \cdot \text{cm}^{-1}$ . The limiting ionic conductivities of  $\text{Ag}^+$  and  $\text{Br}^-$  ions are  $x$  and  $y \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ , respectively. If the molar mass of AgBr is  $188 \text{ g} \cdot \text{mol}^{-1}$ , the solubility of AgBr in  $\text{g} \cdot \text{L}^{-1}$  is:

- (a)  $\frac{\kappa \times 1000 \times 188}{x + y}$
- (b)  $\frac{\kappa \times 1000}{188(x + y)}$
- (c)  $\frac{(x + y) \times 1000}{\kappa \times 188}$
- (d)  $\frac{\kappa \times 188}{(x + y) \times 1000}$

**Problem 1.15****[Sources: Neeraj Kumar Ex-I Q121 / Cengage Solved Ex 3.8]**

Calculate the ionization constant ( $K_a$ ) of acetic acid if its 0.05 M solution has a molar conductivity of  $7.814 \times 10^{-4} \text{ } \Omega^{-1} \cdot \text{m}^2 \cdot \text{mol}^{-1}$  at 25°C. Given that  $\Lambda_m^\circ(\text{CH}_3\text{COOH}) = 3.907 \times 10^{-2} \text{ } \Omega^{-1} \cdot \text{m}^2 \cdot \text{mol}^{-1}$ .

- (a)  $2.0 \times 10^{-5}$
- (b)  $1.8 \times 10^{-5}$
- (c)  $2.08 \times 10^{-5}$
- (d)  $4.0 \times 10^{-4}$

## 1.12.5 Sub-Topic 1.5: Conductometric Titrations

**Problem 1.16**

[Source: Narendra Avasthi Level-1 Q148]

When  $\text{HNO}_3(\text{aq})$  is titrated with  $\text{NaOH}(\text{aq})$  conductometrically, the graphical representation of conductance versus volume of  $\text{NaOH}$  added is best represented by:

- (a) A straight horizontal line followed by a vertical ascent.
- (b) A continuous upward linear curve from the origin.
- (c) A V-shaped curve with a sharp minimum at the equivalence point.
- (d) An inverted V-shaped curve with a maximum at the equivalence point.

**Problem 1.17**

[Source: Narendra Avasthi Level-1 Q150]

During the conductometric titration of an equimolar mixture of  $\text{HCl}$  (strong acid) and  $\text{HCN}$  (very weak acid) against standard aqueous  $\text{NaOH}$ :

- (a) The conductance first falls sharply, then rises moderately, and finally rises steeply.
- (b) The conductance continuously decreases until all acid is consumed.
- (c) Only one equivalence point can be observed.
- (d) The conductance remains strictly constant until both acids are completely neutralized.

## 1.12.6 Sub-Topic 1.6: Advanced Olympiad &amp; Multi-Concept Challenges

**Problem 1.18 (Comprehension Passage)** [Source: Narendra Avasthi Level-3 Passage 5]

A saturated aqueous solution contains two sparingly soluble silver salts  $\text{AgX}$  ( $K_{sp} = 1.0 \times 10^{-12}$ ) and  $\text{AgY}$  ( $K_{sp} = 1.0 \times 10^{-12}$ ). The measured conductivity of the saturated solution is  $8.04 \times 10^{-6} \Omega^{-1} \cdot \text{cm}^{-1}$ . Given the limiting molar ionic conductivities at  $25^\circ\text{C}$ :  $\lambda^\circ(\text{Ag}^+) = 60.0 \Omega^{-1} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ ,  $\lambda^\circ(\text{X}^-) = 90.0 \Omega^{-1} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ . (Assume water conductivity is negligible).

**Part 1:** What is the concentration of  $\text{Ag}^+$  ions in the saturated solution?

- |                                      |                                     |
|--------------------------------------|-------------------------------------|
| (a) $1.0 \times 10^{-6} \text{ M}$   | (c) $2.0 \times 10^{-6} \text{ M}$  |
| (b) $1.414 \times 10^{-6} \text{ M}$ | (d) $1.0 \times 10^{-12} \text{ M}$ |

**Part 2:** What is the limiting molar ionic conductivity of  $\text{Y}^-$  ( $\lambda^\circ(\text{Y}^-)$  in  $\Omega^{-1} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$ )?

- |           |                   |
|-----------|-------------------|
| (a) 290.0 | (c) 2.90          |
| (b) 145.0 | (d) None of these |

**Answer Key: Chapter 1 Practice Problem Bank**

| Problem | Answer | Problem | Answer        | Problem | Answer                   |
|---------|--------|---------|---------------|---------|--------------------------|
| 1.1     | (b)    | 1.7     | (b)           | 1.13    | (d)                      |
| 1.2     | (c)    | 1.8     | (c)           | 1.14    | (a)                      |
| 1.3     | (b)    | 1.9     | (d)           | 1.15    | (c)                      |
| 1.4     | (a)    | 1.10    | (c)           | 1.16    | (c)                      |
| 1.5     | (b)    | 1.11    | (a), (b), (c) | 1.17    | (a)                      |
| 1.6     | (a)    | 1.12    | (b)           | 1.18    | Part 1: (b), Part 2: (a) |

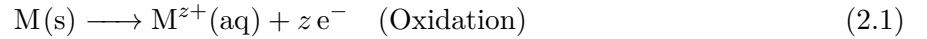
## Chapter 2

# Galvanic Cells, Electrodes & Electrochemical Series

### 2.1 Origin of Electrode Potential & The Electrical Double Layer

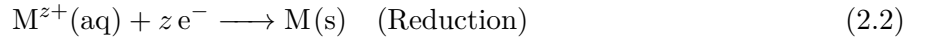
When a strip of metal  $M$  is immersed into an aqueous solution containing its own ions  $M^{z+}$ , two opposing dynamic processes immediately commence at the solid-liquid phase interface:

1. **De-electronation (Dissolution tendency):** Metal atoms on the surface tend to lose valence electrons to the metal lattice and pass into solution as hydrated cations:



This tendency is governed by the **solution pressure** ( $P$ ) of the metal (Nernst's electrolytic solution pressure theory). As positive ions enter the solution, a surplus of electrons is left behind on the metal surface, charging the electrode negatively relative to the solution.

2. **Electronation (Deposition tendency):** Conversely, solvated cations  $M^{z+}$  in solution collide with the electrode surface, capture electrons from the metal, and deposit as neutral metal atoms:



This tendency is driven by the **osmotic pressure** ( $p$ ) of the metal ions in solution. As electrons are extracted from the metal, the electrode acquires a positive charge relative to the solution.

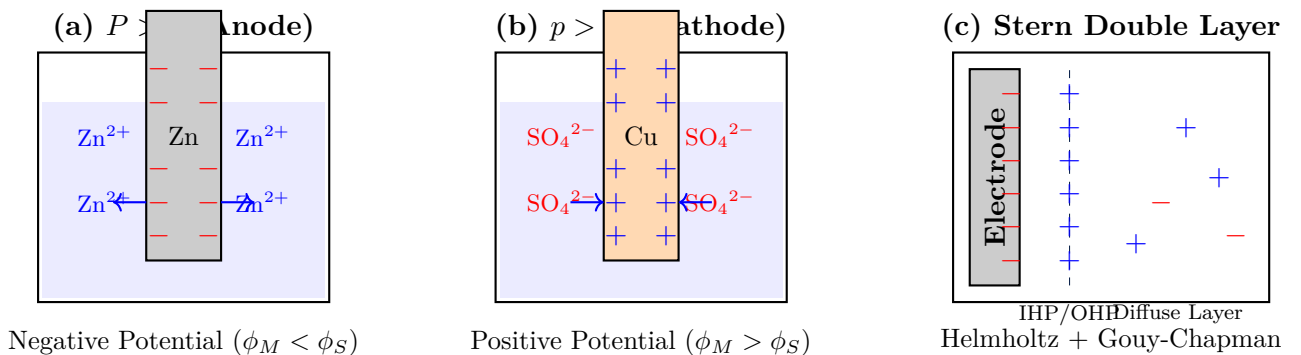


Figure 2.1: Origin of Electrode Potential and the Electrical Double Layer structure.

At dynamic equilibrium, the rate of de-electronation equals the rate of electronation:



A separation of opposite electric charges is established across the interface over molecular dimensions ( $\approx 1 - 10$  nm), forming the **electrical double layer**.

### Definition of Electrode Potential

The electric potential difference established between the metal electrode and the bulk electrolyte solution across the electrical double layer at dynamic equilibrium is termed the **Electrode Potential** ( $E$ ):

$$E = \phi_{\text{metal}} - \phi_{\text{solution}} \quad (2.4)$$

- If  $P > p$  (e.g., Zn in  $\text{ZnSO}_4$ ), oxidation dominates initially, leaving excess electrons on the metal. The electrode is negatively charged relative to the solution ( $E_{\text{ox}} > 0, E_{\text{red}} < 0$ ).
- If  $p > P$  (e.g., Cu in  $\text{CuSO}_4$ ), reduction dominates initially, consuming electrons from the metal. The electrode is positively charged relative to the solution ( $E_{\text{red}} > 0, E_{\text{ox}} < 0$ ).
- When all participating chemical species are in their **thermodynamic standard states** (solute activities  $a = 1$  M, gas fugacities  $f = 1$  bar  $\approx 1$  atm, pure solids/liquids, at  $T = 298.15$  K), the potential difference is termed the **Standard Electrode Potential** ( $E^\circ$ ).

### Atkins Deep Dive: The Stern Model of the Electrochemical Double Layer

According to Otto Stern (1924), the electrical double layer consists of two concentric zones:

1. **The Inner Helmholtz Plane (IHP):** The locus of electrical centers of specifically adsorbed, non-solvated ions on the electrode surface.
2. **The Outer Helmholtz Plane (OHP):** The locus of centers of fully solvated counter-ions held purely by electrostatic forces at the distance of closest approach. The potential drops linearly between the metal surface and the OHP.
3. **The Gouy-Chapman Diffuse Layer:** Beyond the OHP, thermal kinetic agitation prevents ions from forming a rigid sheet; instead, an exponentially decaying diffuse ionic atmosphere extends into the bulk solution. The potential drop across the diffuse shearing plane is the electrokinetic **Zeta Potential** ( $\zeta$ ).

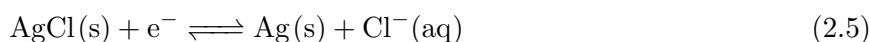
## 2.2 Classification of Electrodes

Electrodes employed in electrochemical cells and equilibrium studies are systematically classified into five major architectural categories:

### 2.2.1 Deep Mathematical Analysis: Metal–Insoluble Salt–Anion Half-Cell

The metal–insoluble salt–anion electrode ( $\text{Ag} | \text{AgCl(s)} | \text{Cl}^-$ ) is one of the most widely tested systems in advanced physical chemistry. Its reduction potential can be derived from two equivalent thermodynamic perspectives:

#### 1. Approach 1 (Direct Heterogeneous Reaction):



Applying the Nernst equation directly ( $a_{\text{Ag(s)}} = 1, a_{\text{AgCl(s)}} = 1$ ):

$$E_{\text{Cl}^- | \text{AgCl} | \text{Ag}} = E_{\text{Cl}^- | \text{AgCl} | \text{Ag}}^\circ - \frac{RT}{F} \ln[\text{Cl}^-] \quad (2.6)$$

#### 2. Approach 2 (Coupled Homogeneous Dissolution + Redox):

The electrode can be viewed as an ordinary  $\text{Ag}^+ | \text{Ag}$  metal-ion electrode, where  $[\text{Ag}^+]$  is strictly constrained by the solubility

| Electrode Type                        | Schematic Representation                                                                                                                                                        | Half-Cell Reduction Reaction                                                                                                                                                                         | Nernst (298 K)                                                                                                                                                                           | Equation |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>1. Metal–Metal Ion</b>             | $M^{z+}(\text{aq})   M(\text{s})$<br>(e.g., $\text{Zn}^{2+}   \text{Zn}$ ,<br>$\text{Cu}^{2+}   \text{Cu}$ )                                                                    | $M^{z+}(\text{aq}) + z e^- \rightleftharpoons M(\text{s})$                                                                                                                                           | $\frac{E}{\frac{0.0591}{z}} = \frac{E^\circ}{\log \frac{1}{[M^{z+}]}}$                                                                                                                   | –        |
| <b>2. Gas–Ion Electrode</b>           | $\text{Pt}, X_2(\text{g})   X^{z\pm}(\text{aq})$<br>(e.g., $\text{Pt}, \text{H}_2   \text{H}^+$ ,<br>$\text{Pt}, \text{Cl}_2   \text{Cl}^-$ )                                   | $2 \text{H}^+(\text{aq}) + 2 e^- \rightleftharpoons \text{H}_2(\text{g})$<br>$\text{Cl}_2(\text{g}) + 2 e^- \rightleftharpoons 2 \text{Cl}^-(\text{aq})$                                             | $\frac{E}{\frac{0.0591}{2}} = \frac{E^\circ}{\log \frac{P_{\text{H}_2}}{[\text{H}^+]^2}}$<br>$\frac{E}{\frac{0.0591}{2}} = \frac{E^\circ}{\log \frac{[\text{Cl}^-]^2}{P_{\text{Cl}_2}}}$ | –        |
| <b>3. Metal–Insoluble Salt–Anion</b>  | $M   \text{MX}_z(\text{s})   X^{z-}(\text{aq})$<br>(e.g., $\text{Ag}   \text{AgCl}(\text{s})   \text{Cl}^-$ ,<br>$\text{Hg}   \text{Hg}_2\text{Cl}_2(\text{s})   \text{Cl}^-$ ) | $\text{AgCl}(\text{s}) + e^- \rightleftharpoons \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq})$<br>$\text{Hg}_2\text{Cl}_2(\text{s}) + 2 e^- \rightleftharpoons 2 \text{Hg}(\text{l}) + 2 \text{Cl}^-$ | $E = E^\circ_{\text{AgCl}/\text{Ag}, \text{Cl}^-}$<br>$0.0591 \log [\text{Cl}^-]$<br>$E = E^\circ_{\text{calomel}}$<br>$\frac{0.0591}{2} \log [\text{Cl}^-]^2$                           | –        |
| <b>4. Oxidation–Reduction (Redox)</b> | $\text{Pt}   M^{z1+}(\text{aq}), M^{z2+}(\text{aq})$<br>(e.g., $\text{Pt}   \text{Fe}^{3+}, \text{Fe}^{2+}$ ,<br>Quinhydrone)                                                   | $\text{Fe}^{3+}(\text{aq}) + e^- \rightleftharpoons \text{Fe}^{2+}(\text{aq})$<br>$\text{Q} + 2 \text{H}^+ + 2 e^- \rightleftharpoons \text{H}_2\text{Q}$                                            | $E = \frac{E^\circ}{\log \frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]}}$<br>$E = \frac{E^\circ_{\text{Q}/\text{H}_2\text{Q}}}{0.0591 \text{ pH}}$                                            | –        |
| <b>5. Amalgam Electrode</b>           | $M(\text{Hg}, a_1)   M^{z+}(\text{aq})$<br>(e.g., $\text{Zn}(\text{Hg})   \text{Zn}^{2+}$ )                                                                                     | $M^{z+}(\text{aq}) + z e^- \rightleftharpoons M(\text{Hg}, a_1)$                                                                                                                                     | $\frac{E}{\frac{0.0591}{z}} = \frac{E^\circ}{\log \frac{a_{\text{M}(\text{Hg})}}{[\text{M}^{z+}]}}$                                                                                      | –        |

Table 2.1: Comprehensive Classification of Half-Cells and their Nernstian EMF expressions.

product equilibrium of AgCl:

$$\text{AgCl}(\text{s}) \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \implies [\text{Ag}^+] = \frac{K_{sp}(\text{AgCl})}{[\text{Cl}^-]} \quad (2.7)$$

The potential of the  $\text{Ag}^+ | \text{Ag}$  couple is:

$$\begin{aligned} E &= E^\circ_{\text{Ag}^+ | \text{Ag}} - \frac{RT}{F} \ln \frac{1}{[\text{Ag}^+]} = E^\circ_{\text{Ag}^+ | \text{Ag}} + \frac{RT}{F} \ln [\text{Ag}^+] \\ &= E^\circ_{\text{Ag}^+ | \text{Ag}} + \frac{RT}{F} \ln \left( \frac{K_{sp}}{[\text{Cl}^-]} \right) \\ &= \left( E^\circ_{\text{Ag}^+ | \text{Ag}} + \frac{RT}{F} \ln K_{sp} \right) - \frac{RT}{F} \ln [\text{Cl}^-] \end{aligned} \quad (2.8)$$

Comparing the two equations yields the **fundamental identity**:

$$\boxed{E^\circ_{\text{Cl}^- | \text{AgCl} | \text{Ag}} = E^\circ_{\text{Ag}^+ | \text{Ag}} + \frac{RT}{F} \ln K_{sp} = E^\circ_{\text{Ag}^+ | \text{Ag}} - \frac{2.303RT}{F} \text{p}K_{sp}} \quad (2.9)$$

At 298.15 K ( $\frac{2.303RT}{F} = 0.0591$  V):

$$\boxed{E^\circ_{\text{Cl}^- | \text{AgCl} | \text{Ag}} = E^\circ_{\text{Ag}^+ | \text{Ag}} - 0.0591 \log \left( \frac{1}{K_{sp}} \right)} \iff \boxed{\log K_{sp} = \frac{E^\circ_{\text{Cl}^- | \text{AgCl} | \text{Ag}} - E^\circ_{\text{Ag}^+ | \text{Ag}}}{0.0591}} \quad (2.10)$$

## 2.3 Standard Reference Electrodes

Because absolute single electrode potentials cannot be measured experimentally in isolation (any voltmeter requires two electrical contact leads, thereby creating a second electrode-solution interface), all electrode potentials are measured relative to an internationally agreed primary reference standard.

### 2.3.1 1. Standard Hydrogen Electrode (SHE)

The primary international reference electrode chosen by IUPAC convention is the **Standard Hydrogen Electrode (SHE)** (or Normal Hydrogen Electrode, NHE).

#### Construction:

- A piece of platinum foil coated with finely divided platinum black (platinized platinum) is sealed in a glass tube.
- Pure, dry hydrogen gas ( $H_2$ ) is continuously bubbled over the platinum surface at a constant pressure of 1 bar ( $10^5 \text{ Pa} \approx 1 \text{ atm}$ ).
- The foil is immersed in an aqueous solution of an acid having a hydrogen ion activity of unity ( $a_{H^+} = 1.0 \text{ M}$ , e.g., 1.18 M HCl).
- **Role of Platinum Black:** (i) Provides a high specific surface area for the adsorption of  $H_2$  gas, (ii) acts as a heterogeneous catalyst to accelerate the reversible electron-transfer equilibrium  $2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$ .

**IUPAC Convention:** The standard reduction potential of the SHE is arbitrarily assigned a value of **exactly zero volts** at all temperatures:

$$E_{H^+|H_2}^\circ \equiv 0.0000 \text{ V} \quad (\text{at all } T) \quad (2.11)$$

**Nernst Expression for Hydrogen Electrode:** For the reduction  $2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$ :

$$E = E^\circ - \frac{0.0591}{2} \log \frac{P_{H_2}}{[H^+]^2} = 0.00 - \frac{0.0591}{2} [\log P_{H_2} - 2 \log [H^+]] \quad (2.12)$$

If  $P_{H_2} = 1 \text{ bar}$ :

$$E = -0.0591 \times (-\log [H^+]) = -0.0591 \text{ pH} \quad (2.13)$$

**Oxidation Potential:**

$$E_{\text{ox}} = +0.0591 \text{ pH} \quad (2.14)$$

#### Practical Limitations of the SHE

Although theoretically fundamental, the SHE is rarely used in routine industrial or analytical laboratories due to:

1. Difficulty in maintaining exactly  $P_{H_2} = 1.00 \text{ bar}$  and  $a_{H^+} = 1.00 \text{ M}$  throughout the measurement.
2. Extreme susceptibility of the platinum black catalyst to **catalytic poisoning** by traces of impurities ( $As_2O_3$ ,  $H_2S$ ,  $CO$ ,  $Hg$  vapors).
3. Inability to be used in solutions containing strong oxidizing agents ( $KMnO_4$ ,  $K_2Cr_2O_7$ ,  $HNO_3$ ) or reducible ions ( $Fe^{3+}$ ,  $Cu^{2+}$ ,  $Ag^+$ ) which react directly with hydrogen gas on the platinum surface.

### 2.3.2 2. Secondary Reference Electrodes

To circumvent the limitations of the SHE, robust and reproducible **secondary reference electrodes** whose potentials have been rigorously calibrated against the SHE are used:

**(a) Saturated Calomel Electrode (SCE)**

- **Notation:**  $\text{Hg(l)} \mid \text{Hg}_2\text{Cl}_2\text{(s)} \mid \text{KCl(saturated)}$
- **Reaction:**  $\text{Hg}_2\text{Cl}_2\text{(s)} + 2\text{e}^- \rightleftharpoons 2\text{Hg(l)} + 2\text{Cl}^-\text{(aq)}$
- **Electrode Potential at 298 K:**

Saturated KCl ( $C \approx 4.0\text{ M}$ ):  $E_{\text{red}} = +0.2415\text{ V vs. SHE}$

1.0 N KCl:  $E_{\text{red}} = +0.2800\text{ V vs. SHE}$

0.1 N KCl:  $E_{\text{red}} = +0.3338\text{ V vs. SHE}$

**(b) Silver–Silver Chloride Electrode**

- **Notation:**  $\text{Ag(s)} \mid \text{AgCl(s)} \mid \text{KCl(saturated)}$
- **Reaction:**  $\text{AgCl(s)} + \text{e}^- \rightleftharpoons \text{Ag(s)} + \text{Cl}^-\text{(aq)}$
- **Potential at 298 K:**  $E_{\text{red}} = +0.2224\text{ V}$  (in 1.0 M  $\text{Cl}^-$ );  $E_{\text{red}} = +0.1988\text{ V}$  (in saturated KCl).

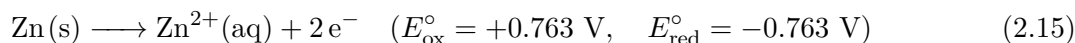
**2.4 Galvanic Cells, IUPAC Conventions & The Salt Bridge**

A galvanic (voltaic) cell converts the free energy change ( $\Delta G < 0$ ) of a spontaneous chemical redox reaction into usable external electrical work.

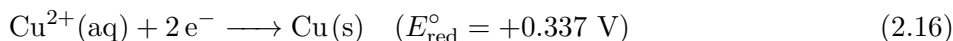
**2.4.1 The Daniell Cell Construction**

The classic archetype is the Daniell Cell (John Frederic Daniell, 1836):

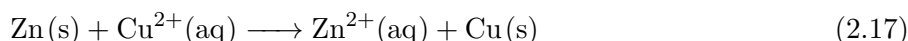
- **Anode Half-Cell (Left):** A zinc rod immersed in 1.0 M  $\text{ZnSO}_4$  solution.



- **Cathode Half-Cell (Right):** A copper rod immersed in 1.0 M  $\text{CuSO}_4$  solution.



- **Overall Cell Redox Reaction:**

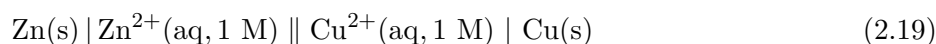


- **Standard Cell Potential:**

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ = (+0.337\text{ V}) - (-0.763\text{ V}) = +1.100\text{ V} \quad (2.18)$$

**2.4.2 IUPAC Cell Representation Rules**

1. The **Anode** (electrode where oxidation occurs) is always placed on the **LEFT**.
2. The **Cathode** (electrode where reduction occurs) is always placed on the **RIGHT**.
3. A single vertical line (|) indicates a phase boundary between an electrode and solution.
4. A double vertical line (||) indicates a salt bridge or porous partition connecting two half-cells with elimination of liquid junction potential.
5. The Daniell cell is represented according to IUPAC notation as:



6. The **LOAN** mnemonic:

Left   •   Oxidation   •   Anode   •   Negative

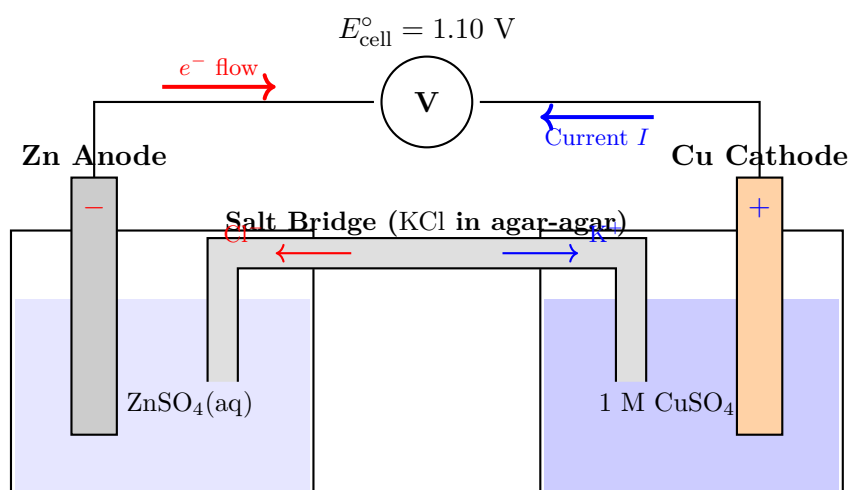


Figure 2.2: Schematic diagram of the Daniell Galvanic Cell.

### 2.4.3 Functions & Chemistry of the Salt Bridge

A salt bridge is a U-shaped glass tube packed with a concentrated solution of an inert electrolyte (KCl, KNO<sub>3</sub>, NH<sub>4</sub>NO<sub>3</sub>) solidified in a gelatinous agar-agar matrix.

#### Four Indispensable Functions of the Salt Bridge

1. **Completes the Electrical Circuit:** Provides an internal electrolytic conduction path allowing ions to migrate between compartments.
2. **Maintains Electrical Neutrality:**
  - At the anode, Zn<sup>2+</sup> ions continuously accumulate, creating a positive charge buildup that would immediately halt further electron release. Anions from the salt bridge (Cl<sup>-</sup>) migrate into the anode beaker to neutralize excess Zn<sup>2+</sup>.
  - At the cathode, Cu<sup>2+</sup> ions are continuously discharged, leaving unneutralized SO<sub>4</sub><sup>2-</sup> anions. Cations from the salt bridge (K<sup>+</sup>) migrate into the cathode beaker to neutralize excess SO<sub>4</sub><sup>2-</sup>.
3. **Eliminates Liquid Junction Potential ( $E_{lj} \approx 0$ ):** When two dissimilar electrolyte solutions are in direct physical contact, ions diffuse across the boundary at different speeds, creating an unwanted boundary potential difference ( $E_{lj}$ ). The salt bridge interposes an inert electrolyte with nearly equal cation and anion mobilities, eliminating  $E_{lj}$ .
4. **Prevents Intermixing of Electrolytes:** Prevents bulk chemical precipitation reactions (e.g., direct spontaneous displacement of Cu<sup>2+</sup> by Zn in bulk liquid).

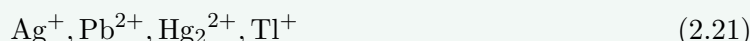
### Critical Criteria for Electrolyte in Salt Bridge

1. **Equal Ionic Mobilities (Criterion of Equal Transference):** The transport numbers of the cation and anion must be virtually identical ( $u_+ \approx u_-$ ):

$$t_+ \approx t_- \approx 0.50 \quad (\text{e.g., for KCl, } t_{\text{K}^+} = 0.49, t_{\text{Cl}^-} = 0.51) \quad (2.20)$$

Consequently, diffusion potential at the anode junction is exactly cancelled by the diffusion potential at the cathode junction.

2. **Chemical Inertness:** The electrolyte must not undergo chemical or precipitation reactions with any component of the cell.
3. **Prohibited Scenarios for KCl:** A KCl salt bridge **cannot** be used if either half-cell contains ions of sparingly soluble chlorides:



because  $\text{AgCl}$ ,  $\text{PbCl}_2$ ,  $\text{Hg}_2\text{Cl}_2$  precipitate out inside the pores of the agar gel, clogging the bridge and breaking electrical contact. For these cells, a  $\text{KNO}_3$  or  $\text{NH}_4\text{NO}_3$  salt bridge must be employed.

## 2.5 The Electrochemical Series

When elements are arranged in increasing order of their standard reduction potentials ( $E^\circ_{\text{red}}$ ) at 298.15 K, the resulting sequence is the **Electrochemical (Electromotive) Series**:

### 2.5.1 Major Applications of the Electrochemical Series

#### 1. Predicting Relative Oxidizing and Reducing Strengths

- Species with the **most negative**  $E^\circ$  have the greatest tendency to lose electrons (undergo oxidation). Hence, metallic lithium ( $\text{Li}$ ) is the strongest reducing agent in aqueous solution.
- Species with the **most positive**  $E^\circ$  have the greatest tendency to accept electrons (undergo reduction). Hence, fluorine gas ( $\text{F}_2$ ) is the strongest oxidizing agent.

#### Atkins Insight: Why is Lithium the Strongest Reducing Agent in Water?

Although the ionization enthalpy of  $\text{Li(g)}$  is higher than that of  $\text{Na(g)}$  or  $\text{K(g)}$ , the standard reduction potential of  $\text{Li}^+/\text{Li}$  is by far the most negative ( $-3.05 \text{ V}$ ). This is explained by the Born-Haber thermodynamic cycle:

$$\Delta H^\circ_{\text{oxidation}} = \Delta H^\circ_{\text{sub}} + \text{IE}_1 + \Delta H^\circ_{\text{hyd}}(\text{Li}^+) \quad (2.22)$$

Because the lithium cation  $\text{Li}^+$  has an exceptionally small ionic radius ( $r = 76 \text{ pm}$ ), it possesses a colossal charge density, releasing an extraordinarily large hydration enthalpy ( $\Delta H^\circ_{\text{hyd}} = -519 \text{ kJ} \cdot \text{mol}^{-1}$ ) upon hydration:

$$|\Delta H^\circ_{\text{hyd}}(\text{Li}^+)| \gg |\Delta H^\circ_{\text{hyd}}(\text{Na}^+)| \quad (2.23)$$

This massive hydration energy completely offsets the high ionization enthalpy, driving  $\Delta G^\circ$  for oxidation to an exceptionally negative value, making lithium the thermodynamically supreme reducing agent in water.

| Element                      | Standard Reduction Half-Reaction                                                                                           | $E_{\text{red}}^{\circ}$ (Volts) | Chemical Character / Reactivity                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------|
| Li                           | $\text{Li}^{+} + \text{e}^{-} \rightleftharpoons \text{Li(s)}$                                                             | -3.05                            | <b>Strongest Reducing Agent</b> in water         |
| K                            | $\text{K}^{+} + \text{e}^{-} \rightleftharpoons \text{K(s)}$                                                               | -2.93                            | Highly electropositive; reacts with water        |
| Ba                           | $\text{Ba}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Ba(s)}$                                                          | -2.90                            | Rapidly displaces $\text{H}_2$ from cold water   |
| Ca                           | $\text{Ca}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Ca(s)}$                                                          | -2.87                            | Strongly electropositive                         |
| Na                           | $\text{Na}^{+} + \text{e}^{-} \rightleftharpoons \text{Na(s)}$                                                             | -2.71                            | Vigorous reaction with water                     |
| Mg                           | $\text{Mg}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Mg(s)}$                                                          | -2.37                            | Displaces $\text{H}_2$ from boiling water/steam  |
| Al                           | $\text{Al}^{3+} + 3 \text{e}^{-} \rightleftharpoons \text{Al(s)}$                                                          | -1.66                            | Displaces $\text{H}_2$ from dilute acids         |
| Zn                           | $\text{Zn}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Zn(s)}$                                                          | -0.76                            | Typical laboratory $\text{H}_2$ generator        |
| Cr                           | $\text{Cr}^{3+} + 3 \text{e}^{-} \rightleftharpoons \text{Cr(s)}$                                                          | -0.74                            | Readily oxidized by $\text{H}^{+}$               |
| Fe                           | $\text{Fe}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Fe(s)}$                                                          | -0.44                            | Readily rusts; displaces $\text{H}_2$ from acids |
| Cd                           | $\text{Cd}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Cd(s)}$                                                          | -0.40                            | Moderately active metal                          |
| Co                           | $\text{Co}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Co(s)}$                                                          | -0.28                            | Displaces $\text{H}_2$ from acids slowly         |
| Ni                           | $\text{Ni}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Ni(s)}$                                                          | -0.25                            | Passive in concentrated nitric acid              |
| Sn                           | $\text{Sn}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Sn(s)}$                                                          | -0.14                            | Weak reducing agent                              |
| Pb                           | $\text{Pb}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Pb(s)}$                                                          | -0.13                            | Weak acid reaction (surface passivation)         |
| $\text{H}_2$                 | $2 \text{H}^{+} + 2 \text{e}^{-} \rightleftharpoons \text{H}_2(\text{g})$                                                  | 0.00                             | <b>Arbitrary Standard Reference (SHE)</b>        |
| Cu                           | $\text{Cu}^{2+} + 2 \text{e}^{-} \rightleftharpoons \text{Cu(s)}$                                                          | +0.34                            | Unreactive toward dilute non-oxidizing acids     |
| $\text{I}_2$                 | $\text{I}_2(\text{s}) + 2 \text{e}^{-} \rightleftharpoons 2 \text{I}^{-}$                                                  | +0.54                            | Mild oxidizing agent                             |
| Fe                           | $\text{Fe}^{3+} + \text{e}^{-} \rightleftharpoons \text{Fe}^{2+}$                                                          | +0.77                            | Stable redox couple                              |
| Hg                           | $\text{Hg}_2^{2+} + 2 \text{e}^{-} \rightleftharpoons 2 \text{Hg(l)}$                                                      | +0.79                            | Noble metal; dissolved only by oxidizing acids   |
| Ag                           | $\text{Ag}^{+} + \text{e}^{-} \rightleftharpoons \text{Ag(s)}$                                                             | +0.80                            | Noble metal; easily reduced                      |
| $\text{Br}_2$                | $\text{Br}_2(\text{l}) + 2 \text{e}^{-} \rightleftharpoons 2 \text{Br}^{-}$                                                | +1.09                            | Strong oxidizing agent                           |
| $\text{Cr}_2\text{O}_7^{2-}$ | $\text{Cr}_2\text{O}_7^{2-} + 14 \text{H}^{+} + 6 \text{e}^{-} \rightleftharpoons 2 \text{Cr}^{3+} + 7 \text{H}_2\text{O}$ | +1.33                            | Powerful analytical oxidant                      |
| $\text{Cl}_2$                | $\text{Cl}_2(\text{g}) + 2 \text{e}^{-} \rightleftharpoons 2 \text{Cl}^{-}$                                                | +1.36                            | Powerful gaseous oxidant                         |
| Au                           | $\text{Au}^{3+} + 3 \text{e}^{-} \rightleftharpoons \text{Au(s)}$                                                          | +1.50                            | Very noble; dissolved only by aqua regia         |
| $\text{MnO}_4^{-}$           | $\text{MnO}_4^{-} + 8 \text{H}^{+} + 5 \text{e}^{-} \rightleftharpoons \text{Mn}^{2+} + 4 \text{H}_2\text{O}$              | +1.51                            | Classic laboratory self-indicating oxidant       |
| $\text{F}_2$                 | $\text{F}_2(\text{g}) + 2 \text{e}^{-} \rightleftharpoons 2 \text{F}^{-}$                                                  | +2.87                            | <b>Strongest Oxidizing Agent</b> in chemistry    |

Table 2.2: The Standard Electrochemical Series at 298.15 K.

## 2. Spontaneity and Direction of Redox Reactions

For any proposed cell reaction:

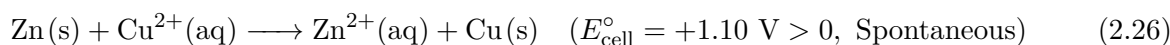
$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} = E_{\text{Right}}^{\circ} - E_{\text{Left}}^{\circ} \quad (2.24)$$

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ} \quad (2.25)$$

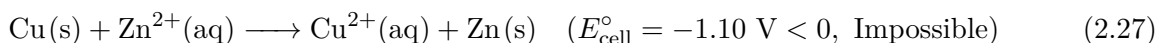
- If  $E_{\text{cell}}^{\circ} > 0 \implies \Delta G^{\circ} < 0$ : The cell reaction is thermodynamically **spontaneous** in the forward direction.
- If  $E_{\text{cell}}^{\circ} < 0 \implies \Delta G^{\circ} > 0$ : The cell reaction is **non-spontaneous** in the forward direction (spontaneous in the reverse direction).
- If  $E_{\text{cell}}^{\circ} = 0 \implies \Delta G^{\circ} = 0$ : The system is at **equilibrium**.

## 3. Feasibility of Metal Displacement Reactions

A metal with a **lower (more negative)**  $E_{\text{red}}^{\circ}$  will displace any metal with a **higher (more positive)**  $E_{\text{red}}^{\circ}$  from its aqueous salt solution:

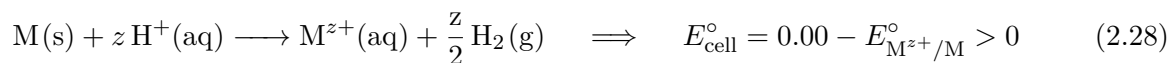


Conversely, copper cannot displace zinc:



## 4. Displacement of Hydrogen Gas from Non-Oxidizing Dilute Acids

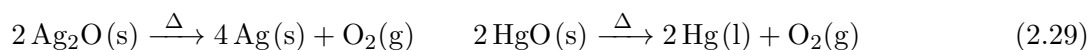
All metals positioned above hydrogen in the reduction potential series ( $E_{\text{red}}^{\circ} < 0.00 \text{ V}$ ) can reduce hydrogen ions from dilute acids ( $\text{HCl}$ , dilute  $\text{H}_2\text{SO}_4$ ) to liberate  $\text{H}_2(\text{g})$ :



Metals below hydrogen ( $E_{\text{red}}^{\circ} > 0.00 \text{ V}$ , such as  $\text{Cu}$ ,  $\text{Ag}$ ,  $\text{Au}$ ,  $\text{Hg}$ ) cannot displace  $\text{H}_2$  gas from non-oxidizing acids.

## 5. Thermal Stability of Metal Oxides

Metals with high positive  $E_{\text{red}}^{\circ}$  (noble metals like  $\text{Ag}$ ,  $\text{Au}$ ,  $\text{Hg}$ ) have very weak affinity for oxygen. Consequently, their oxides are thermally unstable and decompose into pure elemental metal and oxygen gas upon gentle heating:

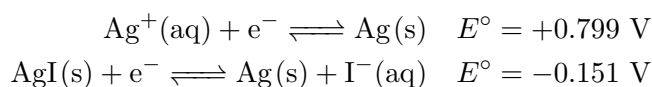


In contrast, oxides of metals with strongly negative  $E^{\circ}$  ( $\text{Al}_2\text{O}_3$ ,  $\text{MgO}$ ,  $\text{CaO}$ ) are extremely stable and cannot be decomposed thermally.

## 2.6 Comprehensive Master Solved Illustrations

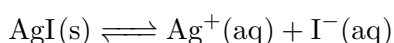
### Illustration 2.1: Coupled Metal–Insoluble Salt Electrode & Solubility Product Calculation

Given the standard reduction potentials at 298 K:

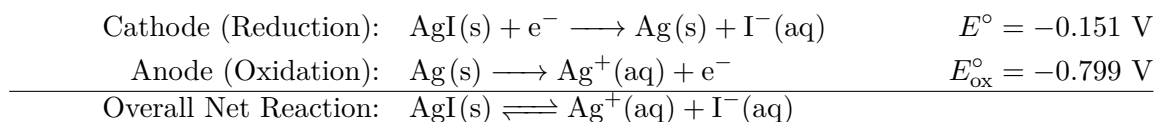


- Formulate a galvanic cell combining these two electrodes such that the cell reaction corresponds to the dissolution equilibrium of AgI(s).
- Calculate the standard cell EMF ( $E^\circ_{\text{cell}}$ ).
- Evaluate the solubility product constant ( $K_{\text{sp}}$ ) of AgI at 298 K.

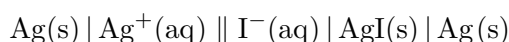
**Part (a): Cell Formulation:** We desire the net reaction to be the dissolution of AgI:



Subtracting the  $\text{Ag}^+/\text{Ag}$  reduction reaction from the  $\text{AgI}/\text{Ag}, \text{I}^-$  reduction reaction:



The cell is represented as:



**Part (b): Standard Cell EMF ( $E^\circ_{\text{cell}}$ ):**

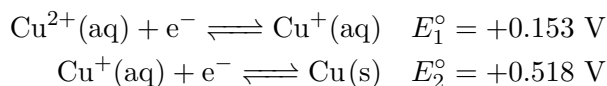
$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = (-0.151 \text{ V}) - (+0.799 \text{ V}) = -0.950 \text{ V}$$

**Part (c): Solubility Product ( $K_{\text{sp}}$ ):** At equilibrium:

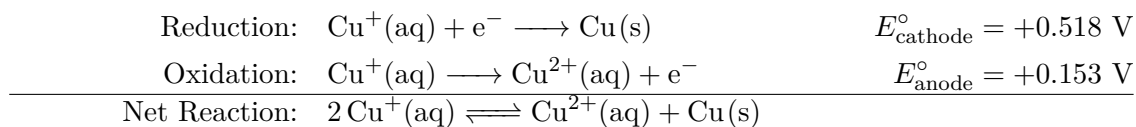
$$\begin{aligned}\Delta G^\circ &= -nFE^\circ_{\text{cell}} = -RT \ln K_{\text{sp}} \\ \log K_{\text{sp}} &= \frac{nE^\circ_{\text{cell}}}{0.0591} = \frac{1 \times (-0.950)}{0.0591} \approx -16.074 \\ K_{\text{sp}} &= 10^{-16.074} = 10^{0.926} \times 10^{-17} \approx 8.43 \times 10^{-17} \text{ mol}^2 \cdot \text{L}^{-2}\end{aligned}$$

**Illustration 2.2: Prediction of Disproportionation Reaction Feasibility**

Given the following standard electrode potentials in aqueous acidic solution:



Determine whether cuprous ion ( $\text{Cu}^{+}$ ) is stable in aqueous solution, or whether it spontaneously disproportionates into  $\text{Cu}^{2+}$  and  $\text{Cu}(\text{s})$ . Calculate the equilibrium constant ( $K_c$ ) for the disproportionation reaction at 298 K.

**Step 1: Write the Disproportionation Half-Reactions:****Step 2: Calculate  $E_{\text{cell}}^{\circ}$ :**

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} = (+0.518 \text{ V}) - (+0.153 \text{ V}) = +0.365 \text{ V}$$

Since  $E_{\text{cell}}^{\circ} > 0$  ( $\Delta G^{\circ} < 0$ ), the forward reaction is **thermodynamically highly spontaneous**. Thus,  $\text{Cu}^{+}$  ions are unstable in aqueous solution and spontaneously **disproportionate** into  $\text{Cu}^{2+}$  and metallic copper.

**Step 3: Calculate the Equilibrium Constant ( $K_c$ ):**

$$\log K_c = \frac{nE_{\text{cell}}^{\circ}}{0.0591} = \frac{1 \times 0.365}{0.0591} = 6.176$$

$$K_c = \frac{[\text{Cu}^{2+}]}{[\text{Cu}^{+}]^2} = 10^{6.176} = 10^{0.176} \times 10^6 \approx 1.50 \times 10^6$$

The immense magnitude of  $K_c$  ( $1.5 \times 10^6$ ) proves that virtually all  $\text{Cu}^{+}$  ions disproportionate in aqueous media unless stabilized by complexation (e.g.,  $[\text{Cu}(\text{CN})_2]^{-}$ ) or precipitation (e.g.,  $\text{CuCl}(\text{s})$ ).

## 2.7 Exhaustive Practice Problem Bank (Chapter 2)

### Organization of Practice Problems

All questions are organized by **Sub-topic** with full origin book citations. In the Complete Solutions Edition, full mathematical and chemical derivations are provided directly beneath each problem.

#### 2.7.1 Sub-Topic 2.1: Cell Fundamentals, Polarity & IUPAC Representation

##### Problem 2.1 [Source: Narendra Avasthi Level-1 Q1]

A cell redox reaction would be thermodynamically spontaneous if the standard cell potential ( $E_{\text{cell}}^{\circ}$ ) and standard Gibbs free energy change ( $\Delta_r G^{\circ}$ ) are respectively:

- |                           |                       |
|---------------------------|-----------------------|
| (a) Positive and negative | (c) Zero and zero     |
| (b) Negative and negative | (d) Positive and zero |

##### Problem 2.2 [Source: Narendra Avasthi Level-1 Q2]

Which of the following statements regarding electrode polarities is correct?

- (a) Cathode is the negative terminal in both galvanic and electrolytic cells.
- (b) Anode is the positive terminal in both galvanic and electrolytic cells.
- (c) Cathode is the negative terminal in an electrolytic cell, and positive in a galvanic cell.
- (d) Cathode is the positive terminal in an electrolytic cell, and negative in a galvanic cell.

##### Problem 2.3 [Sources: Neeraj Kumar Ex-I Q1 / Pearson Ex-I Q1]

A metal rod is immersed in an aqueous solution of its own ions. Its electrode potential is independent of:

- |                                     |                                       |
|-------------------------------------|---------------------------------------|
| (a) Temperature of the solution     | (c) Area of the metal surface exposed |
| (b) Concentration of the metal ions | (d) Nature of the metal               |

#### 2.7.2 Sub-Topic 2.2: Salt Bridge Mechanics & Reference Electrodes

##### Problem 2.4 [Sources: Pearson Ex-II Q14 / Neeraj Kumar Ex-II Q12]

In a galvanic cell, the salt bridge:

- (a) Does not participate chemically in the cell reaction.
- (b) Stops the mutual bulk diffusion of ions from one electrode compartment to another.
- (c) Is necessary for maintaining electrical neutrality and eliminating liquid junction potential.
- (d) Ensures complete mechanical mixing of the two electrolytic solutions.

**Problem 2.5****[Source: Cengage Solved Ex 3.2]**

A KCl salt bridge cannot be used in which of the following electrochemical cells?

- (a)  $\text{Zn} | \text{ZnSO}_4 || \text{CuSO}_4 | \text{Cu}$  (c)  $\text{Ag} | \text{AgNO}_3 || \text{CuSO}_4 | \text{Cu}$   
 (b)  $\text{Fe} | \text{FeSO}_4 || \text{CuSO}_4 | \text{Cu}$  (d)  $\text{Ni} | \text{NiSO}_4 || \text{CuSO}_4 | \text{Cu}$

**Problem 2.6****[Source: Neeraj Kumar Ex-I Q8]**

The potential of a hydrogen electrode placed in a solution of  $\text{pH} = 10$  at  $298 \text{ K}$  and  $P_{\text{H}_2} = 1 \text{ atm}$  is:

- (a)  $+0.591 \text{ V}$  (c)  $0.00 \text{ V}$   
 (b)  $-0.591 \text{ V}$  (d)  $-0.0591 \text{ V}$

**2.7.3 Sub-Topic 2.3: Electrochemical Series & Displacement Phenomena****Problem 2.7****[Source: Pearson Ex-II Section A Q2]**

A student made the following experimental observations in the laboratory:

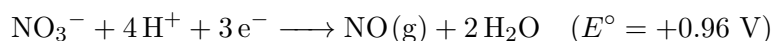
- (I) Clean copper metal did not react when immersed in  $1.0 \text{ M Pb(NO}_3)_2$  solution.  
 (II) Clean lead metal dissolved when placed in  $1.0 \text{ M AgNO}_3$  solution and lustrous crystals of Ag metal deposited on its surface.  
 (III) Clean silver metal did not react when immersed in  $1.0 \text{ M Cu(NO}_3)_2$  solution.

The correct order of decreasing reducing character of the three metals is:

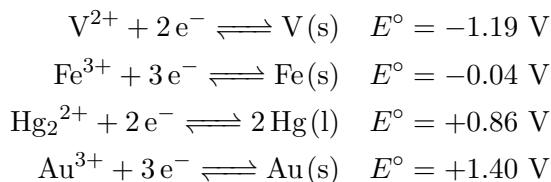
- (a)  $\text{Cu} > \text{Pb} > \text{Ag}$  (c)  $\text{Pb} > \text{Cu} > \text{Ag}$   
 (b)  $\text{Ag} > \text{Cu} > \text{Pb}$  (d)  $\text{Pb} > \text{Ag} > \text{Cu}$

**Problem 2.8****[Source: Pearson Ex-II Q12]**

For the reduction of  $\text{NO}_3^-$  ion in an acidic aqueous solution:



Given the standard reduction potentials of some metal couples:



Which of the following metals can be oxidized by  $\text{NO}_3^-$  in acidic aqueous solution under standard conditions?

- (a) V and Hg (c) Fe and V  
 (b) Hg and Fe (d) All except Au

**Problem 2.9 (Arbitrary Shift of Reference Scale)  
Comprehension I]****[Source: Pearson Ex-II**

Suppose that the Standard Hydrogen Electrode (S.H.E.) was arbitrarily assigned a standard reduction potential of +1.00 V instead of 0.00 V. Given that on the standard scale:  $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$  and  $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$ .

**Part 1:** What would be the new observed standard electrode potential of the  $\text{Cu}^{2+}/\text{Cu}$  couple?

- |             |             |
|-------------|-------------|
| (a) +0.34 V | (c) -0.66 V |
| (b) +1.34 V | (d) +0.66 V |

**Part 2:** What would be the new standard EMF of the complete Zn – Cu Daniell cell?

- |             |                   |
|-------------|-------------------|
| (a) +1.10 V | (c) +0.10 V       |
| (b) +2.10 V | (d) Indeterminate |

## Answer Key: Chapter 2 Practice Problem Bank

| Problem | Answer | Problem | Answer        | Problem | Answer                   |
|---------|--------|---------|---------------|---------|--------------------------|
| 2.1     | (a)    | 2.4     | (a), (b), (c) | 2.7     | (c)                      |
| 2.2     | (c)    | 2.5     | (c)           | 2.8     | (d)                      |
| 2.3     | (c)    | 2.6     | (b)           | 2.9     | Part 1: (b), Part 2: (a) |

## Chapter 3

# Thermodynamics of Cells & The Nernst Equation

### 3.1 Fundamental Thermodynamic Relations of Electrochemical Cells

Electrochemical cells operate under conditions of constant temperature and constant pressure. When a cell discharges reversibly (by opposing its electromotive force  $E$  by an infinitesimal counter-potential from an external potentiometer so that current flow  $I \rightarrow 0$ ), the electrical work performed by the system equals the decrease in Gibbs free energy ( $-\Delta G$ ) of the chemical reaction.

#### 3.1.1 Electrical Work and Free Energy Change

When  $n$  moles of electrons pass through the external circuit under a reversible cell potential  $E_{\text{cell}}$ , the total quantity of electric charge transported is:

$$Q = nF \quad (3.1)$$

where  $F = 96485.3 \text{ C} \cdot \text{mol}^{-1}$  is Faraday's constant. The electrical work performed by the system ( $W_{\text{elec}}$ ) is:

$$W_{\text{elec}} = \text{Charge} \times \text{Potential Difference} = nFE_{\text{cell}} \quad (3.2)$$

According to the fundamental laws of classical thermodynamics, the non-expansion (useful) work done by a closed system under reversible isothermal-isobaric conditions is strictly equal to the decrease in Gibbs free energy:

$$W_{\text{rev, non-PV}} = -\Delta G \quad (3.3)$$

Therefore:

$$\boxed{\Delta G = -nFE_{\text{cell}}} \quad (3.4)$$

Under standard thermodynamic states ( $T = 298.15 \text{ K}$ , all activities  $a_i = 1 \text{ M}$ , partial pressures  $P_i = 1 \text{ bar}$ ):

$$\boxed{\Delta G^\circ = -nFE_{\text{cell}}^\circ} \quad (3.5)$$

### Criteria of Thermodynamic Spontaneity and Directionality

- **Spontaneous Process (Forward Galvanic Cell Discharge):**

$$\Delta G < 0 \iff E_{\text{cell}} > 0 \quad (3.6)$$

- **Non-Spontaneous Process (Requires External Electrolytic Work):**

$$\Delta G > 0 \iff E_{\text{cell}} < 0 \quad (3.7)$$

- **Thermodynamic Equilibrium (Dead Cell):**

$$\Delta G = 0 \iff E_{\text{cell}} = 0 \quad (3.8)$$

**Intensive vs. Extensive Property:** Free energy change  $\Delta G$  is an **extensive property** (measured in kJ or  $\text{kJ} \cdot \text{mol}^{-1}$ ), directly proportional to the stoichiometric coefficients ( $n$ ). Cell potential  $E_{\text{cell}}$  is an **intensive property** (measured in  $\text{V} = \text{J} \cdot \text{C}^{-1}$ ), completely independent of multiplying the reaction equation by any arbitrary stoichiometric multiplier.

## 3.2 Rigorous Derivation of the Nernst Equation

Consider a general reversible redox cell reaction:



The thermodynamic relationship between the Gibbs free energy change ( $\Delta G$ ) under arbitrary concentrations and the standard free energy change ( $\Delta G^\circ$ ) is given by the **Van 't Hoff reaction isotherm**:

$$\Delta G = \Delta G^\circ + RT \ln Q \quad (3.10)$$

where  $R = 8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$  is the universal gas constant,  $T$  is the absolute temperature in Kelvin, and  $Q$  is the **Reaction Quotient**:

$$Q = \frac{a_{\text{C}}^c \cdot a_{\text{D}}^d}{a_{\text{A}}^a \cdot a_{\text{B}}^b} \approx \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b} \quad (3.11)$$

Substituting  $\Delta G = -nFE_{\text{cell}}$  and  $\Delta G^\circ = -nFE_{\text{cell}}^\circ$  into the isotherm:

$$-nFE_{\text{cell}} = -nFE_{\text{cell}}^\circ + RT \ln Q \quad (3.12)$$

Dividing the entire equation by  $-nF$ :

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \ln Q = E_{\text{cell}}^\circ - \frac{2.303RT}{nF} \log_{10} Q \quad (3.13)$$

### 3.2.1 Numerical Values of the Nernst Factor ( $\frac{2.303RT}{F}$ )

At standard reference temperature  $T = 298.15 \text{ K}$  ( $25^\circ\text{C}$ ):

$$\frac{2.303 \times 8.3145 \times 298.15}{96485.3} = 0.05916 \text{ V} \approx 0.0591 \text{ V} \quad (3.14)$$

The Nernst equation at 298 K is therefore:

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.0591}{n} \log_{10} Q = E_{\text{cell}}^\circ - \frac{0.0591}{n} \log_{10} \left( \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b} \right) \quad (3.15)$$

### Variation of the Nernst Factor across Temperatures

For advanced thermodynamic and Olympiad problems where temperatures deviate from 298 K:

$$\begin{aligned}
 T = 273.15 \text{ K } (0^\circ\text{C}) : \quad & \frac{2.303RT}{F} = 0.0542 \text{ V} \\
 T = 298.15 \text{ K } (25^\circ\text{C}) : \quad & \frac{2.303RT}{F} = 0.05916 \text{ V} \approx 0.0591 \text{ V} \\
 T = 300.00 \text{ K } (26.85^\circ\text{C}) : \quad & \frac{2.303RT}{F} = 0.0595 \text{ V} \\
 T = 310.15 \text{ K } (37^\circ\text{C, Body Temp}) : \quad & \frac{2.303RT}{F} = 0.0615 \text{ V} \\
 T = 373.15 \text{ K } (100^\circ\text{C}) : \quad & \frac{2.303RT}{F} = 0.0740 \text{ V}
 \end{aligned}$$

### 3.3 Determination of Equilibrium Constant ( $K_{eq}$ ) from Cell EMF

As a galvanic cell operates, reactants are progressively consumed and products accumulate. Consequently, the reaction quotient  $Q$  continuously increases, causing the cell potential  $E_{\text{cell}}$  to decrease steadily according to the Nernst equation.

Eventually, the system reaches chemical equilibrium:

$$E_{\text{cell}} = 0 \quad \text{and} \quad Q = K_{eq} = K_c \quad (3.16)$$

Substituting these equilibrium conditions into the Nernst equation:

$$0 = E_{\text{cell}}^\circ - \frac{2.303RT}{nF} \log_{10} K_{eq} \quad (3.17)$$

$$E_{\text{cell}}^\circ = \frac{2.303RT}{nF} \log_{10} K_{eq} \quad (3.18)$$

Rearranging for  $\log_{10} K_{eq}$  at 298 K:

$$\log_{10} K_{eq} = \frac{nE_{\text{cell}}^\circ}{0.0591} \iff K_{eq} = 10^{\frac{nE_{\text{cell}}^\circ}{0.0591}} \quad (3.19)$$

### Thermodynamic Significance of $K_{eq} \leftrightarrow E_{\text{cell}}^\circ$ Mapping

- Even a modest positive standard EMF produces an extraordinarily large equilibrium constant due to the exponential mapping. For instance, if  $n = 2$  and  $E_{\text{cell}}^\circ = +1.10 \text{ V}$  (Daniell Cell):

$$\log_{10} K_c = \frac{2 \times 1.10}{0.0591} = 37.22 \implies K_c = 1.68 \times 10^{37}$$

This massive value demonstrates that the Daniell cell reaction proceeds to quantitative completion ( $> 99.9999\%$ ).

- Direct chemical determination of  $K_c > 10^{10}$  or  $K_c < 10^{-10}$  by analytical titration is virtually impossible due to imperceptible trace concentrations at equilibrium. Potentiometric measurement of  $E_{\text{cell}}^\circ$  provides the **most precise experimental route** to evaluate extreme equilibrium constants.

### 3.4 Comprehensive Cell Thermodynamics: $\Delta S$ , $\Delta H$ , $q_{rev}$ , & Efficiency $\eta$

#### 3.4.1 1. Entropy Change ( $\Delta S$ ) and Temperature Coefficient of EMF

From the fundamental thermodynamic identity for Gibbs free energy at constant pressure:

$$dG = Vdp - SdT \implies \left( \frac{\partial \Delta G}{\partial T} \right)_p = -\Delta S \quad (3.20)$$

Substituting  $\Delta G = -nFE_{cell}$ :

$$\left( \frac{\partial(-nFE)}{\partial T} \right)_p = -nF \left( \frac{\partial E}{\partial T} \right)_p = -\Delta S \quad (3.21)$$

Cancelling the negative signs gives the celebrated formula:

$$\boxed{\Delta S = nF \left( \frac{\partial E}{\partial T} \right)_p} \quad (3.22)$$

where  $\left( \frac{\partial E}{\partial T} \right)_p$  is the **Temperature Coefficient of EMF** (in  $V \cdot K^{-1}$ ).

#### Physical Interpretation of the Temperature Coefficient $\left( \frac{\partial E}{\partial T} \right)_p$

- Case 1:**  $\left( \frac{\partial E}{\partial T} \right)_p > 0$  (**Positive Coefficient**):  $\Delta S > 0$ . The cell reaction results in an increase in entropy. The electrical work output ( $nFE$ ) actually **exceeds** the decrease in enthalpy ( $|\Delta G| > |\Delta H|$ ). The cell absorbs heat reversibly from the surrounding thermal reservoir ( $q_{rev} = T\Delta S > 0$ ) to supplement the reaction enthalpy.
- Case 2:**  $\left( \frac{\partial E}{\partial T} \right)_p < 0$  (**Negative Coefficient**):  $\Delta S < 0$ . The cell reaction proceeds with a decrease in entropy. The electrical work output is less than the enthalpy change ( $|\Delta G| < |\Delta H|$ ). A fraction of the reaction enthalpy is released as reversible heat to the surroundings ( $q_{rev} = T\Delta S < 0$ ), causing the cell to warm up during reversible discharge.
- Case 3:**  $\left( \frac{\partial E}{\partial T} \right)_p = 0$  (**Zero Coefficient**):  $\Delta S = 0$ . In this rare scenario,  $\Delta G = \Delta H$ , and no heat is absorbed or released under reversible conditions ( $q_{rev} = 0$ ).

#### 3.4.2 2. Enthalpy Change of Cell Reaction ( $\Delta H$ )

Recall the fundamental definition of Gibbs free energy:

$$\Delta G = \Delta H - T\Delta S \implies \Delta H = \Delta G + T\Delta S \quad (3.23)$$

Substituting  $\Delta G = -nFE$  and  $\Delta S = nF \left( \frac{\partial E}{\partial T} \right)_p$ :

$$\boxed{\Delta H = -nFE + nFT \left( \frac{\partial E}{\partial T} \right)_p = -nF \left[ E - T \left( \frac{\partial E}{\partial T} \right)_p \right]} \quad (3.24)$$

This is the electrochemical form of the **Gibbs-Helmholtz Equation**.

#### 3.4.3 3. Reversible Heat Transferred ( $q_{rev}$ )

For a reversible process:

$$\boxed{q_{rev} = T\Delta S = nFT \left( \frac{\partial E}{\partial T} \right)_p} \quad (3.25)$$

### 3.4.4 4. Thermodynamic Efficiency of Electrochemical Energy Conversion ( $\eta$ )

In an internal combustion engine or thermal power plant, chemical fuel is first converted to heat, which is then converted into mechanical or electrical work subject to the strict upper limit of the **Carnot Efficiency**:

$$\eta_{\text{Carnot}} = \frac{T_H - T_C}{T_H} = 1 - \frac{T_C}{T_H} \quad (\text{typically } 35\% - 45\%) \quad (3.26)$$

In stark contrast, an electrochemical cell or fuel cell converts the chemical free energy of the fuel **directly into electrical energy** without passing through heat as an intermediate:

$$\eta_{\text{thermo}} = \frac{\text{Maximum Useful Electrical Work}}{\text{Total Enthalpy of Combustion}} = \frac{|\Delta G|}{|\Delta H|} = \frac{nFE}{|\Delta H|} \quad (3.27)$$

Substituting  $\Delta G = \Delta H - T\Delta S$ :

$$\eta = 1 - \frac{T\Delta S}{\Delta H} \quad (3.28)$$

For fuel cells like the  $\text{H}_2\text{--O}_2$  cell,  $\eta$  can exceed 80% – 90%, far surpassing all heat engines.

### 3.5 Comprehensive Master Solved Illustrations

#### Illustration 3.1: Determination of Thermodynamic Functions from Temperature-Dependent EMF

The electromotive force of a reversible galvanic cell as a function of temperature (in Kelvin) in the vicinity of 298 K is given by the empirical polynomial:

$$E(T) = 1.01845 - 4.05 \times 10^{-5}(T - 293.15) - 9.50 \times 10^{-7}(T - 293.15)^2 \quad (\text{in Volts})$$

The cell reaction involves the transfer of  $n = 2$  moles of electrons. At  $T = 298.15$  K, calculate:

- The cell potential ( $E$ ).
- The temperature coefficient of EMF,  $\left(\frac{\partial E}{\partial T}\right)_p$ .
- The Gibbs free energy change ( $\Delta G$ ).
- The entropy change ( $\Delta S$ ).
- The enthalpy change of the reaction ( $\Delta H$ ).
- The reversible heat exchange with the surroundings ( $q_{\text{rev}}$ ).

**Part (a): Cell Potential at 298.15 K:**  $\Delta T = 298.15 - 293.15 = 5.0$  K.

$$\begin{aligned} E(298.15) &= 1.01845 - (4.05 \times 10^{-5} \times 5.0) - [9.50 \times 10^{-7} \times (5.0)^2] \\ &= 1.01845 - 0.0002025 - 0.00002375 = 1.01822 \text{ V} \end{aligned}$$

**Part (b): Temperature Coefficient of EMF:** Differentiating  $E(T)$  with respect to  $T$ :

$$\left(\frac{\partial E}{\partial T}\right)_p = -4.05 \times 10^{-5} - 2 \times 9.50 \times 10^{-7}(T - 293.15)$$

At  $T = 298.15$  K ( $\Delta T = 5.0$  K):

$$\left(\frac{\partial E}{\partial T}\right)_p = -4.05 \times 10^{-5} - (1.90 \times 10^{-6} \times 5.0) = -4.05 \times 10^{-5} - 0.95 \times 10^{-5} = -5.00 \times 10^{-5} \text{ V} \cdot \text{K}^{-1}$$

**Part (c): Free Energy Change ( $\Delta G$ ):**

$$\Delta G = -nFE = -2 \times 96485 \times 1.01822 = -196486 \text{ J} \cdot \text{mol}^{-1} = -196.49 \text{ kJ} \cdot \text{mol}^{-1}$$

**Part (d): Entropy Change ( $\Delta S$ ):**

$$\Delta S = nF \left(\frac{\partial E}{\partial T}\right)_p = 2 \times 96485 \times (-5.00 \times 10^{-5}) = -9.6485 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1} \approx -9.65 \text{ J} \cdot \text{K}^{-1}$$

**Part (e): Enthalpy Change ( $\Delta H$ ):**

$$\begin{aligned} \Delta H &= \Delta G + T\Delta S = -196486 + [298.15 \times (-9.6485)] \\ &= -196486 - 2876.7 = -199363 \text{ J} \cdot \text{mol}^{-1} = -199.36 \text{ kJ} \cdot \text{mol}^{-1} \end{aligned}$$

**Part (f): Reversible Heat Exchange ( $q_{\text{rev}}$ ):**

$$q_{\text{rev}} = T\Delta S = -2876.7 \text{ J} \cdot \text{mol}^{-1} = -2.88 \text{ kJ} \cdot \text{mol}^{-1}$$

Because  $q_{\text{rev}} < 0$ , exactly 2.88 kJ of heat is **released** to the surroundings when the cell operates reversibly.

### 3.6 Exhaustive Practice Problem Bank (Chapter 3)

#### Organization of Practice Problems

All questions are categorized by **Sub-topic** with full origin book citations. In the Complete Solutions Edition, full mathematical and chemical derivations are provided directly beneath each problem.

#### 3.6.1 Sub-Topic 3.1: Nernst Equation Calculations (Full & Half Cells)

##### Problem 3.1

[Source: Pearson Ex-II Section A Q6]

For the half-cell reduction process:  $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \longrightarrow \text{Cu}(\text{s})$ , a plot of  $E_{\text{red}}$  versus  $\log_{10}[\text{Cu}^{2+}]$  is a straight line having an intercept on the potential axis equal to +0.34 V. What is the reduction potential of the half-cell containing 0.1 M  $\text{Cu}^{2+}$  ions at 298 K? (Take  $\frac{2.303RT}{F} = 0.06$  V)

- |             |             |
|-------------|-------------|
| (a) -0.31 V | (c) -0.37 V |
| (b) +0.31 V | (d) +0.37 V |

##### Problem 3.2

[Source: Pearson Ex-II Section A Q3]

We have an oxidation-reduction system:  $\text{M}^{4+}(\text{aq}) + \text{e}^{-} \rightleftharpoons \text{M}^{3+}(\text{aq})$  with standard reduction potential  $E^{\circ} = +0.36$  V. The ratio of concentrations of oxidized form to reduced form ( $[\text{M}^{4+}]/[\text{M}^{3+}]$ ) at which the reduction potential of the system becomes +0.24 V at 298 K is: [Given:  $\frac{2.303RT}{F} = 0.06$  V]

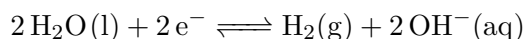
- |            |             |
|------------|-------------|
| (a) 1 : 10 | (c) 1 : 50  |
| (b) 1 : 20 | (d) 1 : 100 |

#### 3.6.2 Sub-Topic 3.2: Equilibrium Constant ( $K_{eq}$ ) from Cell EMF

##### Problem 3.3

[Source: Narendra Avasthi Level-2 Q6]

If the equilibrium constant for the self-ionization of water  $\text{H}^{+}(\text{aq}) + \text{OH}^{-}(\text{aq}) \rightleftharpoons \text{H}_2\text{O}(\text{l})$  is  $10^{13}$  at a certain non-standard temperature, what is the standard reduction potential  $E^{\circ}$  for the half-reaction:



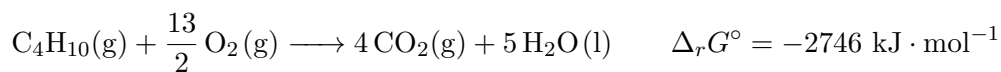
Given that  $\frac{2.303RT}{F} = 0.066$  V at this temperature.

- |              |              |
|--------------|--------------|
| (a) +1.230 V | (c) -0.800 V |
| (b) -0.858 V | (d) -0.827 V |

3.6.3 Sub-Topic 3.3: Cell Thermodynamics:  $\Delta S$ ,  $\Delta H$ , & Efficiency  $\eta$ **Problem 3.4**

[Source: Narendra Avasthi Level-2 Q7]

A fuel cell develops an electrical potential from the combustion of butane gas at 1 bar and 298 K:

What is the standard cell potential ( $E^\circ_{\text{cell}}$ ) of this fuel cell?

- (a) 4.74 V (c) 4.37 V  
(b) 0.547 V (d) 1.094 V

**Problem 3.5**

[Source: Neeraj Kumar Ex-II Subjective Q5]

The EMF of the cell  $\text{Pt} | \text{H}_2(1 \text{ bar}) | \text{HCl}(0.1 \text{ M}) | \text{AgCl}(\text{s}) | \text{Ag}$  is 0.2699 V at 293 K and 0.2660 V at 303 K.

- (a) What is the temperature coefficient of EMF,  $\left(\frac{\partial E}{\partial T}\right)_p$  ?  
(b) Calculate the entropy change ( $\Delta S^\circ$ ) and enthalpy change ( $\Delta H^\circ$ ) of the cell reaction at 298 K.

## Answer Key: Chapter 3 Practice Problem Bank

| Problem | Answer | Problem | Answer | Problem | Answer                                                                                             |
|---------|--------|---------|--------|---------|----------------------------------------------------------------------------------------------------|
| 3.1     | (b)    | 3.3     | (b)    | 3.5     | (a) $-3.90 \times 10^{-4} \text{ V} \cdot \text{K}^{-1}$                                           |
| 3.2     | (d)    | 3.4     | (d)    | 3.5     | (b) $\Delta S^\circ = -37.63 \text{ J} \cdot \text{K}^{-1}$ , $\Delta H^\circ = -37.07 \text{ kJ}$ |

## Chapter 4

# Concentration Cells & Liquid Junction Potential

### 4.1 Classification and Principles of Concentration Cells

In an ordinary chemical galvanic cell (such as the Daniell cell), the electromotive force is generated by an inherent difference in the standard chemical potentials ( $\mu^\circ$ ) of two chemically distinct redox couples ( $E_{\text{cell}}^\circ \neq 0$ ).

In stark contrast, a **Concentration Cell** is an electrochemical device in which:

1. Both half-cells are chemically identical in composition (identical electrode metals and identical electrolyte species).
2. Consequently, the standard cell potential is **identically zero**:

$$E_{\text{cell}}^\circ \equiv 0.0000 \text{ V} \quad (4.1)$$

3. The electromotive force ( $E_{\text{cell}} > 0$ ) arises **solely from a difference in concentration** (or activity  $a$ ) of the electrolyte solutions, or from a difference in the pressure (or activity) of the electrode materials themselves.

Concentration cells are divided into two fundamental classes:

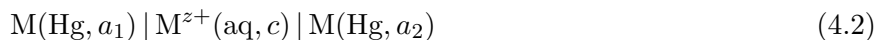
1. **Electrode Concentration Cells:** Electrodes have different concentrations/activities and dip into a common electrolyte solution of uniform concentration.
2. **Electrolyte Concentration Cells:** Identical electrodes dip into solutions containing the same electrolyte at different concentrations ( $c_1 \neq c_2$ ).

—

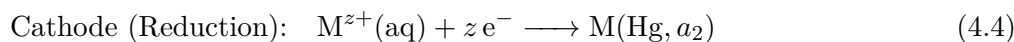
### 4.2 Electrode Concentration Cells

#### 4.2.1 1. Amalgam Concentration Cells

In an amalgam concentration cell, two electrodes consisting of liquid amalgams of the same metal at different activities ( $a_1$  and  $a_2$ , where  $a \propto$  mole fraction of metal dissolved in mercury) dip into a single homogeneous aqueous solution of the metal salt:



The electrode processes are:



Notice that the electrolyte concentration  $c$  cancels out completely from the net cell reaction! The process represents solely the spontaneous transfer of metal from the amalgam of higher activity ( $a_1$ ) to that of lower activity ( $a_2$ ).

Applying the Nernst equation ( $E_{\text{cell}}^{\circ} = 0$ ):

$$E_{\text{cell}} = 0 - \frac{RT}{zF} \ln \frac{a_2}{a_1} = \boxed{\frac{RT}{zF} \ln \frac{a_1}{a_2} = \frac{2.303RT}{zF} \log_{10} \frac{a_1}{a_2}} \quad (4.6)$$

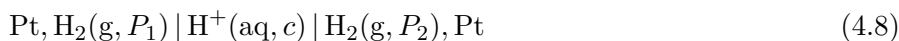
**Spontaneity Criterion:** For  $E_{\text{cell}} > 0$ , we must have:

$$\boxed{a_1 > a_2} \quad (4.7)$$

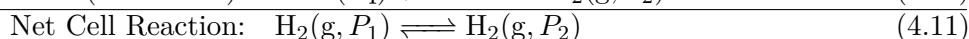
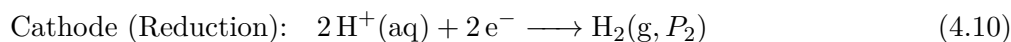
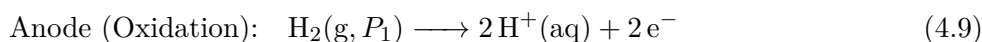
The amalgam with the higher concentration of metal must always act as the **Anode** (oxidation).

## 4.2.2 2. Gas Concentration Cells

A gas concentration cell consists of two gas electrodes operating at different partial pressures ( $P_1$  and  $P_2$ ) immersed in the same electrolyte solution:



The electrode processes are:



The Nernst equation for  $n = 2$  is:

$$\boxed{E_{\text{cell}} = \frac{RT}{2F} \ln \frac{P_1}{P_2} = \frac{0.0591}{2} \log_{10} \frac{P_1}{P_2}} \quad (\text{at } 298 \text{ K}) \quad (4.12)$$

**Spontaneity Criterion:** The cell generates a positive EMF ( $E_{\text{cell}} > 0$ ) when:

$$\boxed{P_1 > P_2} \quad (4.13)$$

The gas at higher pressure expands spontaneously to the region of lower pressure.

## 4.3 Electrolyte Concentration Cells Without Transference

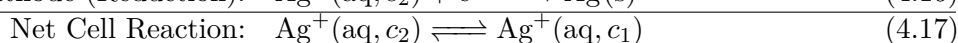
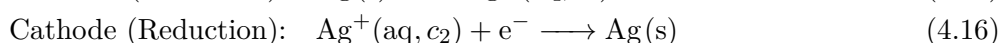
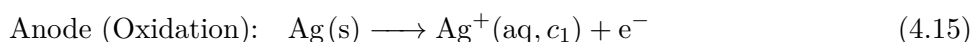
In an electrolyte concentration cell without transference, two half-cells contain the same electrolyte at different concentrations ( $c_1$  and  $c_2$ ) separated by a salt bridge that prevents direct physical contact and liquid junction potentials:

### 4.3.1 1. Reversible with Respect to Cations

Consider two silver electrodes dipping into  $\text{AgNO}_3$  solutions of concentrations  $c_1$  and  $c_2$  ( $c_2 > c_1$ ):



The half-cell reactions are:



The reaction is the spontaneous dilution of silver ions from the more concentrated solution ( $c_2$ ) to the more dilute solution ( $c_1$ ). Applying the Nernst equation ( $E_{\text{cell}}^{\circ} = 0$ ):

$$E_{\text{cell}} = \frac{RT}{F} \ln \frac{c_2}{c_1} = 0.0591 \log_{10} \frac{c_2}{c_1} \quad (\text{at } 298 \text{ K}) \quad (4.18)$$

For a general  $z$ -valent cation ( $M^{z+}$ ):

$$E_{\text{cell}} = \frac{RT}{zF} \ln \frac{c_2}{c_1} = \frac{0.0591}{z} \log_{10} \frac{c_2}{c_1} \quad (4.19)$$

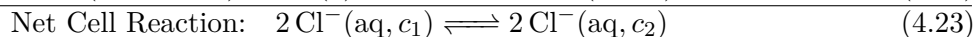
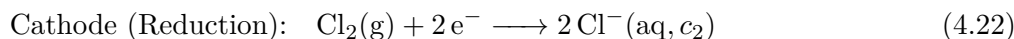
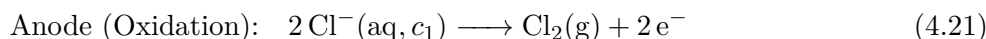
**Rule:** For cation-reversible electrodes, the **more dilute** solution acts as the **Anode** ( $c_1$ ), and the **more concentrated** solution acts as the **Cathode** ( $c_2$ ).

### 4.3.2 2. Reversible with Respect to Anions

Consider two chlorine gas electrodes dipping into KCl solutions of concentrations  $c_1$  and  $c_2$ :



The half-cell reactions are:



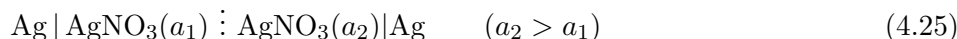
The Nernst equation is:

$$E_{\text{cell}} = \frac{RT}{F} \ln \frac{c_1}{c_2} = 0.0591 \log_{10} \frac{c_1}{c_2} \quad (\text{at } 298 \text{ K}) \quad (4.24)$$

**Rule:** For anion-reversible electrodes, the **more concentrated** solution acts as the **Anode** ( $c_1$ ), and the **more dilute** solution acts as the **Cathode** ( $c_2$ ).

## 4.4 Electrolyte Concentration Cells With Transference & Liquid Junction Potential ( $E_{lj}$ )

When two solutions of the same electrolyte at different concentrations ( $a_1$  and  $a_2$ ) are in direct physical contact across a porous barrier without a salt bridge:



The passage of 1 Faraday of electricity involves not only the electrode reactions, but also the physical migration of ions across the boundary interface (transference).

### 4.4.1 Rigorous Derivation of EMF With Transference ( $E_{wt}$ )

When 1 F of electric charge passes through the cell:

1. **At the Anode:** 1 mole of Ag is oxidized:



2. **Across the Junction:**

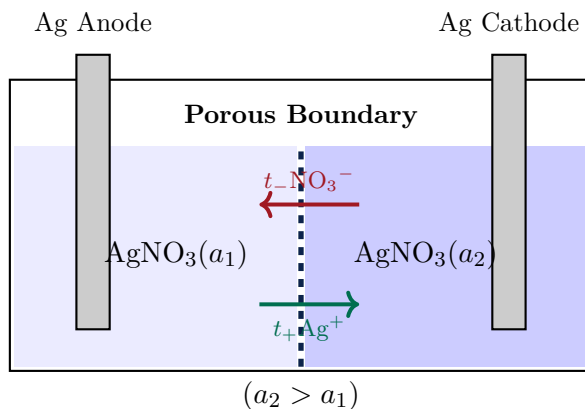


Figure 4.1: Transference of ions across the liquid-liquid junction in a concentration cell with transference.

- $t_+$  moles of  $\text{Ag}^+$  cations migrate from the anode compartment ( $a_1$ ) to the cathode compartment ( $a_2$ ).
- $t_-$  moles of  $\text{NO}_3^-$  anions migrate from the cathode compartment ( $a_2$ ) to the anode compartment ( $a_1$ ).

### 3. Net Change in Anode Compartment ( $a_1$ ):

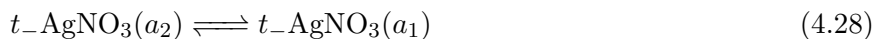
$$\Delta n_{\text{Ag}^+} = 1 - t_+ = t_- \text{ moles of } \text{Ag}^+ \quad (\text{since } t_+ + t_- = 1) \quad (4.27)$$

Along with  $t_-$  moles of  $\text{NO}_3^-$  arriving, exactly  $t_-$  moles of  $\text{AgNO}_3$  are gained in compartment  $a_1$ .

### 4. Net Change in Cathode Compartment ( $a_2$ ):

By symmetry, exactly  $t_-$  moles of  $\text{AgNO}_3$  are lost from compartment  $a_2$ .

Therefore, the net cell process per Faraday of charge passed is:



The free energy change is:

$$\Delta G = t_- \Delta G_{\text{dilution}} = -t_- \cdot 2RT \ln \frac{a_2}{a_1} \quad (4.29)$$

Since  $\Delta G = -1 \cdot F E_{\text{wt}}$ :

$$E_{\text{wt}} = 2t_- \frac{RT}{F} \ln \frac{a_2}{a_1} \quad (4.30)$$

where  $t_-$  is the transport number of the anion.

#### 4.4.2 Derivation of the Liquid Junction Potential ( $E_{lj}$ )

The total potential of the cell with transference ( $E_{\text{wt}}$ ) is the algebraic sum of the electrode potentials without transference ( $E_{\text{wot}}$ ) and the junction potential ( $E_{lj}$ ):

$$E_{\text{wt}} = E_{\text{wot}} + E_{lj} \implies E_{lj} = E_{\text{wt}} - E_{\text{wot}} \quad (4.31)$$

Recall that for the cell without transference:

$$E_{\text{wot}} = \frac{RT}{F} \ln \frac{a_2}{a_1} \quad (4.32)$$

Subtracting  $E_{\text{wot}}$  from  $E_{\text{wt}}$ :

$$E_{lj} = 2t_- \frac{RT}{F} \ln \frac{a_2}{a_1} - \frac{RT}{F} \ln \frac{a_2}{a_1} = (2t_- - 1) \frac{RT}{F} \ln \frac{a_2}{a_1} \quad (4.33)$$

Since  $1 = t_+ + t_- \implies 2t_- - 1 = t_- - t_+$ :

$$E_{lj} = (t_- - t_+) \frac{RT}{F} \ln \frac{a_2}{a_1} = (2t_- - 1) \frac{RT}{F} \ln \frac{a_2}{a_1} \quad (4.34)$$

### Fundamental Insights into the Liquid Junction Potential

- **Physical Origin:**  $E_{lj}$  arises because the cation and anion diffuse across the liquid boundary at different rates ( $u_+ \neq u_-$ ). The faster-moving ion diffuses ahead, setting up a charge separation and potential gradient across the boundary.
- **When  $t_- > t_+$  (e.g., HCl with fast  $H^+$ ? No, for HCl,  $u_+ > u_- \implies t_+ > t_-$ ):** For HCl,  $t_{H^+} \approx 0.83, t_{Cl^-} \approx 0.17$ . Because  $H^+$  diffuses much faster than  $Cl^-$ , the dilute solution becomes positively charged and the concentrated solution negatively charged ( $E_{lj} < 0$ ).
- **When  $t_+ = t_- = 0.50$  (The Salt Bridge Principle):**

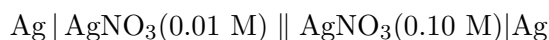
$$E_{lj} = (0.50 - 0.50) \frac{RT}{F} \ln \frac{a_2}{a_1} \equiv 0$$

Because  $K^+$  and  $Cl^-$  have virtually identical mobilities ( $t_{K^+} \approx 0.49, t_{Cl^-} \approx 0.51$ ), the diffusion potentials cancel out, completely eliminating  $E_{lj}$ .

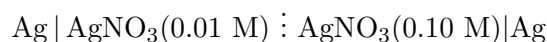
## 4.5 Comprehensive Master Solved Illustrations

### Illustration 4.1: Calculation of Transference EMF and Liquid Junction Potential

At 25°C (298 K), the EMF of the concentration cell without transference:



is found to be 0.0591 V. When the salt bridge is removed and the two solutions are placed in direct contact across a porous partition:



the measured EMF of the cell with transference is 0.0632 V. Calculate:

- The transport number of the nitrate ion ( $t_{\text{NO}_3^-}$ ).
- The transport number of the silver ion ( $t_{\text{Ag}^+}$ ).
- The liquid junction potential ( $E_{lj}$ ) established at the boundary.

**Part (a): Transport Number of  $\text{NO}_3^-$  ( $t_-$ ):** Recall the relationship between the EMF with transference and without transference:

$$E_{\text{wt}} = 2t_- E_{\text{wot}}$$

Given:  $E_{\text{wt}} = 0.0632 \text{ V}$  and  $E_{\text{wot}} = 0.0591 \text{ V}$ .

$$t_- = \frac{E_{\text{wt}}}{2E_{\text{wot}}} = \frac{0.0632}{2 \times 0.0591} = \frac{0.0632}{0.1182} \approx 0.5347 \quad (\approx 0.535)$$

**Part (b): Transport Number of  $\text{Ag}^+$  ( $t_+$ ):**

$$t_+ = 1 - t_- = 1 - 0.5347 = 0.4653 \quad (\approx 0.465)$$

**Part (c): Liquid Junction Potential ( $E_{lj}$ ):**

$$E_{lj} = E_{\text{wt}} - E_{\text{wot}} = 0.0632 \text{ V} - 0.0591 \text{ V} = +0.0041 \text{ V} = +4.1 \text{ mV}$$

**Physical Confirmation:** Since  $t_- > t_+$ , the nitrate anion moves faster than the silver cation. The faster diffusion of  $\text{NO}_3^-$  into the dilute compartment charges it negatively, creating a positive junction potential that accelerates  $\text{Ag}^+$  and retards  $\text{NO}_3^-$ .

## 4.6 Exhaustive Practice Problem Bank (Chapter 4)

### Organization of Practice Problems

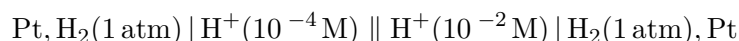
All questions are categorized by **Sub-topic** with origin citations. Complete solutions appear directly beneath each problem in the Complete Solutions Edition.

#### 4.6.1 Sub-Topic 4.1: Electrode & Electrolyte Concentration Cells

##### Problem 4.1

[Source: Narendra Avasthi Level-1 Q45]

Consider the concentration cell:



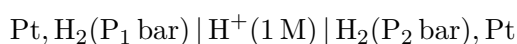
The electromotive force ( $E_{\text{cell}}$ ) of the cell at 298 K is:

- |               |               |
|---------------|---------------|
| (a) +0.0591 V | (c) -0.1182 V |
| (b) +0.1182 V | (d) 0.000 V   |

##### Problem 4.2

[Source: Neeraj Kumar Ex-I Q55]

For the gas concentration cell:



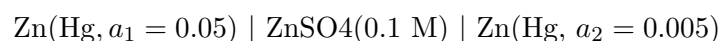
the cell reaction is thermodynamically spontaneous when:

- |                 |                                    |
|-----------------|------------------------------------|
| (a) $P_1 = P_2$ | (c) $P_1 < P_2$                    |
| (b) $P_1 > P_2$ | (d) Independent of $P_1$ and $P_2$ |

##### Problem 4.3

[Source: Cengage Solved Ex 3.12]

An amalgam concentration cell is constructed as:



The cell potential at 298 K is:

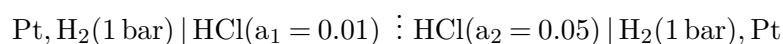
- |               |               |
|---------------|---------------|
| (a) +0.0591 V | (c) -0.0296 V |
| (b) +0.0296 V | (d) +0.0148 V |

#### 4.6.2 Sub-Topic 4.2: Cells With Transference & Liquid Junction Potential ( $E_{lj}$ )

##### Problem 4.4

[Source: Atkins Topic 6C / Physical Chemistry Olympiad]

The EMF of the concentration cell with transference:



is +0.0680 V at 298 K. Given that for HCl, the transport number of  $\text{H}^+$  is  $t_+ = 0.83$  and of  $\text{Cl}^-$  is  $t_- = 0.17$ . Calculate the liquid junction potential ( $E_{lj}$ ) established at the  $\text{HCl}(a_1) : \text{HCl}(a_2)$  interface.

### Answer Key: Chapter 4 Practice Problem Bank

| Problem | Answer | Problem | Answer   | Problem | Answer |
|---------|--------|---------|----------|---------|--------|
| 4.1     | (b)    | 4.3     | (b)      |         |        |
| 4.2     | (b)    | 4.4     | −27.3 mV |         |        |

## Chapter 5

# Advanced Equilibrium Applications of EMF

### 5.1 Determination of Solubility Product ( $K_{sp}$ ) of Sparingly Soluble Salts

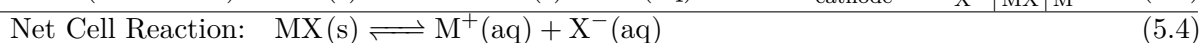
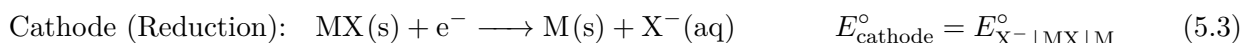
Potentiometry provides the most accurate and sensitive experimental method for determining the solubility product constants ( $K_{sp}$ ) of sparingly soluble salts (AgCl, AgBr, AgI, PbSO<sub>4</sub>, Hg<sub>2</sub>Cl<sub>2</sub>), because ordinary gravimetric or analytical methods are severely limited by the extraordinarily low concentrations of dissolved ions ( $< 10^{-5}$  M).

#### 5.1.1 Thermodynamic Derivation

Consider a sparingly soluble salt MX of a univalent metal M<sup>+</sup> and anion X<sup>-</sup> (such as AgCl). A galvanic cell is constructed combining the metal-metal ion electrode and the corresponding metal-insoluble salt-anion electrode:



The electrode processes are:



Notice that the net cell reaction is precisely the **heterogeneous dissolution equilibrium** of the sparingly soluble salt MX(s)! The standard cell potential is:

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ = E_{\text{X}^- \mid \text{MX} \mid \text{M}}^\circ - E_{\text{M}^+/\text{M}}^\circ \quad (5.5)$$

At dynamic dissolution equilibrium, the standard free energy change relates directly to  $K_{sp}$ :

$$\Delta G^\circ = -nFE_{\text{cell}}^\circ = -RT \ln K_{sp} \quad (5.6)$$

Equating both expressions ( $n = 1$ ):

$$\log_{10} K_{sp} = \frac{n(E_{\text{X}^- \mid \text{MX} \mid \text{M}}^\circ - E_{\text{M}^+/\text{M}}^\circ)}{0.0591} \quad (\text{at } 298 \text{ K}) \quad (5.7)$$

Conversely, the standard reduction potential of any metal-insoluble salt electrode can be calculated theoretically if  $K_{sp}$  and  $E_{\text{M}^+/\text{M}}^\circ$  are known:

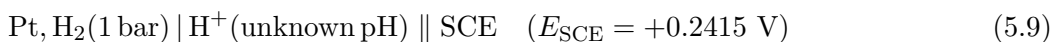
$$E_{\text{X}^- \mid \text{MX} \mid \text{M}}^\circ = E_{\text{M}^+/\text{M}}^\circ + 0.0591 \log_{10} K_{sp} = E_{\text{M}^+/\text{M}}^\circ - 0.0591 \text{p}K_{sp} \quad (5.8)$$

## 5.2 Potentiometric Determination of pH

The hydrogen ion activity ( $\text{pH} = -\log_{10} a_{\text{H}^+}$ ) can be determined with extreme precision by measuring the EMF of an electrochemical cell containing a hydrogen-responsive indicator electrode paired with a standard reference electrode:

### 5.2.1 1. The Hydrogen Electrode Method

An indicator hydrogen electrode dipping into the test solution of unknown pH is coupled with a Saturated Calomel Electrode (SCE):



The cell potential is:

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} = E_{\text{SCE}} - E_{\text{H}^+/\text{H}_2} \quad (5.10)$$

From the Nernst equation for the hydrogen electrode at  $P_{\text{H}_2} = 1 \text{ bar}$ :

$$E_{\text{H}^+/\text{H}_2} = -0.0591 \text{ pH} \quad (5.11)$$

Substituting into the cell EMF:

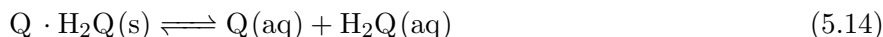
$$E_{\text{cell}} = E_{\text{SCE}} - (-0.0591 \text{ pH}) = E_{\text{SCE}} + 0.0591 \text{ pH} \quad (5.12)$$

Rearranging for pH:

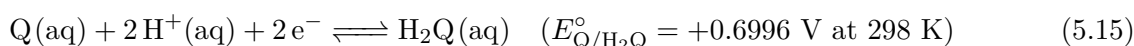
$$\boxed{\text{pH} = \frac{E_{\text{cell}} - E_{\text{SCE}}}{0.0591} = \frac{E_{\text{cell}} - 0.2415}{0.0591}} \quad (\text{at } 298 \text{ K}) \quad (5.13)$$

### 5.2.2 2. The Quinhydrone Electrode Method

Quinhydrone is an equimolar, stoichiometric 1 : 1 molecular addition complex of quinone ( $\text{C}_6\text{H}_4\text{O}_2$ , Q) and hydroquinone ( $\text{C}_6\text{H}_4(\text{OH})_2$ ,  $\text{H}_2\text{Q}$ ), linked by intermolecular hydrogen bonding and charge-transfer interactions:



When a pinch of quinhydrone powder is dissolved in the test solution and an inert platinum wire is immersed, the reversible organic redox equilibrium is established:



Applying the Nernst equation ( $n = 2$ ):

$$E_{\text{Q}} = E_{\text{Q}/\text{H}_2\text{Q}}^\circ - \frac{2.303RT}{2F} \log_{10} \left( \frac{[\text{H}_2\text{Q}]}{[\text{Q}][\text{H}^+]^2} \right) \quad (5.16)$$

Because quinhydrone yields strictly equimolar quantities of quinone and hydroquinone on dissolution,  $[\text{Q}] = [\text{H}_2\text{Q}]$ :

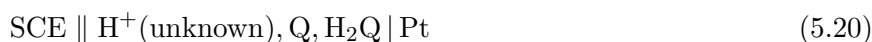
$$\frac{[\text{H}_2\text{Q}]}{[\text{Q}]} = 1 \quad (5.17)$$

The expression simplifies to:

$$E_{\text{Q}} = E_{\text{Q}/\text{H}_2\text{Q}}^\circ - \frac{0.0591}{2} \log_{10} \left( \frac{1}{[\text{H}^+]^2} \right) = E_{\text{Q}/\text{H}_2\text{Q}}^\circ + 0.0591 \log_{10} [\text{H}^+] \quad (5.18)$$

$$\boxed{E_{\text{Q}} = E_{\text{Q}/\text{H}_2\text{Q}}^\circ - 0.0591 \text{ pH} = 0.6996 - 0.0591 \text{ pH}} \quad (5.19)$$

When the quinhydrone electrode is coupled with a Saturated Calomel Electrode:



The cell EMF is:

$$E_{\text{cell}} = E_{\text{Q}} - E_{\text{SCE}} = (E_{\text{Q}/\text{H}_2\text{Q}}^{\circ} - 0.0591 \text{ pH}) - E_{\text{SCE}} \quad (5.21)$$

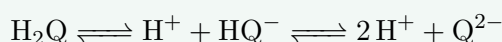
Solving for pH:

$$\text{pH} = \frac{E_{\text{Q}/\text{H}_2\text{Q}}^{\circ} - E_{\text{SCE}} - E_{\text{cell}}}{0.0591} = \frac{0.6996 - 0.2415 - E_{\text{cell}}}{0.0591} = \frac{0.4581 - E_{\text{cell}}}{0.0591} \quad (5.22)$$

### Critical Limitations of the Quinhydrone Electrode

The quinhydrone electrode cannot be used in solutions with  $\text{pH} > 8.5$  due to two breakdown mechanisms:

1. **Acidic Ionization of Hydroquinone:** Hydroquinone is a weak diprotic acid ( $K_{a1} \approx 10^{-10}$ ,  $K_{a2} \approx 10^{-12}$ ). In alkaline media ( $\text{pH} > 8.5$ ), hydroquinone ionizes into  $\text{HQ}^-$  and  $\text{Q}^{2-}$ :



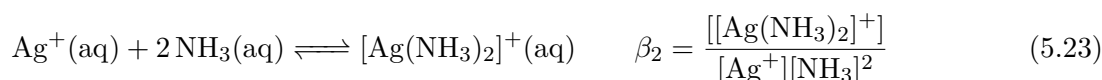
Consequently, the concentration of unionized  $[\text{H}_2\text{Q}]$  falls below  $[\text{Q}]$ , violating the fundamental identity  $[\text{Q}] = [\text{H}_2\text{Q}]$ .

2. **Atmospheric Oxidation of Hydroquinone:** In alkaline solution, hydroquinone is rapidly and irreversibly oxidized to quinone and tarry polymers by dissolved atmospheric oxygen ( $\text{O}_2$ ), permanently destroying the redox equilibrium.

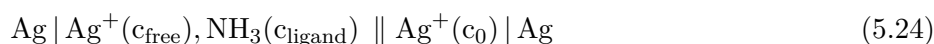
## 5.3 Determination of Complex Formation Constants (Stability Constants, $\beta_n$ )

The stability constant ( $\beta_n$  or  $K_f$ ) of a metal complex ion (e.g.,  $[\text{Ag}(\text{NH}_3)_2]^+$ ) is determined by monitoring the dramatic decrease in free metal cation concentration upon adding the complexing ligand.

Consider the formation of the diamminesilver(I) complex:



A concentration cell is assembled:



The right half-cell contains known  $[\text{Ag}^+] = c_0$  (e.g., 0.10 M). The left half-cell contains total silver  $c_{\text{total}}$  mixed with a large excess of ammonia. Virtually all silver is bound as complex:

$$[[\text{Ag}(\text{NH}_3)_2]^+] \approx c_{\text{total}} \quad (5.25)$$

The cell EMF is:

$$E_{\text{cell}} = 0.0591 \log_{10} \frac{c_0}{[\text{Ag}^+]_{\text{free}}} \quad (5.26)$$

Solving for the residual free silver ion concentration:

$$\log_{10}[\text{Ag}^+]_{\text{free}} = \log_{10} c_0 - \frac{E_{\text{cell}}}{0.0591} \quad (5.27)$$

Substituting  $[\text{Ag}^+]_{\text{free}}$  into the mass-action equilibrium directly yields the formation constant  $\beta_2$ .

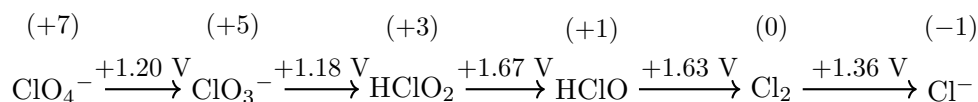
## 5.4 Latimer Diagrams and Frost Diagrams

Latimer and Frost diagrams are the two premier graphical frameworks used in advanced inorganic and physical electrochemistry to survey multiple oxidation states, calculate non-adjacent standard potentials, and predict disproportionation versus comproportionation spontaneity.

### 5.4.1 1. Latimer Diagrams

A Latimer diagram presents the standard reduction potentials connecting consecutive oxidation states of an element, arranged with the highest oxidation state on the far left and the lowest (elemental or negative) on the far right.

For chlorine in acidic solution (pH = 0):



### Calculating Non-Adjacent Standard Potentials

Because  $E^\circ$  is an intensive property, electrode potentials cannot be added directly ( $E_{\text{total}}^\circ \neq E_1^\circ + E_2^\circ$ ). Instead, one must add the extensive free energy changes ( $\Delta G^\circ = -nFE^\circ$ ):

$$\Delta G_{\text{total}}^\circ = \Delta G_1^\circ + \Delta G_2^\circ + \cdots + \Delta G_k^\circ \quad (5.28)$$

$$-n_{\text{total}}FE_{\text{total}}^\circ = -(n_1FE_1^\circ + n_2FE_2^\circ + \cdots + n_kFE_k^\circ) \quad (5.29)$$

Cancelling  $-F$  gives the master formula:

$$E_{\text{total}}^\circ = \frac{\sum_{i=1}^k n_i E_i^\circ}{\sum_{i=1}^k n_i} = \frac{n_1 E_1^\circ + n_2 E_2^\circ + \cdots + n_k E_k^\circ}{n_{\text{total}}} \quad (5.30)$$

### Disproportionation Criterion on a Latimer Diagram

Consider an intermediate oxidation state  $M_{\text{int}}$  situated between a higher state  $M_{\text{high}}$  on its left and a lower state  $M_{\text{low}}$  on its right:



The disproportionation reaction of  $M_{\text{int}}$  is:



The cell EMF for this disproportionation is:

$$E_{\text{disp}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ = E_R^\circ - E_L^\circ \quad (5.32)$$

#### The Latimer Disproportionation Rule

- If the potential to the **RIGHT** is greater than the potential to the **LEFT**:

$$E_R^\circ > E_L^\circ \implies E_{\text{disp}}^\circ > 0 \implies \text{Species undergoes Spontaneous Disproportionation!} \quad (5.33)$$

- If  $E_R^\circ < E_L^\circ$ , the intermediate species is thermodynamically stable against disproportionation; instead,  $M_{\text{high}}$  and  $M_{\text{low}}$  will spontaneously **comproportionate** into  $M_{\text{int}}$ .

### 5.4.2 2. Frost Diagrams

A Frost diagram plots the relative free energy of oxidation states against the oxidation number  $N$ :

$$y\text{-axis: } \frac{\Delta G^\circ}{F} = -nE^\circ \quad \text{relative to the pure element (where } N = 0, \Delta G^\circ = 0) \quad (5.34)$$

$$x\text{-axis: Oxidation Number } N \quad (5.35)$$

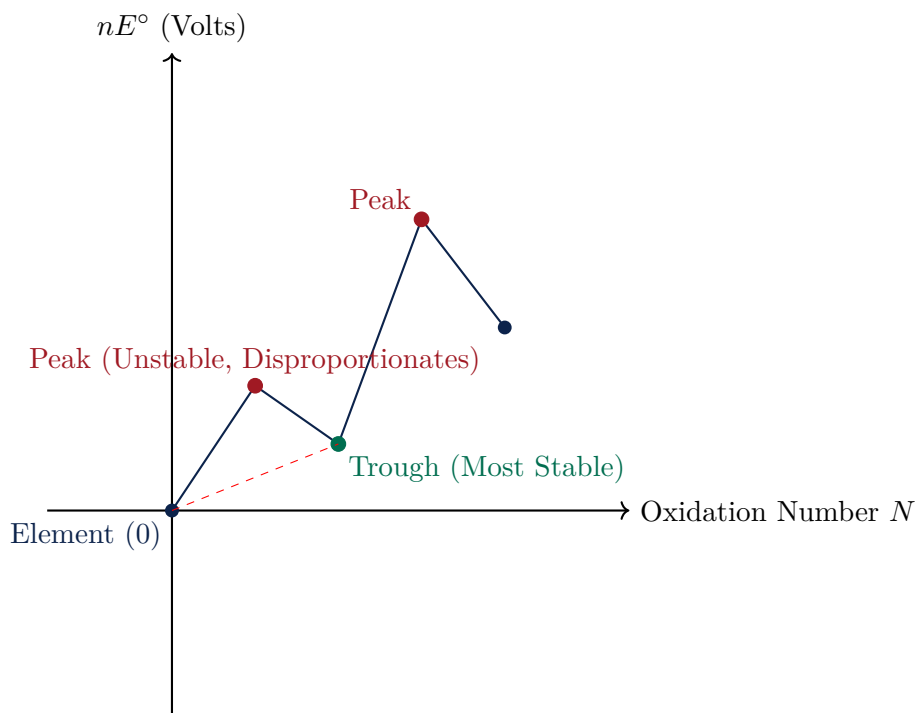


Figure 5.1: Schematic Frost Diagram illustrating stability against disproportionation and comproportionation.

#### Reading a Frost Diagram

1. **Slope of Line:** The slope of the straight line joining any two points on a Frost diagram equals the standard reduction potential ( $E^\circ$ ) connecting those two species:

$$\text{Slope} = \frac{\Delta(nE^\circ)}{\Delta N} = E^\circ \quad (5.36)$$

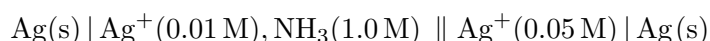
Steeper upward slopes indicate stronger oxidizing couples.

2. **Thermodynamic Stability (Troughs):** The species that lies at the lowest absolute point (deepest trough) on the Frost diagram is the thermodynamically most stable species of that element in the medium.
3. **Disproportionation (Convex Peaks):** Any species that lies **above** the straight chord connecting its two adjacent oxidation states is thermodynamically unstable and will spontaneously **disproportionate** into those flanking states.
4. **Comproportionation (Concave Wells):** Any species that lies **below** the chord connecting two flanking states is stable against disproportionation; the two flanking states will spontaneously react together to **comproportionate** into that intermediate species.

## 5.5 Comprehensive Master Solved Illustrations

### Illustration 5.1: Determination of Complex Formation Constant via Concentration Cell

A galvanic cell is set up at 298 K as:



The measured cell EMF is 0.444 V. Assuming that virtually all silver in the left compartment is sequestered as the diamminesilver(I) complex  $[\text{Ag}(\text{NH}_3)_2]^+$ , and that the concentration of uncomplexed ammonia is essentially unchanged ( $[\text{NH}_3] \approx 1.0 \text{ M}$ ), calculate:

- The concentration of free uncomplexed silver ions ( $[\text{Ag}^+]_{\text{free}}$ ) in the left compartment.
- The overall formation constant ( $\beta_2$  or  $K_f$ ) of  $[\text{Ag}(\text{NH}_3)_2]^+$ .

**Part (a): Free Silver Ion Concentration ( $[\text{Ag}^+]_{\text{free}}$ ):** The cell is a concentration cell reversible with respect to  $\text{Ag}^+$ :

$$E_{\text{cell}} = 0.0591 \log_{10} \left( \frac{[\text{Ag}^+]_{\text{cathode}}}{[\text{Ag}^+]_{\text{anode}}} \right)$$

Given:  $E_{\text{cell}} = 0.444 \text{ V}$  and  $[\text{Ag}^+]_{\text{cathode}} = 0.05 \text{ M}$ .

$$0.444 = 0.0591 \log_{10} \left( \frac{0.05}{[\text{Ag}^+]_{\text{free}}} \right)$$

$$\log_{10} \left( \frac{0.05}{[\text{Ag}^+]_{\text{free}}} \right) = \frac{0.444}{0.0591} \approx 7.5127$$

$$\frac{0.05}{[\text{Ag}^+]_{\text{free}}} = 10^{7.5127} = 10^{0.5127} \times 10^7 \approx 3.256 \times 10^7$$

$$[\text{Ag}^+]_{\text{free}} = \frac{0.05}{3.256 \times 10^7} \approx 1.536 \times 10^{-9} \text{ M}$$

**Part (b): Formation Constant ( $\beta_2$ ):** Since the total silver added was 0.01 M and  $[\text{Ag}^+]_{\text{free}} = 1.54 \times 10^{-9} \text{ M} \ll 0.01 \text{ M}$ :

$$[\text{Ag}(\text{NH}_3)_2]^+ \approx 0.01 \text{ M}$$

Applying the equilibrium expression:

$$\begin{aligned} \beta_2 &= \frac{[\text{Ag}(\text{NH}_3)_2]^+}{[\text{Ag}^+]_{\text{free}}[\text{NH}_3]^2} = \frac{0.01}{(1.536 \times 10^{-9}) \times (1.0)^2} \\ &= \frac{10^{-2}}{1.536 \times 10^{-9}} \approx 6.51 \times 10^6 \text{ M}^{-2} \end{aligned}$$

## 5.6 Exhaustive Practice Problem Bank (Chapter 5)

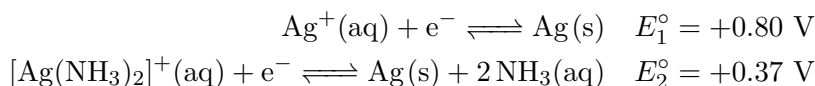
### Organization of Practice Problems

All questions are categorized by **Sub-topic** with full origin citations. Full mathematical derivations appear directly beneath each problem in the Complete Solutions Edition.

#### 5.6.1 Sub-Topic 5.1: Complex Formation Constants & Solubility Products via EMF

##### Problem 5.1 [Sources: Pearson Ex-II Section A Q4 / Neeraj Kumar Ex-II Q28]

The standard reduction potentials for the reactions:



are given at 298 K. What is the overall formation constant ( $K_f$  or  $\beta_2$ ) for the complex ion  $[\text{Ag}(\text{NH}_3)_2]^+$ ? [Given:  $\frac{2.303RT}{F} = 0.06 \text{ V}$ ]

- |                          |                            |
|--------------------------|----------------------------|
| (a) $1.0 \times 10^{-7}$ | (c) $2.15 \times 10^{19}$  |
| (b) $1.0 \times 10^7$    | (d) $4.64 \times 10^{-20}$ |

##### Problem 5.2 [Source: Pearson Ex-II Comprehension II Q4-5]

The standard reduction potential of the  $\text{Ag}^+/\text{Ag}$  electrode at 298 K is +0.80 V. The solubility product constant of AgI is  $K_{sp} = 6.4 \times 10^{-17}$  at 298 K. [Take  $\frac{2.303RT}{F} = 0.06 \text{ V}$ ,  $\log_{10}(6.4) = 0.806$ ]

**Part 1:** What is the potential of an  $\text{Ag}^+/\text{Ag}$  electrode immersed in a saturated aqueous solution of AgI at 298 K?

- |                        |                        |
|------------------------|------------------------|
| (a) $-0.314 \text{ V}$ | (c) $-0.172 \text{ V}$ |
| (b) $+0.314 \text{ V}$ | (d) $+0.172 \text{ V}$ |

**Part 2:** What is the standard reduction potential of the  $\text{I}^- | \text{AgI} | \text{Ag}$  electrode at 298 K?

- |                        |                        |
|------------------------|------------------------|
| (a) $-0.088 \text{ V}$ | (c) $-0.172 \text{ V}$ |
| (b) $+0.088 \text{ V}$ | (d) $+0.172 \text{ V}$ |

#### 5.6.2 Sub-Topic 5.2: Potentiometric pH Determination & Quinhydrone Electrode

##### Problem 5.3 [Source: Narendra Avasthi Level-2 Q8]

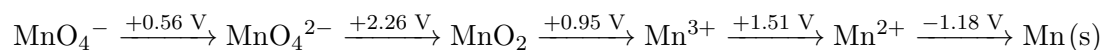
The electrochemical cell:



has an observed EMF of 0.5755 V at 25°C (298 K). If the standard oxidation potential of the calomel electrode is  $-0.2800 \text{ V}$ , the pH of the test solution is:

- |          |                   |
|----------|-------------------|
| (a) 11.0 | (c) 5.5           |
| (b) 4.5  | (d) None of these |

## 5.6.3 Sub-Topic 5.3: Latimer &amp; Frost Diagrams (Multi-Oxidation States)

**Problem 5.4****[Source: Atkins Topic 6D / Olympiad InChO Framework]**The Latimer diagram for manganese species in acidic aqueous solution ( $\text{pH} = 0$ ) is shown below:

- (a) Determine whether manganate ion ( $\text{MnO}_4^{2-}$ , oxidation state +6) is thermodynamically stable against disproportionation in acidic solution. Write the balanced disproportionation reaction.
- (b) Calculate the standard reduction potential for the direct conversion of permanganate to manganese dioxide:  $\text{MnO}_4^- \longrightarrow \text{MnO}_2$ .

## Answer Key: Chapter 5 Practice Problem Bank

| Problem | Answer                   | Problem | Answer                                                     | Pro |
|---------|--------------------------|---------|------------------------------------------------------------|-----|
| 5.1     | (b)                      | 5.3     | (c)                                                        |     |
| 5.2     | Part 1: (b), Part 2: (c) | 5.4     | (a) Disp. Spontaneous ( $E^\circ = +1.70$ V), (b) +1.693 V |     |

## Chapter 6

# Electrolysis, Faraday's Laws & Electrode Kinetics

### 6.1 Mechanism of Electrolysis

**Electrolysis** is the process of using an externally applied electrical current to drive a thermodynamically non-spontaneous chemical reaction at the electrodes of an **electrolytic cell**. Unlike a galvanic cell (which converts chemical energy spontaneously into electrical energy), an electrolytic cell requires an external DC power supply.

#### Electrolytic vs. Galvanic Cell

| Property           | Electrolytic Cell                  | Galvanic Cell                     |
|--------------------|------------------------------------|-----------------------------------|
| Reaction direction | Non-spontaneous ( $\Delta G > 0$ ) | Spontaneous ( $\Delta G < 0$ )    |
| Energy conversion  | Electrical $\rightarrow$ Chemical  | Chemical $\rightarrow$ Electrical |
| External source    | Required                           | Not needed                        |
| Anode polarity     | Positive (+)                       | Negative (–)                      |
| Cathode polarity   | Negative (–)                       | Positive (+)                      |
| Both electrodes    | Oxidation at anode                 | Oxidation at anode                |

#### 6.1.1 Ion Migration in an Electrolytic Cell

When DC voltage is applied:

- **Cations** ( $M^{n+}$ ) migrate to the **cathode** (negative) and are **reduced**.
- **Anions** ( $X^{m-}$ ) migrate to the **anode** (positive) and are **oxidized**.

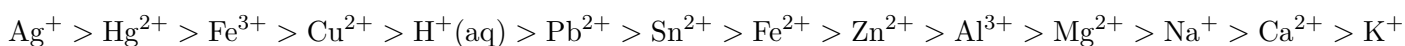
#### 6.1.2 Preferential Discharge Theory

##### Preferential Discharge Principle

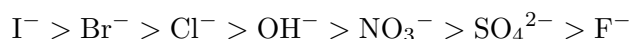
The ion that requires the **least energy** to be discharged is preferentially discharged.

- **Cathode** (reduction): cation with the **highest (most positive)**  $E_{\text{red}}^{\circ}$  is discharged first.
- **Anode** (oxidation): anion that is **easiest to oxidize** (lowest  $E_{\text{red}}^{\circ}$ ) is discharged first.

**Preferred order at cathode** (most  $\rightarrow$  least preferred):



**Preferred order at anode** (most  $\rightarrow$  least preferred):



Metals *below* hydrogen in the electrochemical series deposit preferentially over  $\text{H}_2$  from aqueous solution (e.g.  $\text{Cu}^{2+}$ ,  $\text{Ag}^+$ ). Metals *above* hydrogen cause  $\text{H}_2$  to evolve instead.

### 6.1.3 Overpotential and Overvoltage

The voltage actually required to drive continuous electrolysis always exceeds the thermodynamic back-EMF. The excess is called the **overpotential** ( $\eta$ ):

$$\eta = E_{\text{applied}} - E_{\text{theoretical}} \quad (6.1)$$

#### Causes of Overpotential

1. **Concentration polarization:** depletion or accumulation of ions at the electrode surface creates a concentration-cell back-EMF.
2. **Activation overpotential** (charge-transfer overpotential): energy barrier for the electron-transfer step, described by the **Butler–Volmer equation**:

$$j = j_0 \left[ \exp\left(\frac{\alpha F \eta}{RT}\right) - \exp\left(-\frac{(1-\alpha) F \eta}{RT}\right) \right] \quad (6.2)$$

where  $j_0$  = **exchange current density** (equilibrium reaction rate) and  $\alpha$  = **transfer coefficient** ( $\approx 0.5$ ).

3. **Resistance overpotential** (Ohmic drop):  $iR$  drop across the electrolyte.

#### Hydrogen Overpotential on Various Electrode Materials (at 25°C, 0.1 M $\text{H}_2\text{SO}_4$ )

| Electrode     | $\eta_{\text{H}_2}$ | Electrode    | $\eta_{\text{H}_2}$ |
|---------------|---------------------|--------------|---------------------|
| Platinized Pt | $\approx 0.00$ V    | Mercury (Hg) | $\approx 0.78$ V    |
| Polished Pt   | $\approx 0.09$ V    | Lead (Pb)    | $\approx 0.52$ V    |
| Gold (Au)     | $\approx 0.02$ V    | Zinc (Zn)    | $\approx 0.70$ V    |
| Nickel (Ni)   | $\approx 0.21$ V    | Iron (Fe)    | $\approx 0.40$ V    |
| Graphite (C)  | $\approx 0.60$ V    | Tin (Sn)     | $\approx 0.75$ V    |

#### Key consequences:

- High  $\eta_{\text{H}_2}$  on mercury ( $\approx 0.78$  V) allows deposition of Zn, Cd from *aqueous* solution on Hg cathode, even though thermodynamics favors  $\text{H}_2$  evolution.
- High oxygen overpotential on graphite ( $\approx 0.80$  V) causes  $\text{Cl}_2$  evolution to be preferred over  $\text{O}_2$  in concentrated brine at graphite anodes (chlor-alkali process).
- Small  $\eta_{\text{H}_2}$  on platinized Pt makes it an excellent reference for  $E^\circ = 0.00$  V.

## 6.2 Faraday's Laws of Electrolysis

Faraday's laws quantitatively connect the amount of chemical change at electrodes to the quantity of charge (electricity) passed through the cell.

### 6.2.1 Faraday's First Law

#### Faraday's First Law of Electrolysis

**Statement:** The mass of substance deposited or liberated at an electrode is directly proportional to the quantity of charge passed:

$$w \propto Q$$

$$\boxed{w = Z \cdot Q = Z \cdot I \cdot t} \quad (6.3)$$

- $w$  = mass of substance (g)
- $Z$  = electrochemical equivalent ( $\text{g C}^{-1}$ )
- $Q = I \cdot t$  = charge (C);  $I$  = current (A);  $t$  = time (s)

#### Electrochemical Equivalent ( $Z$ )

The electrochemical equivalent relates to atomic/molecular properties:

$$\boxed{Z = \frac{E}{F} = \frac{M}{nF}} \quad (6.4)$$

- $E = M/n$  = equivalent weight of the substance
- $M$  = molar mass ( $\text{g mol}^{-1}$ )
- $n$  = number of electrons transferred per formula unit
- $F = 96485 \approx 96500 \text{ C mol}^{-1}$  = Faraday's constant

The complete formula combining all parameters:

$$\boxed{w = \frac{M \cdot I \cdot t}{n F}} \quad \text{or equivalently} \quad \boxed{n_{\text{mol}} = \frac{I \cdot t}{n F}} \quad (6.5)$$

### 6.2.2 Faraday's Second Law

#### Faraday's Second Law of Electrolysis

**Statement:** When the same quantity of electricity passes through different electrolytes connected in series, the masses of different substances deposited or liberated at the respective electrodes are proportional to their equivalent weights:

$$\boxed{\frac{w_1}{w_2} = \frac{E_1}{E_2} = \frac{M_1/n_1}{M_2/n_2}} \quad (6.6)$$

**Corollary:** All cells in series receive the same charge  $Q$ , hence the same number of moles of equivalents =  $Q/F$ .

### 6.2.3 Faraday's Constant

$$\boxed{F = N_A \cdot e = (6.022 \times 10^{23} \text{ mol}^{-1})(1.602 \times 10^{-19} \text{ C}) = 96485 \text{ C mol}^{-1} \approx 96500 \text{ C mol}^{-1}} \quad (6.7)$$

One Faraday (96485 C) deposits exactly one gram-equivalent of any substance:

| Substance     | Reaction                                                                        | Deposited per Faraday                  |
|---------------|---------------------------------------------------------------------------------|----------------------------------------|
| Silver (Ag)   | $\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}$                            | $108/1 = 108 \text{ g}$                |
| Copper (Cu)   | $\text{Cu}^{2+} + 2 \text{e}^- \longrightarrow \text{Cu}$                       | $63.5/2 = 31.75 \text{ g}$             |
| Aluminum (Al) | $\text{Al}^{3+} + 3 \text{e}^- \longrightarrow \text{Al}$                       | $27/3 = 9 \text{ g}$                   |
| Hydrogen (H)  | $2 \text{H}^+ + 2 \text{e}^- \longrightarrow \text{H}_2$                        | $1 \text{ g} = 11.2 \text{ L at STP}$  |
| Oxygen (O)    | $\text{O}_2 + 4 \text{H}^+ + 4 \text{e}^- \longrightarrow 2 \text{H}_2\text{O}$ | $8 \text{ g O} = 5.6 \text{ L at STP}$ |

**Illustration 6.1: Three-Cell Series Circuit: Ag, Cu, and H<sub>2</sub>**

**Problem:** A steady current of 2 A flows for 3 hours through three series-connected cells: (i)  $\text{AgNO}_3(\text{aq})/\text{Pt}$ , (ii)  $\text{CuSO}_4(\text{aq})/\text{Pt}$ , (iii) acidified water/ $\text{Pt}$ . Calculate the mass of Ag and Cu deposited and the volume of  $\text{H}_2$  evolved. [ $M_{\text{Ag}} = 108$ ;  $M_{\text{Cu}} = 63.5$ ;  $n_{\text{Ag}} = 1$ ;  $n_{\text{Cu}} = 2$ ;  $n_{\text{H}_2} = 2$ ]

**Solution:**

$$Q = I \times t = 2 \text{ A} \times (3 \times 3600 \text{ s}) = 21600 \text{ C}$$

$$\text{Equivalents passed} = \frac{Q}{F} = \frac{21600}{96500} = 0.2238 \text{ eq}$$

1. **Ag** ( $n = 1$ ,  $E_{\text{Ag}} = 108$ ):  $w_{\text{Ag}} = 0.2238 \times 108 = \mathbf{24.17 \text{ g}}$
2. **Cu** ( $n = 2$ ,  $E_{\text{Cu}} = 31.75$ ):  $w_{\text{Cu}} = 0.2238 \times 31.75 = \mathbf{7.11 \text{ g}}$
3. **H<sub>2</sub>** ( $n = 2$ ,  $E_{\text{H}} = 1$ , moles of  $\text{H}_2 = 0.2238/2 = 0.1119 \text{ mol}$ ):  $V_{\text{H}_2} = 0.1119 \times 22.4 = \mathbf{2.51 \text{ L at STP}}$

### 6.3 Current Efficiency

Side reactions (parallel electrode processes, e.g.  $\text{H}_2$  evolution alongside metal deposition) consume charge without forming the desired product. The **current efficiency** quantifies productive utilization of charge:

**Current Efficiency**

$$\eta_{\text{curr}} = \frac{w_{\text{actual}}}{w_{\text{theoretical}}} \times 100\% \quad (6.8)$$

where  $w_{\text{theoretical}}$  is calculated from Faraday's first law assuming 100% selectivity.

**Illustration 6.2: Current Efficiency in Copper Electroplating**

**Problem:** A 5 A current for 1 h deposits only 5.63 g Cu. [ $M_{\text{Cu}} = 63.5$ ,  $n = 2$ ]

**Solution:**

$$w_{\text{theor}} = \frac{M \cdot I \cdot t}{n \cdot F} = \frac{63.5 \times 5 \times 3600}{2 \times 96500} = 5.932 \text{ g}$$

$$\eta_{\text{curr}} = \frac{5.63}{5.932} \times 100\% = \mathbf{94.9\%}$$

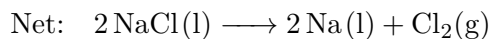
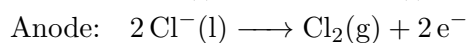
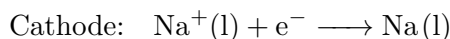
The remaining 5.1% was consumed by  $\text{H}_2$  evolution at the cathode.

### 6.4 Quantitative Electrolysis of Important Systems

Products depend on: (1) electrolyte type (molten vs. aqueous), (2) concentration, (3) electrode material, (4) current density and overpotential.

### 6.4.1 Electrolysis of Molten Salts

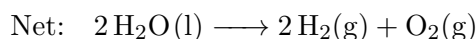
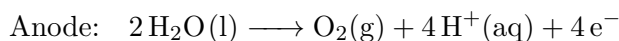
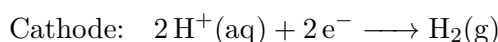
#### Illustration 6.3: Molten NaCl — Downs Cell



$\text{CaCl}_2$  is added to lower the melting point from  $801^\circ\text{C}$  to  $\approx 580^\circ\text{C}$ . A circular iron cathode with a surrounding graphite anode ensures separation of products.

### 6.4.2 Aqueous Electrolysis: Systematic Analysis

#### Dilute $\text{H}_2\text{SO}_4$ or Acidified Water (Inert Pt Electrodes)



Molar ratio  $\text{H}_2 : \text{O}_2 = 2 : 1$  (by moles = by volume at same  $T, P$ ).

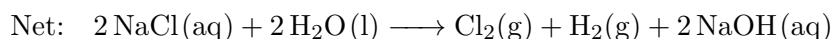
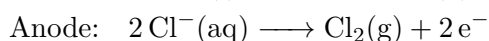
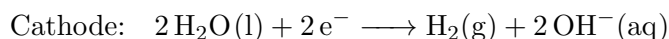
#### Dilute $\text{NaCl}(\text{aq})$ — Pt Electrodes

Low  $[\text{Cl}^-]$ : oxygen overpotential on Pt is not large enough to divert to  $\text{Cl}_2$ .

- Cathode:  $\text{H}_2(\text{g})$  evolved ( $\text{H}^+$  or  $\text{H}_2\text{O}$  reduced)
- Anode:  $\text{O}_2(\text{g})$  evolved ( $\text{OH}^-/\text{H}_2\text{O}$  oxidized, not  $\text{Cl}^-$ )

#### Concentrated $\text{NaCl}$ Brine (Chlor-Alkali Process — Graphite or MMO Anode)

High  $[\text{Cl}^-]$  + high  $\eta_{\text{O}_2}$  on graphite:  $\text{Cl}_2$  preferred at anode.



Three valuable industrial chemicals produced simultaneously. This is the Castner–Kellner cell (Hg cathode variant) or membrane cell (modern variant).

#### $\text{CuSO}_4(\text{aq})$ with Inert Pt Electrodes

- Cathode:  $\text{Cu}^{2+} + 2 \text{e}^- \longrightarrow \text{Cu}(\text{s})$  ( $E^\circ = +0.34 \text{ V}$ , strongly preferred over  $\text{H}^+$ )
- Anode:  $2 \text{H}_2\text{O} \longrightarrow \text{O}_2(\text{g}) + 4 \text{H}^+ + 4 \text{e}^-$  ( $\text{SO}_4^{2-}$  not oxidized on Pt)
- Effect:  $[\text{Cu}^{2+}]$  decreases;  $[\text{H}_2\text{SO}_4]$  increases; solution acidifies.

#### $\text{CuSO}_4(\text{aq})$ with Active Copper Anode (Electrorefining)

- Cathode:  $\text{Cu}^{2+} + 2 \text{e}^- \longrightarrow \text{Cu}(\text{s})$  — pure Cu deposited
- Anode:  $\text{Cu}(\text{s}) \longrightarrow \text{Cu}^{2+}(\text{aq}) + 2 \text{e}^-$  — impure Cu dissolves
- $[\text{CuSO}_4]$  remains constant; precious metal impurities (Ag, Au) fall as **anode mud**

**AgNO<sub>3</sub>(aq)**

- Inert Pt cathode:  $\text{Ag}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s})$  (basis of silver plating)
- Inert Pt anode:  $2\text{H}_2\text{O} \longrightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$  ( $\text{NO}_3^-$  not discharged; solution acidifies)
- Active Ag anode:  $\text{Ag}(\text{s}) \longrightarrow \text{Ag}^+(\text{aq}) + \text{e}^-$  (electrolyte maintained)

**NaOH(aq) — Pt Electrodes**

Cathode:  $\text{H}_2(\text{g})$ . Anode:  $\text{O}_2(\text{g})$ . Net: water electrolyzed;  $[\text{NaOH}]$  increases.

**K<sub>2</sub>SO<sub>4</sub>(aq) — Pt Electrodes**

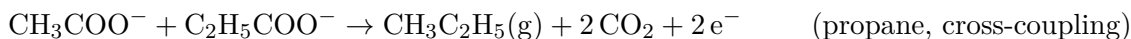
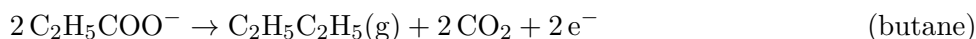
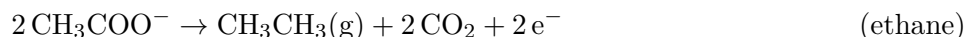
Cathode:  $\text{H}_2(\text{g})$ . Anode:  $\text{O}_2(\text{g})$ . Net: equivalent to dilute  $\text{H}_2\text{SO}_4$ .

**6.4.3 Kolbe's Electrolytic Synthesis**

Oxidative decarboxylation of carboxylate anions at a smooth Pt anode forms alkyl radicals, which couple:

**Illustration 6.4: Kolbe Cross-Coupling Products**

Electrolysis of a mixture of  $\text{CH}_3\text{COONa}$  and  $\text{C}_2\text{H}_5\text{COONa}$  yields **three alkanes**:



General rule:  $n$  different carboxylate anions produce  $\binom{n+1}{2}$  different alkanes.

**6.5 Decomposition Potential and Back-EMF**

The **decomposition potential** (decomposition voltage) is the minimum applied voltage to sustain continuous electrolysis at an observable rate:

$$E_{\text{decomp}} = E_{\text{back-EMF}} + |\eta_{\text{cathode}}| + |\eta_{\text{anode}}| + iR \quad (6.10)$$

$E_{\text{back-EMF}} = E_{\text{cell}}^\circ (\text{galvanic}) = \text{EMF of the cell that the products would form spontaneously.}$

**Illustration 6.5: Decomposition Potential of Dilute H<sub>2</sub>SO<sub>4</sub>**

$$E^\circ(\text{H}^+/\text{H}_2) = 0.00 \text{ V} \quad E^\circ(\text{O}_2/\text{H}_2\text{O}) = +1.23 \text{ V}$$

$$E_{\text{back}} = 1.23 - 0.00 = 1.23 \text{ V (theoretical)}$$

With  $\eta_{\text{H}_2}$  and  $\eta_{\text{O}_2}$  on polished Pt electrodes, experimentally  $E_{\text{decomp}} \approx 1.67 \text{ V}$ .

**6.6 Important Industrial Electrolytic Processes****6.6.1 Electrolytic Refining of Metals**

- **Setup:** Impure metal = anode; pure metal sheet = cathode; aqueous salt of same metal = electrolyte.

- **Process:** Impure anode dissolves; pure metal deposits on cathode. Impurities less noble than the metal pass into solution; noble metal impurities (Ag, Au, Pt) fall as **anode mud** and are economically valuable.
- **Applied to:** Cu, Ag, Au, Pb, Sn, Ni, Zn.

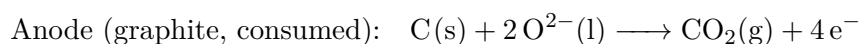
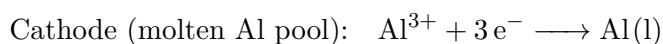
### 6.6.2 Electroplating

The object to be plated forms the cathode; the plating metal (or an inert anode) forms the anode; electrolyte = salt of the plating metal.

| Plating            | Electrolyte                                        | Anode    | Purpose                  |
|--------------------|----------------------------------------------------|----------|--------------------------|
| Nickel             | NiSO <sub>4</sub> (aq)                             | Ni metal | Corrosion resistance     |
| Chromium           | Acidic CrO <sub>3</sub>                            | Inert Pb | Hardness, shine          |
| Silver             | AgCN complex                                       | Ag metal | Cutlery, jewelry         |
| Gold               | [Au(CN) <sub>2</sub> ] <sup>-</sup>                | Au metal | Electronics, jewelry     |
| Zinc (galvanizing) | ZnSO <sub>4</sub> (aq)                             | Zn metal | Rust prevention on steel |
| Copper             | CuSO <sub>4</sub> + H <sub>2</sub> SO <sub>4</sub> | Cu metal | PCB circuits             |

### 6.6.3 Hall–Héroult Process (Aluminium Production)

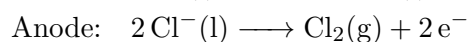
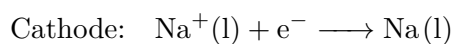
Purified Al<sub>2</sub>O<sub>3</sub> (alumina) is dissolved in molten cryolite (Na<sub>3</sub>AlF<sub>6</sub>) at  $\approx 950^\circ\text{C}$  to reduce the melting point and improve conductivity:



Graphite anodes are progressively consumed and require periodic replacement.

### 6.6.4 Downs Cell (Sodium and Chlorine Production)

Electrolysis of molten NaCl/CaCl<sub>2</sub> mixture at  $\approx 580^\circ\text{C}$ :



## 6.7 Summary Table: Electrode Products for Common Electrolytic Systems

| Electrolyte                                     | Electrode    | Cathode Product                             | Anode Product                   |
|-------------------------------------------------|--------------|---------------------------------------------|---------------------------------|
| Molten NaCl                                     | Pt           | Na metal                                    | Cl <sub>2</sub> gas             |
| Dilute H <sub>2</sub> SO <sub>4</sub> (aq)      | Pt           | H <sub>2</sub> gas                          | O <sub>2</sub> gas              |
| Dilute NaCl(aq)                                 | Pt           | H <sub>2</sub> gas                          | O <sub>2</sub> gas              |
| Conc. NaCl(aq)                                  | Graphite     | H <sub>2</sub> gas + NaOH                   | Cl <sub>2</sub> gas             |
| CuSO <sub>4</sub> (aq)                          | Pt (inert)   | Cu metal                                    | O <sub>2</sub> gas              |
| CuSO <sub>4</sub> (aq)                          | Cu (active)  | Cu metal                                    | Cu dissolves                    |
| AgNO <sub>3</sub> (aq)                          | Pt (inert)   | Ag metal                                    | O <sub>2</sub> gas              |
| AgNO <sub>3</sub> (aq)                          | Ag (active)  | Ag metal                                    | Ag dissolves                    |
| Molten Al <sub>2</sub> O <sub>3</sub> /cryolite | Graphite     | Al metal                                    | CO <sub>2</sub> gas             |
| NiSO <sub>4</sub> (aq)                          | Ni (active)  | Ni metal                                    | Ni dissolves                    |
| NaOH(aq)                                        | Pt           | H <sub>2</sub> gas                          | O <sub>2</sub> gas              |
| K <sub>2</sub> SO <sub>4</sub> (aq)             | Pt           | H <sub>2</sub> gas                          | O <sub>2</sub> gas              |
| Molten Al <sub>2</sub> O <sub>3</sub> alone     | C            | Al(l)                                       | O <sub>2</sub> /CO <sub>2</sub> |
| ZnSO <sub>4</sub> (aq)                          | Hg (cathode) | Zn metal (due to high $\eta_{\text{H}_2}$ ) | O <sub>2</sub> gas              |

## 6.8 Exhaustive Practice Problem Bank (Chapter 6)

### Chapter 6 Problem Bank

All problems sourced from: [GRB] pp. 1–10, [CNG-TH] pp. 76–92, [NA] pp. 1–20, [NRJ] Ex-I & II, [PRSN] Sec. A–B, [CNG-DPP] DPP-1–3. Full solutions in Complete Solutions Edition.

### 6.8.1 6.1 Faraday's Laws — Numerical

#### Problem 6.1

[GRB Ex-I Q3 / NA Level-1 Q1]

A current of 1.5 A is passed through a solution of  $\text{AgNO}_3$  for 25 minutes. Find the mass of silver deposited. [ $M_{\text{Ag}} = 108$ ,  $F = 96500 \text{ C mol}^{-1}$ ]

- |            |            |
|------------|------------|
| (a) 1.26 g | (c) 3.15 g |
| (b) 2.53 g | (d) 0.63 g |

#### Problem 6.2

[CNG-TH Illustration 5 / GRB Ex-I Q4]

A certain current liberates 0.504 g of hydrogen in 2 hours. Find the current (in amperes). [ $M_{\text{H}_2} = 2$ ,  $n = 2$ ,  $F = 96500$ ]

- |            |            |
|------------|------------|
| (a) 3.34 A | (c) 6.68 A |
| (b) 1.67 A | (d) 0.84 A |

#### Problem 6.3

[NA Level-1 Q3 / NRJ Ex-I Q2]

How many Faradays of charge are required to deposit 0.1 mol of each of the following: (i) Al from  $\text{AlCl}_3$ , (ii) Cu from  $\text{CuSO}_4$ , (iii) Cr from  $\text{Cr}_2(\text{SO}_4)_3$ , (iv) Au from  $\text{AuCl}_4^-$  ( $n = 3$ )?

#### Problem 6.4

[PRSN Ex-II Q7 / NRJ Ex-II Q6]

A series of three cells are connected in series: (A)  $\text{AgNO}_3(\text{aq})$ , (B)  $\text{CuSO}_4(\text{aq})$ , (C)  $\text{Al}_2(\text{SO}_4)_3(\text{aq})$ . The same charge passes through all three cells. If 2.7 g of Al is deposited in cell C, what masses of Ag and Cu are deposited? [ $M_{\text{Ag}} = 108$ ,  $M_{\text{Cu}} = 63.5$ ,  $M_{\text{Al}} = 27$ ,  $n_{\text{Ag}} = 1$ ,  $n_{\text{Cu}} = 2$ ,  $n_{\text{Al}} = 3$ ]

#### Problem 6.5

[NA Level-2 Q5]

How long would it take to deposit 10 g of gold from a solution of  $\text{HAuCl}_4$  ( $\text{Au}^{3+}$  is reduced) at a current of 2 A? [ $M_{\text{Au}} = 197$ ]

### 6.8.2 6.2 Electrolysis Products and Reactions

#### Problem 6.6

[GRB Ex-I Q8 / CNG-TH Concept App Q3]

In the electrolysis of aqueous  $\text{CuSO}_4$  with Pt electrodes, which of the following correctly describes the changes at each electrode?

- (a) Cathode: Cu deposited; Anode:  $\text{SO}_4^{2-}$  oxidised to  $\text{S}_2\text{O}_8^{2-}$
- (b) Cathode:  $\text{H}_2$  evolved; Anode:  $\text{O}_2$  evolved
- (c) Cathode: Cu deposited; Anode:  $\text{O}_2$  evolved
- (d) Cathode:  $\text{Cu}^{2+}$  reduced; Anode: Cu dissolves

**Problem 6.7****[CNG-DPP DPP-1 Q4]**

What products are obtained at cathode and anode when an aqueous solution of NaCl is electrolysed using (i) dilute, (ii) concentrated conditions, with Pt electrodes?

**Problem 6.8****[NRJ Ex-I Q8 / GRB Q15]**

The electrolysis of molten  $\text{Al}_2\text{O}_3$  dissolved in cryolite (Hall–Héroult process) uses:

- (a) Pt anode, steel cathode
- (b) Graphite anode, molten Al cathode
- (c) Graphite anode, Pt cathode
- (d) Inert anode, Pt cathode

**Problem 6.9****[PRSN Q10 / NA Level-2 Q12]**

**[Multiple Correct]** In which of the following cases does the composition of the electrolytic solution **NOT change** appreciably during electrolysis?

- (a)  $\text{CuSO}_4(\text{aq})$  with Pt electrodes
- (b)  $\text{CuSO}_4(\text{aq})$  with Cu electrodes
- (c)  $\text{AgNO}_3(\text{aq})$  with Ag electrodes
- (d) Dilute  $\text{H}_2\text{SO}_4(\text{aq})$  with Pt electrodes

**6.8.3 6.3 Current Efficiency and Advanced Numericals****Problem 6.10****[NA Level-2 Q14 / NRJ Ex-II Q12]**

An industrial electrolytic cell operates to deposit Zn from  $\text{ZnSO}_4(\text{aq})$ . A current of 200 A deposits 220 g of Zn in 30 minutes. Calculate the current efficiency. [ $M_{\text{Zn}} = 65.4$ ,  $n = 2$ ]

**Problem 6.11****[NRJ Ex-II Q14 / PRSN Ex-II Q9]**

**[Numerical Value]** The volume of  $\text{O}_2$  gas (at STP) liberated at the anode when 9.65 A passes through dilute  $\text{H}_2\text{SO}_4$  for 10000 s is \_\_\_\_\_ litres.

**Problem 6.12****[CNG-TH Advanced Q / NRJ Ex-II Q20]**

**[Comprehension]** In a Kolbe electrolysis, a solution containing both  $\text{CH}_3\text{COONa}$  and  $\text{C}_3\text{H}_7\text{COONa}$  (sodium butanoate) is electrolysed.

(A) How many different alkane products are obtained?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

(B) What is the heaviest alkane product formed?

- (a) Ethane
- (b) Butane
- (c) Hexane
- (d) Heptane

**Problem 6.13****[NA Level-3 Q7 / NRJ Ex-II Q22]****[Assertion–Reason]****Statement-1:** The hydrogen overpotential on mercury is very high ( $\approx 0.78$  V), making it possible to deposit zinc from aqueous  $\text{ZnSO}_4$  at a Hg cathode.**Statement-2:** Thermodynamically,  $\text{H}_2$  should evolve before Zn deposits from aqueous solution.

- (a) Both statements are true; Statement-2 is the correct explanation of Statement-1.
- (b) Both statements are true; Statement-2 is NOT the correct explanation of Statement-1.
- (c) Statement-1 is true; Statement-2 is false.
- (d) Statement-1 is false; Statement-2 is true.

**Problem 6.14****[GRB Numerical Q / CNG-TH Q]****[Numerical Value]** How many grams of copper can be deposited by passing 0.1 mol of electrons through a  $\text{CuSO}_4$  solution? [ $M_{\text{Cu}} = 63.5$ ]**Problem 6.15****[PRSN Ex-I Q11 / CNG-DPP DPP-2 Q6]**What volume (in mL) of  $\text{O}_2$  gas at STP is produced at the anode during electrolysis of water if 1 g of  $\text{H}_2$  is simultaneously evolved at the cathode?

## Chapter 7

# Commercial Cells, Fuel Cells & Corrosion

### 7.1 Primary Batteries (Non-Rechargeable Cells)

Primary batteries are **galvanic cells** that convert chemical energy into electrical energy through a spontaneous redox reaction. They are non-rechargeable because the electrode materials and electrolyte are consumed irreversibly during discharge.

#### 7.1.1 The Dry Cell (Leclanché Cell)

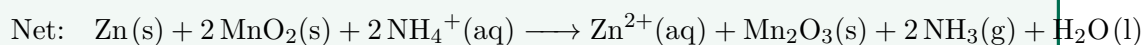
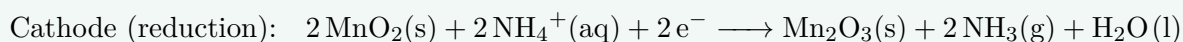
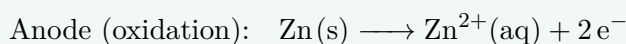
The dry cell is the most common primary battery (standard cylindrical AA, AAA, C, D batteries). It was invented by Georges Leclanché in 1866.

##### Dry Cell (Leclanché Cell) — Construction and Chemistry

###### Construction:

- **Anode:** Zinc container (Zn metal)
- **Cathode:** Carbon (graphite) rod surrounded by  $\text{MnO}_2$  paste
- **Electrolyte:** Moist paste of  $\text{NH}_4\text{Cl} + \text{ZnCl}_2$  (ammonium chloride / zinc chloride)

###### Electrode Reactions:



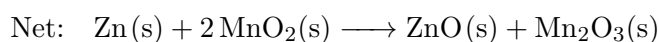
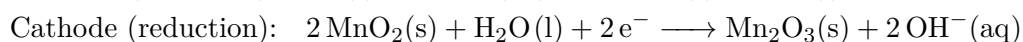
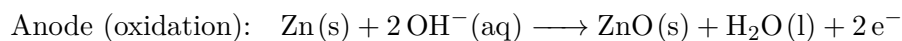
**Ammonia** is immobilized by complexing with  $\text{Zn}^{2+}$ :  $\text{Zn}^{2+} + 4\text{NH}_3 \longrightarrow [\text{Zn}(\text{NH}_3)_4]^{2+}$

**EMF:**  $\approx 1.5\text{ V}$  (fresh cell); drops with discharge.

**Limitations:** Voltage drops under load; Zn can corrode locally even when not in use; cannot be recharged.

#### 7.1.2 The Alkaline Battery

An improved version of the dry cell using an alkaline (KOH) electrolyte instead of the acidic  $\text{NH}_4\text{Cl}$  paste:

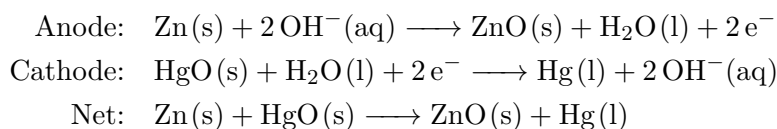


**Advantages over Leclanché:** Longer shelf life; more stable voltage under load; greater capacity; can be stored in a wider temperature range.

**EMF:**  $\approx 1.5\text{ V}$  (nominally same as Leclanché but more stable).

### 7.1.3 The Mercury Button Cell (Ruben–Mallory Cell)

Used in hearing aids, watches, and electronic instruments requiring a **precisely constant voltage**:



**Electrolyte:** Concentrated KOH or NaOH solution (absorbed in paste).

**EMF:**  $\approx 1.35\text{ V}$  (very stable throughout discharge because neither electrode material changes significantly in activity).

**Note:** Mercury cells are now banned in many countries due to mercury toxicity and environmental concerns.

## 7.2 Secondary (Rechargeable) Batteries

Secondary batteries can be restored to their original state by passing current through them in the reverse direction (recharging). The cell reactions are reversible.

### 7.2.1 The Lead–Acid Storage Battery

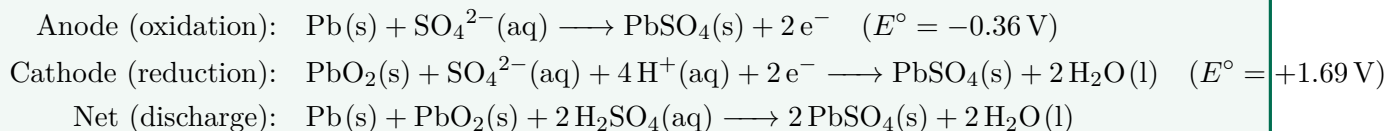
The lead–acid battery is the oldest and most widely used rechargeable battery, found in virtually every automobile.

#### Lead–Acid Battery: Discharge Reactions (Cell Delivers Current)

##### Construction:

- **Anode:** Lead (Pb) plates (negative during discharge)
- **Cathode:** Lead(IV) oxide ( $\text{PbO}_2$ ) plates (positive during discharge)
- **Electrolyte:** Dilute sulfuric acid ( $\text{H}_2\text{SO}_4$ ), density  $\approx 1.28\text{ g mL}^{-1}$  when fully charged

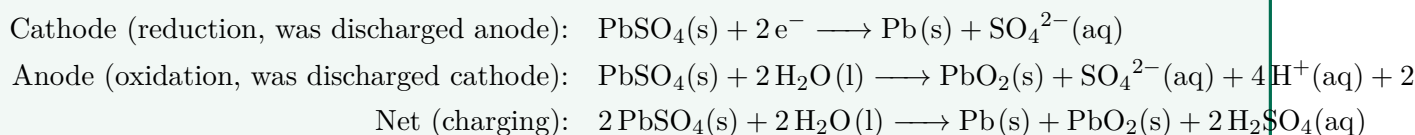
##### Discharge Electrode Reactions:



$E^\circ_{\text{cell}} = 1.69 - (-0.36) = 2.05\text{ V}$  per cell. A 12-V battery contains 6 cells in series.

#### Lead–Acid Battery: Charging Reactions (External Power Restores Cell)

During charging, an external EMF  $> 2.05\text{ V}$  is applied, reversing the discharge reactions:



**Key observations and quantitative analysis:**

- Both electrodes form  $\text{PbSO}_4(\text{s})$  during discharge — the “double sulphate” rule.
- During discharge:  $\text{H}_2\text{SO}_4$  is consumed and water is produced  $\rightarrow$  density of electrolyte **decreases**.
- During charging:  $\text{H}_2\text{SO}_4$  is regenerated  $\rightarrow$  density **increases**. Density of electrolyte (measured with a hydrometer) is a direct indicator of **state of charge**:
  - Fully charged:  $\rho \approx 1.28 \text{ g mL}^{-1}$
  - Half discharged:  $\rho \approx 1.20 \text{ g mL}^{-1}$
  - Fully discharged:  $\rho \approx 1.10 \text{ g mL}^{-1}$
- **Ampere-hour capacity:** A 12-V, 60 Ah battery can deliver 60 A for 1 h (or 1 A for 60 h).  
Energy stored =  $V \times \text{Ah} = 12 \times 60 = 720 \text{ Wh} = 2.592 \text{ MJ}$ .

**Illustration 7.1: Lead-Acid Battery: Mass and Density Change**

**Problem:** A lead-acid battery discharges for 2 h at 10 A. Calculate (a) the mass of Pb dissolved, (b) the mass of  $\text{PbO}_2$  consumed, (c) the mass of  $\text{H}_2\text{SO}_4$  consumed, and (d) the mass of  $\text{PbSO}_4$  deposited on each electrode. [ $M_{\text{Pb}} = 207$ ,  $M_{\text{PbO}_2} = 239$ ,  $M_{\text{H}_2\text{SO}_4} = 98$ ,  $M_{\text{PbSO}_4} = 303$ ]

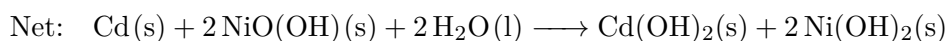
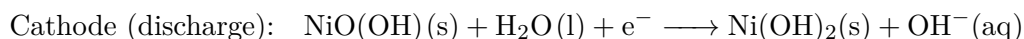
**Solution:**

$$Q = 10 \text{ A} \times 2 \times 3600 \text{ s} = 72000 \text{ C}$$

$$\text{mol e}^- = \frac{72000}{96500} = 0.7461 \text{ mol}$$

For each electrode reaction ( $n = 2$  per formula unit):  $\text{mol reacted} = 0.7461/2 = 0.3730 \text{ mol}$

1.  $w_{\text{Pb}} = 0.3730 \times 207 = \mathbf{77.2 \text{ g}}$  dissolved at anode
2.  $w_{\text{PbO}_2} = 0.3730 \times 239 = \mathbf{89.1 \text{ g}}$  consumed at cathode
3.  $w_{\text{H}_2\text{SO}_4} = 0.3730 \times 2 \times 98 = \mathbf{73.1 \text{ g}}$  consumed (2 mol per mol of cell reaction)
4.  $w_{\text{PbSO}_4}$  at each electrode =  $0.3730 \times 303 = \mathbf{113.0 \text{ g}}$  (anode) + same amount at cathode

**7.2.2 Nickel–Cadmium (Nicad) Battery**

**Electrolyte:**  $\text{KOH}(\text{aq})$ . **EMF:**  $\approx 1.25 \text{ V}$  per cell.

**Advantages:** Rechargeable hundreds of times; stable voltage; performs well at low temperatures.

**Disadvantages:** Memory effect (capacity decreases if not fully discharged before recharging); Cd is highly toxic.

**7.2.3 Lithium-Ion Battery**

The lithium-ion battery is the dominant rechargeable battery in consumer electronics and electric vehicles.

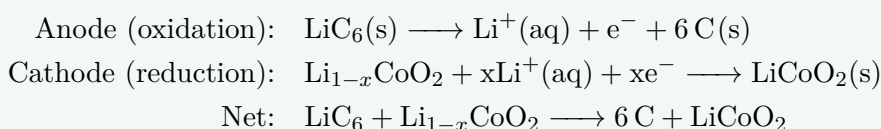
### Lithium-Ion Battery: Intercalation Mechanism

**Key principle:** Lithium ions ( $\text{Li}^+$ ) are reversibly “intercalated” (inserted/extracted) into layered crystalline structures of the electrode materials — the electrodes themselves are not consumed.

**Electrode Materials (most common):**

- **Cathode:**  $\text{LiCoO}_2$  (lithium cobalt oxide) — layered structure
- **Anode:** Graphite (C) —  $\text{Li}^+$  inserts between graphene layers to form  $\text{LiC}_6$
- **Electrolyte:** Lithium hexafluorophosphate ( $\text{LiPF}_6$ ) in organic carbonate solvents (non-aqueous)

**Discharge Reactions:**



During charging, the reverse occurs.

**Advantages:** High energy density ( $\approx 150\text{--}250 \text{ Wh kg}^{-1}$ ); no memory effect; low self-discharge; wide operating temperature range; high cell voltage ( $\approx 3.6 \text{ V}$  per cell).

**Disadvantages:** Risk of thermal runaway if overcharged or physically damaged; Co is scarce and expensive; requires sophisticated battery management systems (BMS).

**Other cathode materials:**  $\text{LiFePO}_4$  (LFP — safer, more stable),  $\text{LiMn}_2\text{O}_4$  (LMO),  $\text{Li}(\text{NiMnCo})\text{O}_2$  (NMC),  $\text{Li}(\text{NiCoAl})\text{O}_2$  (NCA).

## 7.3 Fuel Cells

A fuel cell is a galvanic cell that generates electricity by the **continuous** electrochemical oxidation of a fuel (supplied externally). Unlike a battery, a fuel cell does not run down as long as fuel and oxidant are supplied.

### Fuel Cell vs. Battery

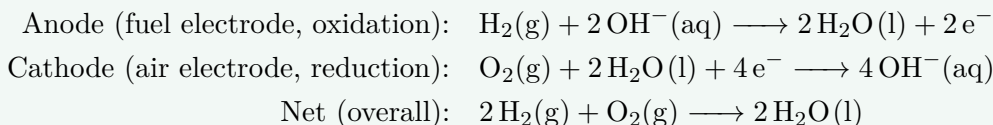
- **Battery:** Closed system; stores energy in electrode materials; runs down when material consumed.
- **Fuel Cell:** Open system; converts chemical energy of externally supplied fuel; runs continuously.
- Both are galvanic cells with spontaneous overall reactions.

### 7.3.1 Alkaline Hydrogen–Oxygen Fuel Cell (Apollo Mission Cell)

This cell was used to power the Apollo spacecraft. It operates in alkaline KOH electrolyte:

### Alkaline H<sub>2</sub>–O<sub>2</sub> Fuel Cell Reactions

**Electrolyte:** Hot concentrated KOH(aq) at  $\approx 200^\circ\text{C}$ , 20–40 atm



$$E^\circ_{\text{cell}} = E^\circ(\text{O}_2/\text{OH}^-) - E^\circ(\text{H}_2/\text{OH}^-) = 0.40 - (-0.83) = 1.23 \text{ V}$$

A bonus: the water produced is **drinkable** — it served as drinking water for the Apollo astronauts!

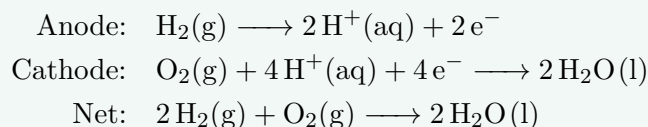
### 7.3.2 Proton Exchange Membrane (PEM) Fuel Cell

The PEM fuel cell is the leading technology for automobiles and portable power:

#### PEM Fuel Cell

**Electrolyte:** Solid polymer membrane (Nafion<sup>®</sup>) — conducts H<sup>+</sup> ions only, not electrons.

**Operating temperature:** 60–90°C. **Electrode material:** Pt or Pt alloy catalysts.



**Proton pathway:** H<sup>+</sup> ions move through the solid Nafion membrane from anode to cathode.

**Electron pathway:** Electrons flow through the external circuit (producing usable current).

### 7.3.3 Thermodynamic Efficiency of Fuel Cells

A conventional heat engine is limited by the Carnot efficiency:

$$\eta_{\text{Carnot}} = 1 - \frac{T_{\text{cold}}}{T_{\text{hot}}} = \frac{T_{\text{hot}} - T_{\text{cold}}}{T_{\text{hot}}} \quad (7.1)$$

A fuel cell converts Gibbs free energy directly to electrical work, bypassing the thermal step, and has a thermodynamic efficiency:

$$\eta_{\text{FC}} = \frac{\Delta G}{\Delta H} = \frac{-nFE_{\text{cell}}}{\Delta H} \quad (7.2)$$

#### Illustration 7.2: Efficiency of the H<sub>2</sub>–O<sub>2</sub> Fuel Cell

For  $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{H}_2\text{O}(\text{l})$  at 298 K:

$$\begin{aligned}\Delta H^\circ &= -572 \text{ kJ mol}^{-1} \\ \Delta G^\circ &= -nFE^\circ = -4 \times 96500 \times 1.23 = -474.8 \text{ kJ mol}^{-1} \\ \eta &= \frac{474.8}{572} = \mathbf{83\%}\end{aligned}$$

Compare with Carnot efficiency of a typical heat engine operating between 650 K and 300 K:  
 $\eta_{\text{Carnot}} = 1 - 300/650 = 53.8\%$ .

The fuel cell achieves significantly higher thermodynamic efficiency because it avoids the intermediate thermal step.

## 7.4 Corrosion: Electrochemical Mechanism and Prevention

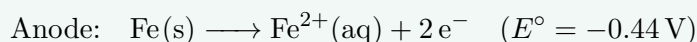
Corrosion is the **spontaneous electrochemical deterioration** of metals by reaction with their environment. It is responsible for enormous economic losses worldwide ( $\approx 3\text{--}4\%$  of GDP in industrialized nations).

### 7.4.1 Electrochemical Mechanism of Rusting of Iron

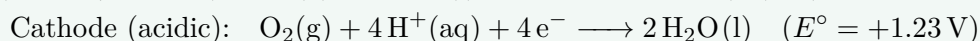
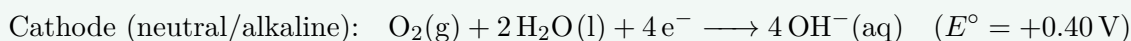
Rusting of iron is an electrochemical process that requires both water and oxygen (an electrolyte and an oxidant). Pure water without dissolved oxygen does not cause significant rusting. The process involves local galvanic cells formed at different points on the iron surface (due to grain boundaries, surface impurities, stress differences, dissolved salts, etc.):

#### Electrochemical Mechanism of Rusting

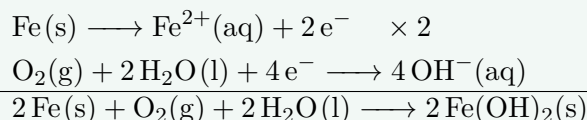
**At the anodic region** (where iron is in contact with less oxygenated water):



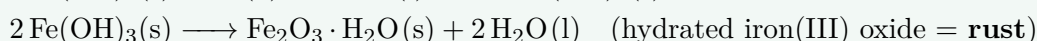
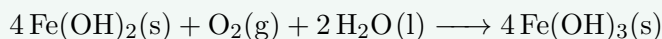
**At the cathodic region** (where iron is in contact with more oxygenated water):



**Overall reaction forming iron(II) hydroxide:**



**Further oxidation by dissolved oxygen** converts  $\text{Fe(OH)}_2$  to rust ( $\text{Fe}_2\text{O}_3$  hydrate):



**Rust composition:**  $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$  (hydrated iron(III) oxide), also written as  $\text{Fe(OH)}_3$  or  $\text{FeO(OH)}$ .

$E_{\text{cell}}^{\circ} = +0.40 - (-0.44) = 0.84\text{ V}$  (spontaneous,  $\Delta G < 0$ ).

### 7.4.2 Factors Promoting Corrosion

1. **Presence of electrolytes:** Salt water (sea spray, road salt) greatly accelerates corrosion by increasing electrolyte conductivity.
2. **Dissolved oxygen:** Necessary as the cathodic oxidant; higher  $[\text{O}_2]$  increases rate.
3. **Acidity (lower pH):** Provides  $\text{H}^{+}$  as an alternative cathodic reactant.
4. **Temperature:** Higher temperature increases reaction rates.
5. **Contact with a more noble metal:** Galvanic coupling; the less noble metal (higher  $E^{\circ}$  for oxidation) corrodes preferentially (e.g., Fe in contact with Cu corrodes faster).
6. **Physical stress/non-uniformity:** Stressed regions or grain boundaries act as anodes.
7.  **$\text{CO}_2$ :** Dissolves in water to form carbonic acid ( $\text{H}_2\text{CO}_3$ ), reducing pH.
8.  **$\text{SO}_2$ :** Industrial pollutant; forms  $\text{H}_2\text{SO}_4$  in solution — acid rain.

### 7.4.3 Corrosion Prevention Techniques

#### 1. Barrier (Passive) Protection

Physical coating isolates metal from the environment:

- Painting, lacquering, varnishing
- Coating with polymers (PVC coating on reinforcing bars)
- Enameling and vitreous enamel coatings
- Electroplating with Ni, Cr, Sn, Au (noble metals)
- Galvanizing (zinc coating, see below)
- Anodizing aluminum (electrochemically thickening the natural  $\text{Al}_2\text{O}_3$  layer)

#### 2. Sacrificial (Anodic) Protection — Galvanization

A more active (less noble) metal is placed in electrical contact with the metal to be protected. The more active metal acts as the **sacrificial anode** and corrodes preferentially while protecting the base metal:

##### Galvanization (Zinc Coating on Steel)

Zinc ( $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$ ) is more active than iron ( $E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.44 \text{ V}$ ).

When galvanized steel (Zn-coated steel) is scratched:

- Zn acts as the **anode**:  $\text{Zn} \longrightarrow \text{Zn}^{2+} + 2 \text{e}^-$
- Fe acts as the **cathode**:  $\text{O}_2 + 2 \text{H}_2\text{O} + 4 \text{e}^- \longrightarrow 4 \text{OH}^-$
- Fe is protected; Zn sacrifices itself.

**Note:** If Fe is coated with Sn (tinned steel, used in food cans), and the coating is scratched, Sn ( $E^\circ = -0.14 \text{ V}$ ) is *less* active than Fe, so Fe acts as the anode and corrodes *faster* than if uncoated. This is why damaged tin cans rust quickly.

Other sacrificial anodes used industrially:

- Zinc blocks attached to ship hulls (protect the hull and propeller)
- Magnesium rods in underground pipelines and hot water tanks
- Zinc or Mg blocks on offshore drilling platforms

#### 3. Cathodic Protection with Impressed Current

An external DC current is applied to make the metal to be protected the **cathode** of an electrolytic cell:

- Used for pipelines, ship hulls, oil rigs, reinforced concrete structures.
- The metal structure is connected to the negative terminal of a DC power supply.
- An inert anode (graphite or titanium) is buried in the soil nearby.
- The impressed cathodic current prevents the metal from acting as an anode (oxidizing).

## 4. Alloying

Incorporating corrosion-resistant elements into the metal:

- **Stainless steel:** Fe + 10.5% Cr (+ Ni, Mo). Chromium forms a thin, adherent, self-healing  $\text{Cr}_2\text{O}_3$  passive film.
- **Monel metal:** Ni + Cu alloy; resistant to seawater.
- **Cupronickel:** Cu + 10–30% Ni; used in coins and marine engineering.

## 5. Corrosion Inhibitors

Chemical substances added to the corrosive medium to retard corrosion:

- **Anodic inhibitors:** Increase anodic overpotential (e.g., chromates  $\text{CrO}_4^{2-}$ , nitrites  $\text{NO}_2^-$ , phosphates). They promote passivation.
- **Cathodic inhibitors:** Slow the cathodic reaction (e.g., arsenic, bismuth compounds; increase  $\eta_{\text{H}_2}$ ).
- **Mixed inhibitors:** Organic amines and surfactants that adsorb on both electrode surfaces.

### 7.4.4 Standard Reduction Potential and Corrosion

The ease of corrosion is directly related to the metal's position in the electrochemical series:

- Metals with **very negative**  $E^\circ$  (Na, K, Mg, Al) are highly reactive and corrode easily.
- Metals with **positive**  $E^\circ$  (Au, Pt, Ag) are noble and do not corrode under normal conditions.
- Iron ( $E^\circ = -0.44\text{ V}$ ) is in the middle; it corrodes at a significant rate in the presence of water and oxygen.

#### Pilling–Bedworth Ratio: Passivation vs. Cracking

The **Pilling–Bedworth (PB) ratio** determines whether an oxide film is protective:

$$R_{\text{PB}} = \frac{V_{\text{oxide}}}{V_{\text{metal}}} = \frac{M_{\text{oxide}} \cdot \rho_{\text{metal}}}{n \cdot M_{\text{metal}} \cdot \rho_{\text{oxide}}} \quad (7.3)$$

- $R_{\text{PB}} < 1$ : Oxide is porous (e.g., MgO,  $R = 0.81$ ): not protective; metal corrodes.
- $1 < R_{\text{PB}} < 2$ : Oxide is protective and adherent (e.g.,  $\text{Al}_2\text{O}_3$ ,  $R = 1.28$ ;  $\text{Cr}_2\text{O}_3$ ,  $R = 2.02$ ).
- $R_{\text{PB}} > 2$ : Oxide cracks and spalls off due to compressive stress (e.g.,  $\text{Fe}_2\text{O}_3$ ,  $R = 2.14$ ): not protective.

Rust ( $\text{Fe}_2\text{O}_3$ ) has  $R_{\text{PB}} > 2$ , so it spalls off and exposes fresh Fe to further corrosion — this is why rusting is a progressive, self-accelerating process unlike aluminum corrosion.

## 7.5 Comparison: Primary, Secondary, and Fuel Cells

| Property        | Primary Cell         | Secondary Cell       | Fuel Cell                                               |
|-----------------|----------------------|----------------------|---------------------------------------------------------|
| Rechargeability | Not rechargeable     | Rechargeable         | Continuous (not stored)                                 |
| Fuel source     | Internal (electrode) | Internal (electrode) | External (continuous)                                   |
| Examples        | Dry cell, Hg cell    | Lead-acid, Li-ion    | H <sub>2</sub> –O <sub>2</sub> , PEM                    |
| Energy density  | Moderate             | Moderate–High        | Very High                                               |
| Reaction type   | Irreversible         | Reversible           | Continuous                                              |
| By-products     | Chemical waste       | Chemical waste       | Only H <sub>2</sub> O (H <sub>2</sub> /O <sub>2</sub> ) |

## 7.6 Exhaustive Practice Problem Bank (Chapter 7)

### Chapter 7 Problem Bank

All problems sourced from: [GRB] pp. 37–45, [CNG-TH] pp. 91–110, [NA] pp. 21–43, [NRJ] Ex-I & II, [PRSN] Sec. C, [CNG-DPP] DPP-4–5. Full solutions in Complete Solutions Edition.

#### 7.6.1 7.1 Primary Batteries

**Problem 7.1****[GRB Ex-I Q22 / CNG-TH Concept App]**

In the dry cell (Leclanché cell), the cathode is:

- (a) Zinc container
- (b) Graphite rod
- (c) Ammonium chloride paste
- (d) Manganese(IV) oxide alone

**Problem 7.2****[CNG-DPP DPP-4 Q2]**

In the mercury button cell, the net cell reaction is:

- (a)  $\text{Zn} + \text{HgO} \longrightarrow \text{ZnO} + \text{Hg}$
- (b)  $\text{Hg} + \text{ZnO} \longrightarrow \text{ZnHgO}$
- (c)  $\text{Zn} + \text{Hg} \longrightarrow \text{ZnHg}$  (amalgam)
- (d)  $\text{ZnO} + \text{Hg} \longrightarrow \text{Zn} + \text{HgO}$

**Problem 7.3****[NRJ Ex-I Q18]****[Assertion–Reason]**

**Statement-1:** The alkaline battery has a longer shelf life than the Leclanché (dry) cell.

**Statement-2:** In an alkaline battery, the electrolyte is  $\text{KOH}(\text{aq})$ , which prevents corrosion of the zinc anode by reducing the rate of local cell action at impurity sites.

- (a) Both are true; S-2 is the correct explanation.
- (b) Both are true; S-2 is NOT the correct explanation.
- (c) S-1 is true; S-2 is false.
- (d) S-1 is false; S-2 is true.

#### 7.6.2 7.2 Lead-Acid Battery — Quantitative

**Problem 7.4****[NA Level-2 Q22 / CNG-TH Illustration 9]**

During discharge of a lead-acid battery, the density of sulfuric acid decreases from  $1.28 \text{ g mL}^{-1}$  to  $1.10 \text{ g mL}^{-1}$ . The cell delivers 10 A for  $t$  hours. If the total mass of  $\text{H}_2\text{SO}_4$  consumed per cycle is 29.4 g, find  $t$ .

[ $M_{\text{H}_2\text{SO}_4} = 98$ ; for the cell reaction:  $\text{Pb} + \text{PbO}_2 + 2\text{H}_2\text{SO}_4 \longrightarrow 2\text{PbSO}_4 + 2\text{H}_2\text{O}$ ;  $n = 2$  per mole of cell reaction]

**Problem 7.5**

[GRB Numerical Q8 / NRJ Ex-II Q24]

In a lead-acid battery, during charging, what happens to the density of the sulfuric acid solution? Also, write the net charging reaction.

**Problem 7.6**

[NA Level-3 Q15]

[Matrix Match] Match the cell with its electrode reaction:

| Column I (Cell)           | Column II (Anode Reaction)                                                                   |
|---------------------------|----------------------------------------------------------------------------------------------|
| (A) Dry cell              | (p) $\text{Zn} + 2\text{OH}^- \longrightarrow \text{ZnO} + \text{H}_2\text{O} + 2\text{e}^-$ |
| (B) Mercury cell          | (q) $\text{Zn} \longrightarrow \text{Zn}^{2+} + 2\text{e}^-$                                 |
| (C) Lead-acid (discharge) | (r) $\text{Pb} + \text{SO}_4^{2-} \longrightarrow \text{PbSO}_4 + 2\text{e}^-$               |
| (D) Nicad (discharge)     | (s) $\text{Cd} + 2\text{OH}^- \longrightarrow \text{Cd}(\text{OH})_2 + 2\text{e}^-$          |

**7.6.3 7.3 Fuel Cells and Efficiency****Problem 7.7**

[ATK Topic 6C / NRJ Ex-II Q28]

For the alkaline hydrogen–oxygen fuel cell, the overall reaction is  $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{H}_2\text{O}(\text{l})$  with  $\Delta H^\circ = -286 \text{ kJ mol}^{-1}$  and  $E^\circ_{\text{cell}} = 1.23 \text{ V}$ . Calculate the thermodynamic efficiency of this fuel cell.

**Problem 7.8**

[CNG-TH Advanced Q / GRB Ex-I Q29]

In a PEM fuel cell, protons travel from:

- (a) Cathode through external circuit to anode
- (b) Anode through the Nafion membrane to cathode
- (c) Anode through the external circuit to cathode
- (d) Cathode through the Nafion membrane to anode

**7.6.4 7.4 Corrosion****Problem 7.9**

[GRB Ex-I Q31 / CNG-DPP DPP-5 Q3]

Rusting of iron requires the simultaneous presence of:

- (a) Oxygen only
- (b) Water only
- (c) Both oxygen and water
- (d) Carbon dioxide only

**Problem 7.10**

[NA Level-2 Q30 / NRJ Ex-I Q22]

A piece of iron is connected to a piece of copper by a wire and immersed in salt water. Which of the following is correct?

- (a) Iron acts as cathode and is protected
- (b) Copper corrodes because it is less active
- (c) Iron acts as anode and corrodes preferentially
- (d) Both metals corrode at equal rates

**Problem 7.11**

[CNG-TH Q / GRB Ex-I Q34]

[Multiple Correct] Which of the following methods can prevent corrosion of iron?

- (a) Coating with zinc (galvanizing)
- (b) Connecting to a copper wire
- (c) Connecting to a magnesium block
- (d) Applying cathodic protection with impressed current

**Problem 7.12**

[NRJ Ex-II Q32 / PRSN Ex-II Q17]

Write the overall reaction for rusting of iron in neutral water and calculate the standard EMF of the corrosion cell. [ $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$ ;  $E^\circ(\text{O}_2/\text{OH}^-) = +0.40 \text{ V}$ ]

**Problem 7.13**

[ATK Physical Chemistry / ESS Advanced]

[Assertion–Reason]

**Statement-1:** Aluminum resists corrosion in air, despite having a very negative standard reduction potential ( $E^\circ = -1.66 \text{ V}$ ).

**Statement-2:** Aluminum forms a thin, adherent, self-repairing oxide layer ( $\text{Al}_2\text{O}_3$ ) on its surface that acts as a passivating film, preventing further oxidation.

- (a) Both true; S-2 is the correct explanation.
- (b) Both true; S-2 is NOT the correct explanation.
- (c) S-1 true; S-2 false.
- (d) S-1 false; S-2 true.

**Problem 7.14****[NA Level-3 Q18 / CNG-TH Advanced]**

**[Comprehension]** A ship hull is made of iron. To prevent corrosion, zinc blocks are attached to the hull below the water line.

(A) What role does the zinc block play?

- (a) It acts as a cathodic protection using impressed current.
- (b) It acts as a sacrificial anode.
- (c) It acts as a barrier coating.
- (d) It increases the electrical resistance of the hull.

(B) The zinc blocks need to be replaced periodically. Why?

- (a) They become contaminated with rust from the iron.
- (b) They are gradually consumed (corrode away) while protecting the iron.
- (c) The seawater dissolves them chemically without electrochemical action.
- (d) They passivate after prolonged exposure.

# Master Problem Bank

# Global Exercise Bank — Subjective Problems

## Subjective / Descriptive Problems (All Chapters)

Deep derivations, multi-step quantitative problems and open-ended questions. Sources: [ATK], [NRJ] Ex-II Subjective, [PRSN] Ex-II.

### S-1. Conductance and Transport

**S.1** [ATK Topic 6D / NRJ Subjective Q3]  
Derive the Debye–Hückel–Onsager (DHO) equation  $\Lambda_m = \Lambda_m^\circ - b\sqrt{C}$ , identifying the physical origin of the two contributions to the slope  $b$ . State all assumptions made in the derivation and their range of validity.

**S.2** [NRJ Ex-II Subjective Q5 / PRSN Ex-II Q14]  
A conductometric titration is performed: 25 mL of 0.1 M acetic acid ( $\text{CH}_3\text{COOH}$ ) is titrated with 0.1 M NaOH. Sketch the conductance-vs-volume curve, indicate the equivalence point, and explain the trend in each region. Which ionic species cause the conductance to change in each region?

### S-2. Cell Thermodynamics

**S.3** [ATK Topic 6C.4 / NRJ Subjective Q8]  
For the cell  $\text{Pt} | \text{H}_2(\text{g}, 1 \text{ bar}) | \text{HCl}(\text{aq}) | \text{AgCl}(\text{s}) | \text{Ag}(\text{s})$ , the EMF varies with temperature as  $E = 0.2224 - 6.4 \times 10^{-4}(T - 298) \text{ V}$ . At 298 K: calculate (a)  $\Delta G$ , (b)  $\Delta S$ , (c)  $\Delta H$ , (d)  $q_{\text{rev}}$  (heat reversibly exchanged with surroundings). [ $F = 96500$ ,  $n = 1$ ]

**S.4** [NRJ Subjective Q10 / PRSN Advanced]  
A concentration cell (without transference) is constructed:



(a) Write the cell reaction and identify which side is the anode. (b) Calculate  $E_{\text{cell}}$  at 298 K. (c) Show that the cell will spontaneously drive silver from the dilute side to the concentrated side.

### S-3. Faraday's Laws and Industrial Electrolysis

**S.5****[NRJ Subjective Q14 / PRSN Ex-II Q20]**

In the Hall–Héroult process, a cell operates at 5 V and 100 kA to produce aluminium. (a) Calculate the mass of Al produced per hour. (b) Calculate the electrical energy consumed per kg of Al. (c) If electricity costs \$0.05/kWh, what is the electricity cost per tonne of Al? [ $M_{\text{Al}} = 27$ ,  $n = 3$ ,  $F = 96500$ ]

# Global Exercise Bank — Single Correct MCQs

## Single Correct Option MCQs (All Chapters)

Sources: [CNG-TH] Concept App., [GRB] Ex-I, [NA] Level-1, [ESS], [CNG-DPP].

### SC.1

[GRB Ex-I / ESS Q]

The unit of molar conductance ( $\Lambda_m$ ) is:

- (a)  $\Omega^{-1}$
- (b)  $\text{S cm}^2 \text{mol}^{-1}$
- (c)  $\text{S cm}^{-1}$
- (d)  $\Omega \text{ cm}^2 \text{mol}^{-1}$

### SC.2

[CNG-TH Concept App. / GRB]

For a weak electrolyte at infinite dilution, which of Kohlrausch's law gives  $\Lambda_m^\circ$ ?

- (a) By extrapolating the Kohlrausch plot ( $\Lambda_m$  vs  $\sqrt{c}$ ) to  $c = 0$
- (b) By adding  $\lambda^\circ$  of constituent ions algebraically
- (c) Both (a) and (b)
- (d) (a) only for strong electrolytes; (b) only for weak electrolytes

### SC.3

[NA Level-1 Q6]

The molar conductance of 0.1 M  $\text{CH}_3\text{COOH}$  is  $5.2 \text{ S cm}^2 \text{mol}^{-1}$  and  $\Lambda_m^\circ(\text{CH}_3\text{COOH}) = 390.7 \text{ S cm}^2 \text{mol}^{-1}$ . The degree of dissociation  $\alpha$  is:

- (a) 0.0133
- (b) 0.133
- (c) 0.75
- (d) 0.33

### SC.4

[GRB Ex-I / CNG-TH]

In the standard hydrogen electrode (SHE), the standard conditions are:

- (a)  $[\text{H}^+] = 1 \text{ mol L}^{-1}$ ,  $P_{\text{H}_2} = 1 \text{ atm}$ ,  $T = 298 \text{ K}$
- (b)  $[\text{H}^+] = 1 \text{ mol L}^{-1}$ ,  $P_{\text{H}_2} = 1 \text{ bar}$ ,  $T = 298 \text{ K}$
- (c)  $a_{\text{H}^+} = 1$ ,  $f_{\text{H}_2} = 1 \text{ bar}$ ,  $T = 298 \text{ K}$
- (d)  $[\text{H}^+] = 7 \text{ M}$ ,  $P_{\text{H}_2} = 1 \text{ bar}$ ,  $T = 298 \text{ K}$

**SC.5****[CNG-DPP DPP-2 / GRB Q]**

Which of the following couples has the **highest** tendency to act as a reducing agent?

- (a)  $\text{Li}^+/\text{Li}$  ( $E^\circ = -3.05 \text{ V}$ ) (c)  $\text{Cu}^{2+}/\text{Cu}$  ( $E^\circ = +0.34 \text{ V}$ )  
(b)  $\text{Zn}^{2+}/\text{Zn}$  ( $E^\circ = -0.76 \text{ V}$ ) (d)  $\text{Ag}^+/\text{Ag}$  ( $E^\circ = +0.80 \text{ V}$ )

**SC.6****[ESS §24 / GRB Q]**

The transference number of  $\text{K}^+$  in KCl at infinite dilution is approximately 0.49. This means:

- (a) 49% of the current is carried by  $\text{K}^+$  ions  
(b) 49% of the ions are  $\text{K}^+$   
(c)  $\text{K}^+$  moves 49% faster than  $\text{Cl}^-$   
(d) 49% of KCl is dissociated

**SC.7****[NA Level-1 Q12 / GRB]**

The EMF of the cell  $\text{Zn} | \text{ZnSO}_4(0.01 \text{ M}) || \text{CuSO}_4(1 \text{ M}) | \text{Cu}$  at 298 K is: [ $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$ ,  $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$ ]

- (a) 1.10 V (c) 1.04 V  
(b) 1.16 V (d) 0.98 V

**SC.8****[CNG-TH Concept App / NA Q14]**

Given  $K_{sp}(\text{AgCl}) = 1.8 \times 10^{-10}$  at 298 K and  $E^\circ(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$ , the standard reduction potential of the  $\text{Ag} | \text{AgCl(s)} | \text{Cl}^-$  electrode is:

- (a) +0.22 V (c) -0.22 V  
(b) +0.50 V (d) +0.80 V

**SC.9****[GRB / CNG-DPP DPP-3]**

In a Daniell cell, the salt bridge contains saturated KCl. Its primary function is to:

- (a) Provide  $\text{K}^+$  and  $\text{Cl}^-$  as reactants in the cell reaction  
(b) Prevent mixing of the two electrolytes  
(c) Maintain electrical neutrality in both half-cells and eliminate liquid junction potential  
(d) Increase the cell potential by providing a third electrolyte

**SC.10****[NA Level-1 Q20]**

The equilibrium constant for the cell reaction of the Daniel cell ( $\text{Zn} | \text{Zn}^{2+} || \text{Cu}^{2+} | \text{Cu}$ ) at 298 K is: [ $E^\circ_{\text{cell}} = 1.10 \text{ V}$ ,  $n = 2$ ]

- (a)  $10^{37.2}$  (c)  $e^{85.6}$   
(b)  $10^{3.72}$  (d)  $10^{18.6}$

# Global Exercise Bank — Multiple Correct MCQs

## One or More Than One Correct Options (All Chapters)

Sources: [NRJ] Ex-II Multi, [PRSN] Ex-II, [CNG-TH], [NA] Level-3.

### MC.1

[NRJ Ex-II Q5 / PRSN Ex-II Q3]

Which of the following statements about the molar conductance of electrolytes are **correct**?

- (a) For strong electrolytes,  $\Lambda_m$  increases linearly with  $1/\sqrt{c}$  according to Kohlrausch's law.
- (b) For weak electrolytes,  $\Lambda_m$  cannot be obtained by extrapolating the  $\Lambda_m$  vs  $\sqrt{c}$  plot.
- (c) The specific conductance ( $\kappa$ ) always decreases on dilution.
- (d) At the same concentration,  $\Lambda_m$  of a strong electrolyte is always greater than that of a weak electrolyte of the same formula type.

### MC.2

[PRSN Ex-II Q7 / NRJ Multi Q8]

Which of the following cells have  $E^\circ_{\text{cell}} > 0$  (spontaneous under standard conditions)?

- (a)  $\text{Zn} | \text{Zn}^{2+} || \text{Cu}^{2+} | \text{Cu}$ ;  $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76$ ,  $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$
- (b)  $\text{Cu} | \text{Cu}^{2+} || \text{Ag}^+ | \text{Ag}$ ;  $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34$ ,  $E^\circ(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$
- (c)  $\text{Ag} | \text{Ag}^+ || \text{Cu}^{2+} | \text{Cu}$  (reverse of above)
- (d)  $\text{Pt}, \text{H}_2 | \text{H}^+(\text{pH}=7) | \text{O}_2(1 \text{ bar}), \text{Pt}$ ;  $E^\circ(\text{O}_2/\text{H}_2\text{O}) = +1.23 \text{ V}$

### MC.3

[NA Level-3 Q5 / NRJ Ex-II Multi Q12]

In the electrolysis of an aqueous NaCl solution with inert platinum electrodes, which of the following statements are **correct**?

- (a) In dilute solution,  $\text{H}_2$  is produced at the cathode.
- (b) In dilute solution,  $\text{O}_2$  is produced at the anode.
- (c) In concentrated solution,  $\text{Cl}_2$  is produced at the anode.
- (d) Na metal is deposited at the cathode in both dilute and concentrated solutions.

**MC.4****[NRJ Ex-II Multi Q16]**

Which of the following are correct statements about corrosion of iron?

- (a) It is an electrochemical process involving local anodic and cathodic regions.
- (b) Salt water accelerates corrosion by increasing ionic conductivity.
- (c) Connecting iron to a copper plate slows corrosion of iron.
- (d) Rusting requires both oxygen and water simultaneously.

**MC.5****[PRSN Ex-II Q12 / CNG-TH Advanced]**

For the half-reaction  $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$  ( $E^\circ = +1.51\text{ V}$ ), which of the following are true?

- (a) Lowering the pH increases  $E$  for this half-reaction.
- (b) Raising  $[\text{Mn}^{2+}]$  decreases  $E$ .
- (c) This half-reaction can oxidize  $\text{Cl}^-$  to  $\text{Cl}_2$  under standard conditions.
- (d)  $E^\circ$  for the reverse reaction is  $-1.51\text{ V}$ .

# Global Exercise Bank — Assertion–Reason

## Assertion–Reason Questions (All Chapters)

Use the following key:

- (a) Both A and R are true; R is the correct explanation of A.
- (b) Both A and R are true; R is NOT the correct explanation of A.
- (c) A is true; R is false.
- (d) A is false; R is true.

### AR.1

[GRB / ESS]

**A:** The specific conductance ( $\kappa$ ) of an electrolyte solution always decreases on dilution.

**R:** On dilution, the number of ions per unit volume decreases even though the degree of dissociation increases.

### AR.2

[CNG-TH / NRJ]

**A:** The standard hydrogen electrode (SHE) is used as the universal reference electrode.

**R:** The standard reduction potential of the SHE is arbitrarily defined as exactly zero at all temperatures.

### AR.3

[ATK / GRB]

**A:** The temperature coefficient of EMF,  $(\partial E/\partial T)_P$ , can be positive, negative, or zero.

**R:** The sign of  $(\partial E/\partial T)_P$  depends on the sign of the entropy change  $\Delta S$  for the cell reaction.

### AR.4

[GRB / CNG-DPP]

**A:** During the discharge of a lead-acid battery, the density of sulfuric acid decreases.

**R:** During discharge, sulfuric acid is consumed at both electrodes to form  $\text{PbSO}_4$ , and water is produced.

### AR.5

[CNG-TH / NRJ]

**A:**  $\text{KCl}$ ,  $\text{KNO}_3$ , and  $\text{NH}_4\text{NO}_3$  are preferred filling materials for salt bridges.

**R:** These salts have nearly equal ionic mobilities for cation and anion ( $t_+ \approx t_-$ ), which minimizes liquid junction potential.

### AR.6

[ATK / CNG-TH Advanced]

**A:** The mobility of  $\text{H}^+$  in aqueous solution ( $3.62 \times 10^{-7} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ ) is approximately 5 times larger than a typical ion like  $\text{Na}^+$  ( $5.19 \times 10^{-8} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ ).

**R:** The anomalously high mobility of  $\text{H}^+$  is due to the Grotthuss (proton-hopping) mechanism, where  $\text{H}^+$  is transferred sequentially along a chain of hydrogen-bonded water molecules rather than migrating as a physical particle.

**AR.7****[GRB / CNG-DPP DPP-5]**

**A:** Galvanized (zinc-coated) steel remains protected from rusting even when the zinc coating is scratched.

**R:** Tin-coated steel (tinned iron / food-grade steel) corrodes faster than uncoated iron when the tin layer is scratched.

# Global Exercise Bank — Linked Comprehension Passages

## Linked Comprehension / Case-Study Questions (All Chapters)

Paragraph-based and scenario-linked problem sets with multi-step analytical reasoning. Sources: [NA] Level-3 Comprehensions, [NRJ] Ex-II Comprehension, [PRSN] Advanced Passages.

### Passage 1: Debye–Hückel–Onsager Theory and High-Field Effects

The Debye–Hückel–Onsager (DHO) equation for a 1 : 1 univalent electrolyte in dilute aqueous solution at 25°C is given by:

$$\Lambda_m = \Lambda_m^\circ - (A + B\Lambda_m^\circ)\sqrt{c}$$

where  $A$  accounts for the electrophoretic drag of the solvent countercurrent and  $B$  accounts for the relaxation (asymmetry) field retardance caused by the finite relaxation time  $\tau$  of the ionic atmosphere. Under ordinary conditions, the external electric field is small ( $E \sim 1\text{--}10\text{ V cm}^{-1}$ ), so the velocity of ion migration is small compared to the thermal motion, allowing the ionic atmosphere to be treated as perturbed linearly.

However, when an extremely high potential gradient is applied ( $E > 10^5\text{ V cm}^{-1}$ ), an ion moves so rapidly that it travels a distance greater than the thickness of the ionic cloud ( $1/\kappa$ ) within the relaxation time  $\tau$ . Consequently, the ion escapes its retarding counter-ion cloud, and both asymmetry and electrophoretic effects diminish to zero (the **Wien effect**). Furthermore, at very high alternating frequencies ( $\nu > 10^6\text{ Hz}$ ), the direction of the field alternates faster than the relaxation time  $\tau$ , so the ionic atmosphere remains essentially spherically symmetric, yielding an increased conductivity (the **Debye–Falkenhagen effect**).

#### CP.1.1

[NA Level-3 / ATK]

Which of the following statements correctly explains the Wien effect?

- (a) High electric field increases the thermal ionization energy of solvent molecules.
- (b) Under high electric field gradient, an ion traverses a distance greater than the radius of its ionic atmosphere within the relaxation time, stripping off the cloud.
- (c) High voltage destroys the hydration shell of the ions permanently.
- (d) Viscosity of water drops to zero at high electric potential gradients.

**CP.1.2****[ATK / PRSN]**

The Debye–Falkenhagen effect predicts that when the frequency of the alternating current is increased beyond  $\sim 10$  MHz, the molar conductance of a strong electrolyte solution:

- (a) Decreases to zero because ions cannot follow the field oscillations.
- (b) Remains unchanged because relaxation time is independent of frequency.
- (c) Increases because the ionic atmosphere has no time to form an asymmetric retardance behind the oscillating ion.
- (d) First decreases and then approaches infinity.

**CP.1.3****[NRJ Ex-II / NA]**

For a 0.0004 M solution of KCl at 25°C,  $\Lambda_m = 148.2 \text{ S cm}^2 \text{ mol}^{-1}$  and  $\Lambda_m^\circ = 149.8 \text{ S cm}^2 \text{ mol}^{-1}$ . The experimental Onsager slope  $b_{\text{obs}}$  in  $\text{S cm}^2 \text{ mol}^{-1} (\text{mol L}^{-1})^{-1/2}$  is:

- (a) 80
- (b) 40
- (c) 160
- (d) 0.80

## Passage 2: Concentration Cells and Liquid Junction Potential

A concentration cell consists of two half-cells with identical electrodes and electrolytes but differing electrolyte activities or electrode pressures. Concentration cells can be classified into:

- **Electrode concentration cells:** Differing gas pressures or amalgam activities with uniform solution.
- **Electrolyte concentration cells:** Identical electrodes dipping into solutions of the same electrolyte at different mean activities  $a_1$  and  $a_2$  ( $a_2 > a_1$ ).

When the two solutions of different concentrations are in direct liquid-liquid contact without a salt bridge, a liquid junction potential ( $E_{lj}$ ) develops at the interface due to the unequal diffusion rates of cations and anions. For a 1 : 1 uni-univalent electrolyte ( $\text{M}^+\text{A}^-$ ), the EMF with transference ( $E_{wt}$ ) and without transference ( $E_{wot}$ ) are given by:

$$E_{wt} = 2t_- \frac{RT}{F} \ln \frac{a_2}{a_1}, \quad E_{wot} = \frac{RT}{F} \ln \frac{a_2}{a_1}$$

where  $t_-$  is the transport number of the anion. The liquid junction potential is defined as  $E_{lj} = E_{wt} - E_{wot} = (2t_- - 1) \frac{RT}{F} \ln \frac{a_2}{a_1}$ . A salt bridge containing KCl or  $\text{NH}_4\text{NO}_3$  in agar-agar gel is employed because  $t_+ \approx t_- \approx 0.5$  for these electrolytes, practically nullifying  $E_{lj}$ .

**CP.2.1****[ATK Topic 6C / NA]**

Consider the cell:  $\text{Pt} \mid \text{H}_2(1 \text{ atm}) \mid \text{HCl}(a_1) \parallel \text{HCl}(a_2) \mid \text{H}_2(1 \text{ atm}) \mid \text{Pt}$ , where a salt bridge eliminates the liquid junction. If  $a_2 = 10 a_1$  at 298 K, the cell EMF is:

- (a) 0.0591 V
- (b) 0.1182 V
- (c) 0.0296 V
- (d) 0 V

**CP.2.2****[PRSN / NRJ]**

If the cell with direct boundary  $\text{Pt}, \text{H}_2(1 \text{ atm}) \mid \text{HCl}(a_1) : \text{HCl}(a_2) \mid \text{H}_2(1 \text{ atm}), \text{Pt}$  is constructed without a salt bridge, and the transport number of  $\text{H}^+$  is  $t_+ = 0.82$ , then the liquid junction potential  $E_{lj}$  for  $a_2/a_1 = 10$  at 25°C is:

- (a) +0.0378 V
- (b) -0.0378 V
- (c) +0.0591 V
- (d) -0.0189 V

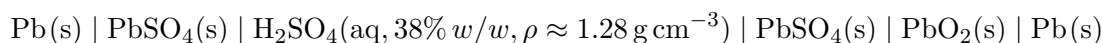
**CP.2.3****[NA Level-3 / CNG-TH]**

Why is  $\text{KNO}_3$  or  $\text{NH}_4\text{NO}_3$  chosen over  $\text{NaCl}$  for salt bridges in general electrochemistry?

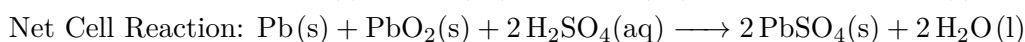
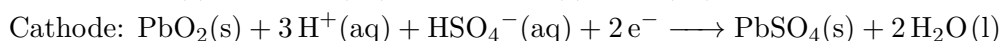
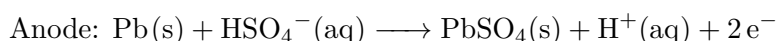
- (a)  $\text{Na}^+$  reacts with most metal cations to precipitate hydroxides.
- (b)  $\text{K}^+$  and  $\text{NO}_3^-$  have nearly equal ionic mobilities, so  $t_+ \approx t_- \approx 0.5$ , minimizing  $E_{lj}$  to under 1–2 mV.
- (c)  $\text{NaCl}$  is insoluble in agar-agar gel.
- (d) Nitrates undergo redox oxidation at the cathode during discharge.

### Passage 3: Thermodynamics of the Lead-Acid Accumulator

The lead-acid storage battery is the ubiquitous secondary electrochemical energy source in automobiles. Its cell representation in the fully charged state is:



During discharge, the electrode half-reactions and overall reversible cell process are:



For this reaction,  $n = 2$ ,  $E_{\text{cell}}^\circ \approx 2.05 \text{ V}$  at 298 K, and  $\left(\frac{\partial E}{\partial T}\right)_P \approx +1.5 \times 10^{-4} \text{ V K}^{-1}$ . During continuous discharge,  $\text{H}_2\text{SO}_4$  is consumed and  $\text{H}_2\text{O}$  is produced, causing the electrolyte density to fall towards  $\rho \approx 1.15 \text{ g cm}^{-3}$ , providing an accurate measure of the battery's state of charge via a hydrometer.

**CP.3.1****[GRB / PRSN]**

What is the entropy change  $\Delta S$  for the discharge of the lead-acid cell per mole of Pb consumed at 298 K?

- (a)  $+28.95 \text{ J K}^{-1}$
- (b)  $-28.95 \text{ J K}^{-1}$
- (c)  $+14.48 \text{ J K}^{-1}$
- (d)  $+57.90 \text{ J K}^{-1}$

**CP.3.2****[NA Level-3 / NRJ]**

A lead-acid battery delivers a steady current of 5.0 A for 10.0 hours. Assuming 100% current efficiency, the total mass of  $\text{H}_2\text{SO}_4$  consumed from the electrolyte is: ( $M_{\text{H}_2\text{SO}_4} = 98.08 \text{ g mol}^{-1}$ )

- (a) 183 g
- (b) 91.5 g
- (c) 366 g
- (d) 245 g

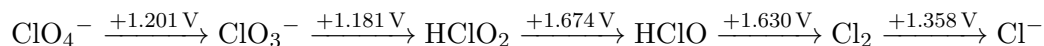
**CP.3.3****[CNG-TH / GRB]**

During the recharging process of a discharged lead storage cell, which of the following processes occurs at the anode (electrode connected to the positive terminal of the external DC source)?

- (a)  $\text{PbSO}_4\text{(s)}$  is reduced to spongy  $\text{Pb(s)}$ .
- (b)  $\text{PbSO}_4\text{(s)}$  is oxidized to  $\text{PbO}_2\text{(s)}$ .
- (c)  $\text{H}_2\text{SO}_4$  is consumed, lowering the electrolyte density further.
- (d) Hydrogen gas is evolved briskly from the  $\text{PbO}_2$  grid.

## Passage 4: Frost and Latimer Diagrams of Chlorine

The standard reduction potentials (in Volts) for various chlorine species in acidic aqueous solution ( $\text{pH} = 0$ ) are represented by the following Latimer diagram:

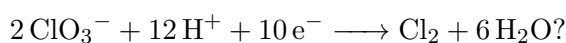


For any half-reaction connecting oxidation state  $N_1$  to  $N_2$  ( $\Delta N = N_1 - N_2$ ), the standard Gibbs energy is  $\Delta G^\circ = -\Delta N \cdot F \cdot E^\circ$ . On a Frost diagram, the quantity  $N \cdot E^\circ$  (which is proportional to  $\Delta G^\circ/F$  relative to the element in its standard state  $N = 0$ ) is plotted on the vertical axis against the oxidation number  $N$  on the horizontal axis. A species is thermodynamically unstable with respect to disproportionation if it lies *above* the chord connecting its two adjacent oxidation states, and stable against disproportionation if it lies *below* that chord.

### CP.4.1

[ATK Topic 6B / NA]

What is the standard reduction potential  $E^\circ$  for the direct conversion of chlorate to chlorine gas:



- (a) +1.47 V                      (b) +1.29 V                      (c) +1.38 V                      (d) +1.61 V

### CP.4.2

[ATK / NRJ]

From the Latimer diagram, does hypochlorous acid ( $\text{HClO}$ ) tend to undergo spontaneous disproportionation into  $\text{HClO}_2$  and  $\text{Cl}_2$  in 1 M acid at 298 K?

- (a) Yes, because  $E_{\text{right}}^\circ(1.630\text{ V}) < E_{\text{left}}^\circ(1.674\text{ V})$ .  
 (b) No, because  $E_{\text{cell}}^\circ = 1.630 - 1.674 = -0.044\text{ V} < 0$ .  
 (c) Yes, because all chlorine oxyacids are universally explosive.  
 (d) No, because oxidation states cannot change without a catalyst.

### CP.4.3

[ATK / NA]

On a Frost diagram, the slope of the line segment joining two oxidation states represents:

- (a) The equilibrium constant  $K$  of the reaction.  
 (b) The standard reduction potential  $E^\circ$  of the couple connecting those two states.  
 (c) The activation energy  $E_a$  of the redox interconversion.  
 (d) The rate of electron transfer across the electrical double layer.

# Global Exercise Bank — Matrix Match Questions

## Multi-Concept Matrix Matching Problems (All Chapters)

$4 \times 4$  and  $4 \times 5$  cross-topic matching matrices testing structural, mechanistic, and numerical principles. Sources: [NA] Level-2 Matchings, [PRSN] Advanced Matchings, [NRJ] Matrix-Match.

### MM.1

[NA Level-2 / PRSN]

Match the electrode system in Column I with its characteristic redox half-cell reaction or potential equation in Column II:

| Column I (Electrode System) | Column II (Characteristics / Formula)                                                              |
|-----------------------------|----------------------------------------------------------------------------------------------------|
| (A) Calomel Electrode       | (P) $E = E^\circ - \frac{2.303RT}{F} \text{pH}$                                                    |
| (B) Quinhydrone Electrode   | (Q) Reversible with respect to chloride ion; $E^\circ \approx +0.2422 \text{ V}$ for saturated KCl |
| (C) Glass Electrode         | (R) Involves keto-enol / quinone-hydroquinone organic redox equilibrium                            |
| (D) Hydrogen Electrode      | (S) Potential depends on exchange of alkali metal cations across a hydrated silicate membrane      |
|                             | (T) Reference electrode with $E^\circ = 0.000 \text{ V}$ at all temperatures by convention         |

**MM.2****[NRJ Matrix-Match / NA]**

Match the conductometric titration in Column I with the shape of the conductance curve / behavior of conductance prior to the equivalence point in Column II (titrant added from burette into analyte solution in beaker):

| Column I (Titration System)                         | Column II (Conductance Curve Characteristics)                                                                   |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| (A) HCl(aq) vs NaOH(aq)                             | (P) Sharp drop in conductance before equivalence point due to removal of highly mobile $\text{H}^+$ ions        |
| (B) $\text{CH}_3\text{COOH}(\text{aq})$ vs NaOH(aq) | (Q) Initial slight decrease followed by steady increase before equivalence point due to buffer formation        |
| (C) HCl(aq) vs $\text{NH}_4\text{OH}(\text{aq})$    | (R) Conductance remains almost constant after equivalence point due to weak ionization of excess titrant        |
| (D) $\text{AgNO}_3(\text{aq})$ vs KCl(aq)           | (S) Conductance remains almost constant before equivalence point, then rises sharply after                      |
|                                                     | (T) Conductance increases steeply after equivalence point due to excess unneutralized mobile $\text{OH}^-$ ions |

**MM.3****[PRSN / CNG-TH]**

Match the thermodynamic and kinetic electrochemical quantities in Column I with their correct expressions and dimensional relations in Column II:

| Column I (Quantity)                             | Column II (Expression / Identity)                                                                             |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| (A) Reversible cell reaction entropy $\Delta S$ | (P) $nFT \left( \frac{\partial E}{\partial T} \right)_P - nFE$                                                |
| (B) Reaction enthalpy $\Delta H$                | (Q) $nF \left( \frac{\partial E}{\partial T} \right)_P$                                                       |
| (C) Heat absorbed reversibly $q_{\text{rev}}$   | (R) $T\Delta S = nFT \left( \frac{\partial E}{\partial T} \right)_P$                                          |
| (D) Maximum thermodynamic efficiency $\eta$     | (S) $\frac{\Delta G}{\Delta H} = 1 - \frac{T\Delta S}{\Delta H} = \frac{E - T(\partial E / \partial T)_P}{E}$ |
|                                                 | (T) $-\frac{\Delta H}{T}$                                                                                     |

**MM.4****[GRB / NA Level-2]**

Match the electrolytic system in Column I with the dominant anodic and cathodic electrode processes in Column II:

| Column I (Electrolysis System) |                                                                         | Column II (Products / Reactions) |                                                                                                                                    |
|--------------------------------|-------------------------------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| (A)                            | Aqueous NaCl with inert Pt electrodes                                   | (P)                              | Cathode: $\text{Na}^+(\text{aq}) + \text{e}^- + \text{Hg} \longrightarrow \text{Na-Hg}(\text{amalgam})$                            |
| (B)                            | Aqueous NaCl with liquid mercury cathode                                | (Q)                              | Anode: $2 \text{Cl}^-(\text{aq}) \longrightarrow \text{Cl}_2(\text{g}) + 2 \text{e}^-$                                             |
| (C)                            | Aqueous $\text{CuSO}_4$ with active Cu electrodes                       | (R)                              | Cathode: $2 \text{H}_2\text{O}(\text{l}) + 2 \text{e}^- \longrightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$           |
| (D)                            | Concentrated $\text{H}_2\text{SO}_4(\text{aq})$ at high current density | (S)                              | Anode: $\text{Cu}(\text{s}) \longrightarrow \text{Cu}^{2+}(\text{aq}) + 2 \text{e}^-$ (anode dissolution)                          |
|                                |                                                                         | (T)                              | Anode: $2 \text{HSO}_4^-(\text{aq}) \longrightarrow \text{S}_2\text{O}_8^{2-}(\text{aq}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^-$ |

**MM.5****[CNG-TH / PRSN]**

Match the battery/cell in Column I with its active chemical constituents and electrolyte in Column II:

| Column I (Cell / Battery) |                                                | Column II (Active Materials / Electrolyte) |                                                                                                                     |
|---------------------------|------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| (A)                       | Leclanché Dry Cell                             | (P)                                        | Anode: Pb; Cathode: $\text{PbO}_2$ ; Electrolyte: 38% $\text{H}_2\text{SO}_4$                                       |
| (B)                       | Nickel-Cadmium (Nicad) Cell                    | (Q)                                        | Anode: Zn; Cathode: $\text{MnO}_2$ + Carbon; Electrolyte: moist paste of $\text{NH}_4\text{Cl}$ and $\text{ZnCl}_2$ |
| (C)                       | Lead-Acid Accumulator                          | (R)                                        | Anode: Cd; Cathode: $\text{NiO}(\text{OH})$ ; Electrolyte: aqueous KOH                                              |
| (D)                       | Alkaline $\text{H}_2$ - $\text{O}_2$ Fuel Cell | (S)                                        | Porous carbon electrodes impregnated with Pt/Pd catalyst; Electrolyte: concentrated hot KOH                         |
|                           |                                                | (T)                                        | Nominal single-cell operating potential $\approx 1.2$ – $1.4$ V                                                     |

# Global Exercise Bank — Numerical Value Questions

## Numerical Value Type Questions (Integer & Decimal Answers)

Exact quantitative problems testing rigorous computation, precision constants, and multi-step derivations. Sources: [NA] Level-2 & Level-3 Numerical, [NRJ] Integer Problems, [PRSN] Advanced Numerical.

### NV.1

[NA Level-2 Q14]

The resistance of a 0.01 M solution of acetic acid at 25°C measured in a cell with cell constant  $0.366 \text{ cm}^{-1}$  is  $2220 \Omega$ . If  $\Lambda_m^\circ(\text{HCl}) = 426 \text{ S cm}^2 \text{ mol}^{-1}$ ,  $\Lambda_m^\circ(\text{NaCl}) = 126 \text{ S cm}^2 \text{ mol}^{-1}$ , and  $\Lambda_m^\circ(\text{CH}_3\text{COONa}) = 91 \text{ S cm}^2 \text{ mol}^{-1}$ , calculate the acid dissociation constant  $K_a$  of acetic acid in the format  $X \times 10^{-5} \text{ mol L}^{-1}$ . What is the value of  $X$  rounded to two decimal places?

### NV.2

[NRJ Ex-II / NA]

For the cell  $\text{Zn(s)} \mid \text{Zn}^{2+}(\text{aq}, 0.1 \text{ M}) \parallel \text{Ag}^+(\text{aq}, 0.01 \text{ M}) \mid \text{Ag(s)}$ , the standard potentials are  $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$  and  $E^\circ(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$ . If  $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.0591}{n} \log_{10} Q$ , calculate  $E_{\text{cell}}$  in Volts at 298 K rounded to three decimal places.

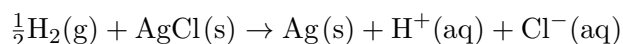
### NV.3

[ATK / NA Level-3]

The EMF of the cell  $\text{Pt} \mid \text{H}_2(1 \text{ bar}) \mid \text{HCl}(0.1 \text{ m}) \mid \text{AgCl(s)} \mid \text{Ag(s)}$  is measured as a function of temperature:

$$E(T) = 0.2366 - 4.86 \times 10^{-4}(T - 298) - 3.4 \times 10^{-6}(T - 298)^2 \quad (\text{in Volts})$$

Calculate the standard enthalpy of reaction  $\Delta H^\circ$  in  $\text{kJ mol}^{-1}$  for the cell reaction:

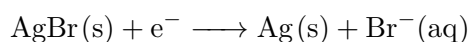


at  $T = 298 \text{ K}$ . (Take  $F = 96485 \text{ C mol}^{-1}$ , give answer rounded to one decimal place).

### NV.4

[NA Level-3 / PRSN]

The solubility product  $K_{sp}$  of  $\text{AgBr}$  is  $5.0 \times 10^{-13}$  at 25°C. Given  $E^\circ(\text{Ag}^+/\text{Ag}) = +0.799 \text{ V}$  and  $\frac{2.303RT}{F} = 0.0591 \text{ V}$ , calculate the standard reduction potential of the silver-silver bromide electrode:



in Volts, rounded to three decimal places.

**NV.5****[GRB / NRJ Ex-II]**

An aqueous solution of  $\text{CuSO}_4$  is electrolyzed between inert platinum electrodes with a current of 2.50 A for 96.5 minutes. The current efficiency is 80.0%. What mass of copper (in grams) is deposited at the cathode? (Molar mass of Cu =  $63.55 \text{ g mol}^{-1}$ ,  $F = 96500 \text{ C mol}^{-1}$ ). Give answer rounded to two decimal places.

**NV.6****[NA Level-3 / PRSN]**

A quinhydrone electrode is immersed in a solution of unknown pH at  $25^\circ\text{C}$  and coupled with a saturated calomel electrode ( $E_{\text{SCE}} = +0.242 \text{ V}$ ). The cell EMF is measured to be  $0.154 \text{ V}$  with the quinhydrone electrode functioning as the positive terminal (cathode). Given  $E^\circ(\text{Q}, 2\text{H}^+/\text{H}_2\text{Q}) = +0.699 \text{ V}$ , calculate the pH of the solution rounded to two decimal places.

**NV.7****[ATK / NA Level-2]**

The limiting molar conductivity of  $\text{K}^+$  in water at  $298 \text{ K}$  is  $\lambda^\circ(\text{K}^+) = 73.52 \text{ S cm}^2 \text{ mol}^{-1}$ . Calculate the absolute ionic mobility of the potassium ion  $u^\circ(\text{K}^+)$  in  $\text{cm}^2 \text{ V}^{-1} \text{ s}^{-1}$  in the format  $Y \times 10^{-4}$ . What is the value of  $Y$  rounded to two decimal places? (Use  $F = 96485 \text{ C mol}^{-1}$ ).

**NV.8****[NRJ / CNG-TH]**

How many total liters of gas ( $\text{H}_2$  and  $\text{O}_2$  combined) measured at STP ( $0^\circ\text{C}$ ,  $1 \text{ atm}$ , molar volume =  $22.4 \text{ L mol}^{-1}$ ) are produced by the electrolysis of acidified water when a constant current of  $4.0 \text{ A}$  is passed for  $1.0 \text{ hour}$ ? Assume 100% current efficiency. (Give answer rounded to two decimal places).

# Global Exercise Bank — JEE Archives (PYQs)

## JEE Main & JEE Advanced Archive (1995–2023)

Selected authentic previous years' questions illustrating IIT-JEE exam patterns, boundary conditions, and conceptual depth.

### Section A: JEE Advanced (IIT-JEE) Problems

#### PYQ.1

[JEE Advanced 2023]

For the cell  $\text{Pt(s)} \mid \text{H}_2(\text{g}, 1 \text{ bar}) \mid \text{H}^+(\text{aq}, 1 \text{ M}) \parallel \text{M}^{4+}(\text{aq}), \text{M}^{2+}(\text{aq}) \mid \text{Pt(s)}$ , the standard cell potential is 0.151 V at 298 K and the temperature coefficient  $\left(\frac{\partial E^\circ}{\partial T}\right)_P = -5.0 \times 10^{-4} \text{ V K}^{-1}$ . The value of  $\Delta H^\circ$  for the cell reaction in  $\text{kJ mol}^{-1}$  is: (Given:  $F = 96500 \text{ C mol}^{-1}$ )

- (a)  $-57.8$                       (b)  $+57.8$                       (c)  $-29.1$                       (d)  $-43.5$

#### PYQ.2

[JEE Advanced 2021]

Consider a 70% ( $w/w$ ) aqueous solution of  $\text{H}_2\text{SO}_4$  (density =  $1.6 \text{ g cm}^{-3}$ ). When 100 mL of this solution is electrolyzed using inert electrodes for 965 seconds with a steady current of 10.0 A, water is electrolyzed to evolve  $\text{H}_2$  and  $\text{O}_2$ . What is the mass percentage of  $\text{H}_2\text{SO}_4$  in the solution after electrolysis? (Assume only water is decomposed,  $F = 96500 \text{ C mol}^{-1}$ ).

- (a) 70.4%                      (b) 72.1%                      (c) 75.0%                      (d) 71.0%

#### PYQ.3

[JEE Advanced 2019 (Multiple Correct)]

For the electrochemical cell  $\text{Zn(s)} \mid \text{Zn}^{2+}(\text{aq}, 1 \text{ M}) \parallel \text{Cu}^{2+}(\text{aq}, 1 \text{ M}) \mid \text{Cu(s)}$ , which of the following statements are correct? (Given:  $E_{\text{Zn}^{2+}/\text{Zn}}^\circ = -0.76 \text{ V}$ ,  $E_{\text{Cu}^{2+}/\text{Cu}}^\circ = +0.34 \text{ V}$ )

- (a) The cell potential increases when the concentration of  $\text{Cu}^{2+}$  is increased.  
(b) The cell potential decreases when the concentration of  $\text{Zn}^{2+}$  is increased.  
(c) When an external opposing voltage  $E_{\text{ext}} > 1.10 \text{ V}$  is applied, electrons flow from Cu to Zn.  
(d) When  $E_{\text{ext}} = 1.10 \text{ V}$ , no current flows through the circuit and no chemical reaction occurs.

**PYQ.4****[JEE Advanced 2017]**

The conductance of a 0.0015 M aqueous solution of a weak monobasic acid was determined by using a conductivity cell with cell constant  $105 \text{ m}^{-1}$ . The conductance is  $5.0 \times 10^{-5} \text{ S}$ . The molar conductivity of the acid at infinite dilution is  $4.0 \times 10^{-2} \text{ S m}^2 \text{ mol}^{-1}$ . The degree of dissociation ( $\alpha$ ) of the acid is:

- (a) 0.0875                      (b) 0.0525                      (c) 0.125                      (d) 0.250

**Section B: JEE Main Archive****PYQ.5****[JEE Main 2023]**

Given below are two statements:

- **Statement I:** The limiting molar conductivity of weak electrolytes cannot be determined directly by extrapolation of molar conductivity plots ( $\Lambda_m$  vs  $\sqrt{c}$ ) to zero concentration.
- **Statement II:** Kohlrausch's law of independent migration of ions helps in calculating the limiting molar conductivity of weak electrolytes from the limiting molar conductivities of individual ions.

In the light of the above statements, choose the most appropriate answer:

- (a) Both Statement I and Statement II are correct.  
(b) Both Statement I and Statement II are incorrect.  
(c) Statement I is correct but Statement II is incorrect.  
(d) Statement I is incorrect but Statement II is correct.

**PYQ.6****[JEE Main 2022]**

How long (in seconds) must a current of 5.0 A be passed through a solution of gold salt  $\text{Au}^{3+}$  to plate out 1.97 g of gold on an ornament? (Atomic mass of Au =  $197 \text{ g mol}^{-1}$ ,  $F = 96500 \text{ C mol}^{-1}$ ).

- (a) 579 s                      (b) 193 s                      (c) 289.5 s                      (d) 1158 s

**PYQ.7****[JEE Main 2021]**

For a cell  $\text{Cu(s)} \mid \text{Cu}^{2+}(\text{aq}, 0.001 \text{ M}) \parallel \text{Ag}^{+}(\text{aq}, 0.01 \text{ M}) \mid \text{Ag(s)}$ , the standard potentials are  $E_{\text{Ag}^{+}/\text{Ag}}^{\circ} = +0.80 \text{ V}$  and  $E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = +0.34 \text{ V}$ . The cell potential at 298 K is: (Given:  $\frac{2.303RT}{F} = 0.059 \text{ V}$ )

- (a) 0.460 V                      (b) 0.430 V                      (c) 0.489 V                      (d) 0.401 V